MIL-C-81511E(AS)
23 May 1977
SUPERSEDING
MIL-C-81511C
5 June 1970
MIL-C-0081511D(NAVY)
18 October 1971

MILITARY SPECIFICATION

CONNECTORS, ELECTRICAL, CIRCULAR, HIGH DENSITY, QUICK DISCONNECT, ENVIRONMENT RESISTING: AND ACCESSORIES GENERAL SPECIFICATION FOR

This specification is approved for use by Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The MIL-C-81511 connectors have been declared NOT FOR USE IN FUTURE DESIGN after 30 April 1975 by decision of the Department of Defense. This does not preclude their use for maintenance, repair, or resupply purposes, nor does it preclude use in design circumstances wherein the connectors must interface with the connectors being supplied with Government Furnished Equipment.

1. SCOPE

1.1 Scope. This specification covers four series of electrical connectors (plugs and receptacles) with removable crimp and nonremovable solder contacts and accessories (see 6.1). Electrical, mechanical, and environmental features of these connectors include:

(1) Environment resisting at sea level and high altitude.
(2) Quick disconnect.
(3) RFI/EMI protection (includes shell to shell grounding spring members).
(4) High density insert arrangements.
(5) Low level circuit capacilities.
(6) Scoop proof.
(7) Fluid resistant class provided.
(8) High temperature class provided.
(9) Several voltage service ratings available.
(10) Individual contact release from the rear of the connector (Series 3 and 4 only).

1.1.1 Series description. The connectors are identified as series 1, 2, 3 or 4. All series are designed to prevent inadvertent electrical contact and to provide contact protection during mating. They can be mounted in the same size panel cutout. They are identified as follows:

Series 1: Long Shell. (Shell size 8 through 24). Scoop-proof when pins are installed in either the plug or the receptacle. Plugs contain the grounding spring members. Gang contact release system (GCR). (See figure 1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 83, Lakehurst, NJ 08733) by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Series 2: Lightweight, Short Shell. (Shell size 8 through 18.) Scoop-proof when pins are installed in the receptacle, not scoop-proof when pins are installed in the plug. Receptacles contain the grounding spring members. Gang contact release system (GCR). (See figure 1).

Series 3: Long Shell. (Shell size 8 through 24). Scoop-proof when pins are installed in either the plug or the receptacle. Plugs contain the grounding spring members. Individual contact release system (ICR). (See figure 1).

Series 4: Lightweight, Short Shell. (Shell size 8 through 18). Scoop-proof when pins are installed in the receptacle; not scoop-proof when pins are installed in the plug. Receptacles contain the grounding spring members. Individual contact release system (ICR). (See figure 1).

1.1.1.1 Intermateability. Series 1 is intermateable with series 3, and series 2 is intermateable with series 4. Series 1 and 3 are not intermateable with series 2 and 4.

1.1.1.2 Contact retention. Contacts are inserted and removed from the connector rear. (However, in series 1 and 2 connectors, the removal tool must be applied from the front.) See figure 1. Series 1 and 2 connectors have a "ganged" contact retention and release mechanism. Series 3 and 4 connectors have individual contact retention and release.

1.1.1.3 Weight and size. The individual contact release connectors (series 3 and 4) are shorter and lighter than the "gang release" connectors (series 1 and 2) respectively.

1.1.2 Temperature. These connectors are rated for operation within a temperature range specified for each class. The upper temperature limit is the maximum internal hot-spot temperature resulting from any combination of electrical load and ambient temperature. The low temperature limit is -65°C for all classes.

1.1.2.1 Variation of insulation resistance. Insulation resistance varies with temperature as indicated in table I and figure 2.

TABLE I. Variation of insulation resistance.

Class	Hot-spot temperature	Insulation resistance megohms (Min)
All	25°C	5000
E, P	150°C	2000
H	150°C	500
F, T, W	175°C	2000
D, G, L	175°C	500
A, C	200°C	2000
B	200°C	500

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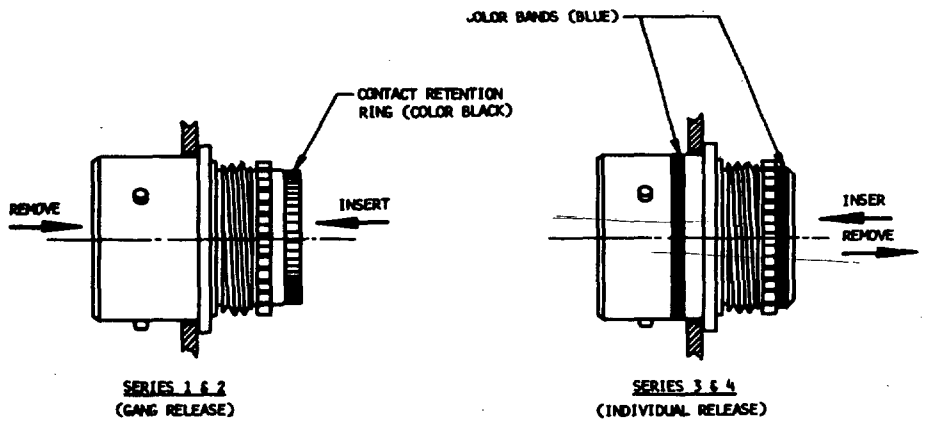


FIGURE 1. CONTACT INSERTION AND REMOVAL METHODS.

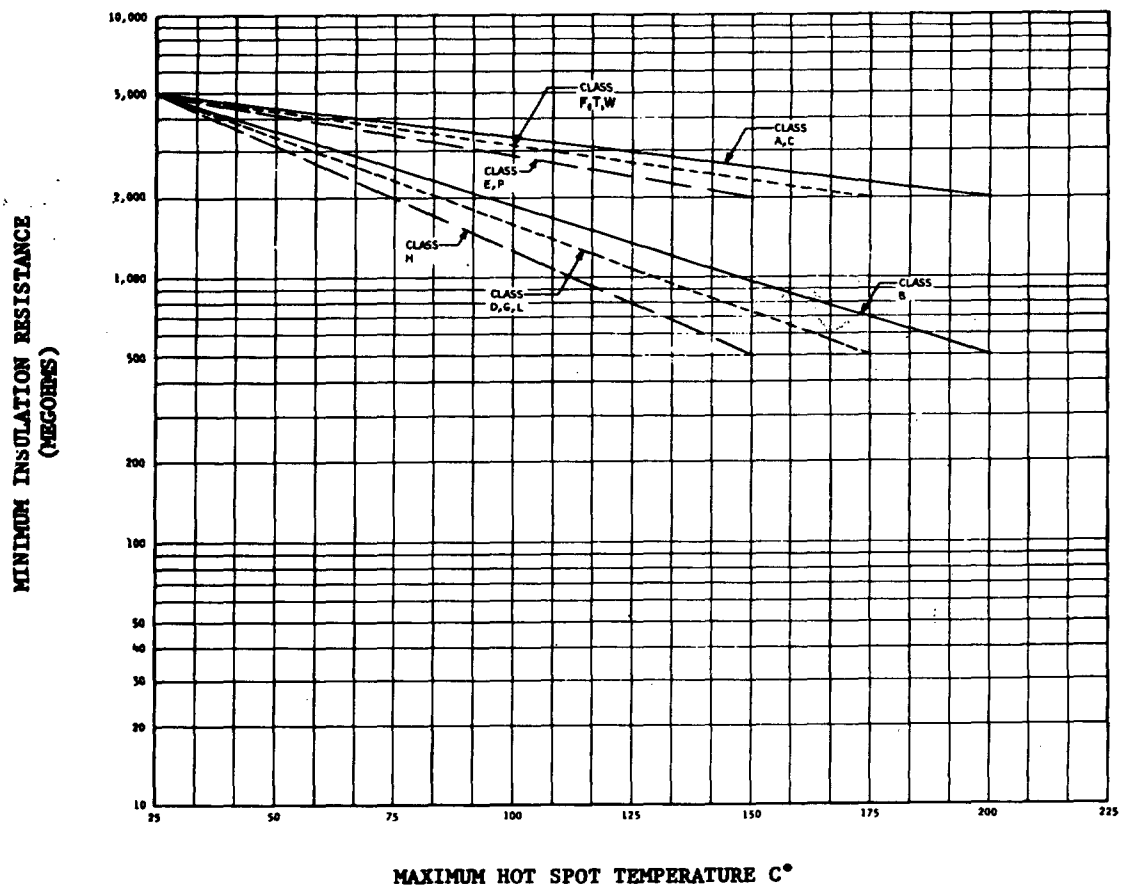


FIGURE 2. MINIMUM INSULATION RESISTANCE VS HOT SPOT TEMPERATURE.

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1.1.2.2 Variation of service life. Service life varies with temperature as indicated in figure 3 and table II.

TABLE II. Variation of service life.

Class	Hot-spot Temperature	Service life
E,H,P	25°C 150°C Max.	Continuous 1000 hours
D,F,G L,T,W	25°C 175°C Max	Continuous 1000 hours
A,B,C	25°C 200°C Max	Continuous 1000 hours

1/ class E,H, and P are inactive for new design.

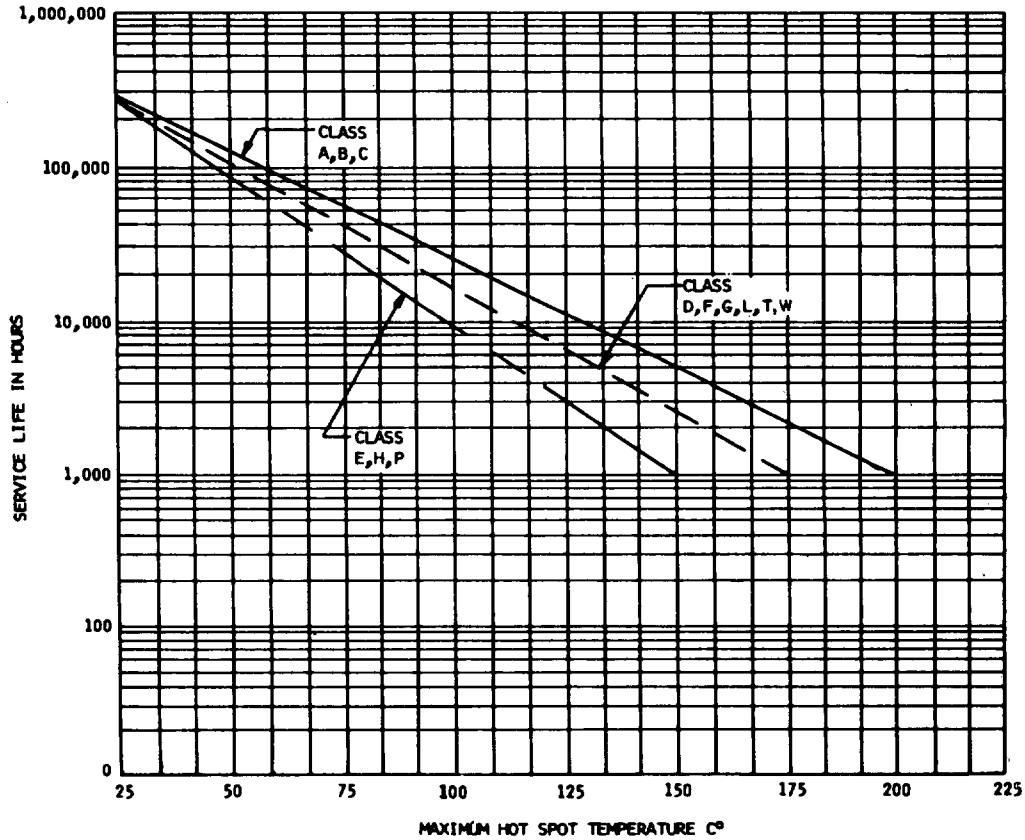


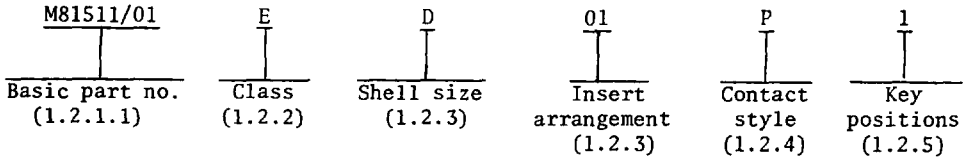
FIGURE 3. SERVICE LIFE VS HOT SPOT TEMPERATURE.

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1.2 Classification.

1.2.1 Military part number.

1.2.1.1 Military part number for connectors. The connector part number shall consist of the letter "M", the basic number of the applicable specification sheet, and the applicable alpha-numeric characters (see 3.1), formulated as in the following example; M81511/01ED01P1



1.2.1.2 Military part number for connector accessories. The part number for the accessories shall consist of the letter "M", the basic number of the specification sheet and assigned dash numbers as indicated on the applicable specification sheet (see 3.1) (example: M81511/14-08-01).

1.2.2 Class. The class of the connector shall be identified as shown in table III.

TABLE III. Connector class and series.

Class	Description	Applicable series	
		1 and 2	3 and 4
E	Grommet seal connectors (150°C operating temperature).	X	
F	Grommet seal connectors (175°C operating temperature), fluid resistant.	X	X
A	Grommet seal connectors (200°C operating temperature), fluid resistant.	X	X
H	Hermetic seal connectors (150°C operating temperature), solder terminated, ferrous alloy shell.	X	
D	Hermetic seal connectors (175°C operating temperature), solder terminated, ferrous alloy shell, fluid resistant.		X
G	Hermetic seal connectors (175°C operating temperature), solder terminated, corrosion resistant steel shell, fluid resistant.	X	
L	Hermetic seal connectors (175°C operating temperature), crimp terminated, ferrous alloy shell, fluid resistant.		X
B	Hermetic seal connectors (200°C operating temperature), solder terminated, corrosion resistant steel shell, fluid resistant.	X	
P	Potting seal connectors (150°C operating temperature).	X	
T	Potting seal connectors (175°C operating temperature), fluid resistant.	X	
C	Potting seal connectors (200°C operating temperature), fluid resistant.	X	
W	Grommet seal connectors (175°C operating temperature), fluid resistance, 500 hour salt spray.		X

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1.2.2.1 Fluid resistance. The fluid resistance capabilities for the various connector class are shown below:

Class	Fluid resistance capabilities
E, H, and P	MIL-H-5606 and MIL-L-23699
A, B, C, D, F, G, L, T, and W	MIL-L-7808, MIL-L-23699, MIL-H-5606, MIL-T-5624, (Grade JP-5), Hydraulic Fluid, MIL-A-8243, MIL-C-25769, MIL-G-3056 (Type 1), Coolant Dielectric Fluid, Synthetic Silicate Ester Base, Solvents (a), (b), and (c) of MIL-STD-202, Method 215.

1.2.3 Shell size and insert arrangement. The shell size and insert arrangement shall be identified by a two-part code. The first part shall be a single letter indicating the insert arrangement in accordance with the applicable insert arrangement MS standard. Paragraph 6.8 provides a summary of insert arrangements and shell sizes.

TABLE IV. Shell size code.

Shell size	Code letter
8	A
10	B
14	D
16	E
18	F
20	G
22	H
24	J

1.2.4 Contact designator.

1.2.4.1 Connector with contacts. The following designators are used to indicate a full complement of applicable power contacts (see 3.4.1):

P - Pin contacts
S - Socket contacts
C - Feedthrough contacts

1.2.4.2 Connector without contacts. The following designators are used to indicate a connector less contacts. They will be used only when other than a full complement of power contacts are to be installed in the connector. Examples of this are shielded, thermocouple and coaxial contacts (see 6.2).

A - Less pin contacts
B - Less socket contacts

1.2.5 Key position. Key or keyway position shall be identified by a digit 1 thru 6 (see 3.4.4.1).

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1.2.6 Accessories.

F adapter
Type I - Straight cable clamp, for open wiring.
Type II - 90° Cable clamp, for open wiring.

J adapter
Type I - Straight adapter for jacketed cable.
Type II - 45° adapter for jacketed cable.
Type III - 90° adapter for jacketed cable.

Stowage receptacle -- Dummy receptacle shell for stowing connector plugs.
Protective covers -- Cover for protecting connector plugs or receptacles.

1.3 Wire range accommodations. The wire ranges given in Table V shall be accommodated by the connectors and contact wire barrels as indicated.

TABLE V-A. Wire range accommodations, Series 1 and 2.

Wire barrel size	Wire size	OD of finished wire, (inch) 1/ (normal size)		Oversize wire 2/	Wire type (reference) 3/
28	32 30 28	0.030 min 0.054 max	.042 nominal	0.064 max	
22	26 24 22	0.030 min 0.054 max	.042 nominal	0.064 max	MS21985 MIL-W-81381/1 MIL-W-16878, Type E
20	24 22 20	0.041 min 0.081	.062 nominal	0.085 max	MS21985 MS21985 MIL-W-81044/5
16	18 16	0.066 min 0.101 max	.083 nominal	0.109 max	MS21985 MIL-W-81044/5
12	14 12	0.097 min 0.135 max	.106 nominal	0.139 max	MS18113 MIL-W-81044/5

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TABLE V-B. Wire range accommodations, Series 3 and 4.

Wire barrel	Wire size	O.D. of finished wire (inch) 1/	Wire type (reference) 3/
28	32 30 28	0.030 minimum 0.054 maximum	
22	28 26 24 22	0.030 minimum 0.054 maximum	MS21985 MIL-W-81381/1 MIL-W-16878, Type E
20	24 22 20	0.040 minimum 0.074 maximum	MS21985 MIL-W-81044/5 MIL-W-16878, Type E and EE
16	20 18 16	0.060 minimum 0.103 maximum	MS21985 MIL-W-81044/5 MIL-W-16878, Type E and EE
12	14 12	0.097 minimum 0.135 maximum	MS18113 MIL-W-81044/5 MIL-W-16878, Type and EE

1/ All contacts may be assembled with either minimum or maximum normal diameter wire as shown or in any combination thereof.
2/ 50 percent of the contacts in a given insert arrangement may be assembled with oversize wire up to and including the maximum diameter shown; however, the remaining 50 percent shall not exceed the nominal finished wire diameter. When using groupings of oversize wire, they shall be randomly distributed.
3/ Wire numbers are for reference only and are not intended to restrict the use of acceptable wire-types with these connectors.

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

QQ-A-225	-	Aluminum Alloy Bar, Rod, Wire of Special Shapes, Rolled, Drawn, or Cold Finished, General Specification for.
QQ-A-367	-	Aluminum Alloy Forging, Heat Treated.
QQ-A-591	-	Aluminum Alloy Die Castings.
QQ-P-35	-	Passivation Treatments for Austenitic, Ferritic, and Martensitic Corrosion-Resisting Steel (Fastening Devices).
QQ-P-416	-	Plating, Cadmium (Electrodeposited).
UU-P-553	-	Paper, Wrapping Tissue.
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-676	-	Boxes, Setup.
PPP-T-60	-	Tape: Packaging, Waterproof.
PPP-T-76	-	Tape, Packaging, Paper (for Carton Sealing).

MILITARY

MIL-P-116	-	Preservation-Packaging, Methods of.
MIL-G-3056	-	Gasoline, Automotive, Combat
MIL-H-5606	-	Hydraulic Fluid, Petroleum Base, Aircraft, Missile and Ordnance.
MIL-T-5624	-	Turbine Fuel, Aviation Grades JP-4 and JP-5.
MIL-S-7742	-	Screw Threads, Standard, Optimum Selected Series, General Specification for.
MIL-L-7808	-	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base.
MIL-A-8243	-	Anti-icing and Deicing-Defrosting Fluid.
MIL-L-8937	-	Lubricant, Solid Film, Heat Cured.
MIL-T-10727	-	Tin Plating: Electrodeposited or Hot-Dipped, for Ferrous and Nonferrous Metals.
MIL-W-16878	-	Wire, Electrical, Insulated, High Temperature.
MIL-I-17214	-	Indicator, Permeability: Low-Mu (Go-No-Go).
MIL-C-22520	-	Tool, Crimping, Contact, Electric Connector.
MIL-C-23216	-	Contact, Electric Connector.
MIL-C-25769	-	Cleaning Compound, Aircraft Surface, Alkaline Waterbase.
MIL-C-26074	-	Coating, Electroless Nickel.
MIL-C-39029	-	Connector, Electrical Contacts, General Specification for.
MIL-L-23699	-	Lubricating Oil, Aircraft Turbine Engines, Synthetic Base.
MIL-C-45662	-	Calibration System Requirements.
MIL-G-45204	-	Gold Plating, Electrodeposited.
MIL-W-81044	-	Wire, Electric Crosslined Polyaldene Insulated, Copper.

(See supplement 1 for list of applicable specification sheets).

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STANDARDS

MILITARY

MIL-STD-104	-	Limits for Electrical Insulation Color.
MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of U. S. Military Property.
MIL-STD-147	-	Palletized and Containerized Unit Loads 40" x 48" 4-way (Partial) Pallet Skids, Runners, or Pattet-Type Base.
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.
MIL-STD-1285	-	Marking of Electrical and Electronic Parts.
MIL-STD-1344	-	Test Methods for Electrical Connectors.
MS3159	-	Tool, Extract, Unwired, Contact, Electric Connector, Size 23, 22, 20, 26 and 12.
MS3160	-	Tool, Insert Extract, Electric Connector.
MS3191	-	Tool, Hand, Crimp, Class 1 - for Contacts of Electric Connector (MIL-T-22520).
MS3197	-	Gage Pin, for Socket Contact Engagement Test.
MS3198	-	Tool, Hand, Crimp, Miniature, Class 1 - for Electric Connector Contacts.
MS3321	-	Test Fixture, Probe Damage, Socket Contact, Connector, Electric.
MS3322	-	Gasket, Conductive, Electrical Connector.
MS3323	-	Tool, Insertion, Contact, Connector, Electrical.
MS3342	-	Tool, Removal, Pin Contact, Connector, Electrical.
MS3343	-	Contact, Electric, Socket, Crimp, Removable.
MS3344	-	Tool, Removal, Socket Contact, Connector, Electrical.
MS3434	-	Test Gage, Contact Location and Retention Feature, MIL-C-0081511 Series 1 and 2 Connectors.
MS3435	-	Test Gage, Contact Location and Retention Feature, MIL-C-0081511 Series 3 and 4 Connectors.
MS3462	-	Test Gage, MIL-C-0026482 Series 2 (Class N), MIL-C-81707 Series 3 (Class N) and MIL-C-0081511 Series 3 and 4 (Class L) Contact Retention Feature.
MS9068	-	Packing, Preformed - AMS 3304, O Ring.
MS21985	-	Wire, Electric, Extruded TFE - Fluorcarbon-insulated, Silver Coated Copper, 600 Volt.
MS90460	-	Contact, Electric, Socket, Crimp, Removable.
MS90461	-	Contact, Electric, Pin, Crimp, Removable.
MS90484	-	Gasket, Flange Mount, Connector, Electric.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSITUTTE, INC.

ANS B46.1 - 1961 Surface Texture (Surface Roughness, Waviness, and Lay).
ANS Y14.5 - 1966 Dimensioning and Tolerancing for Engineering Drawings.

(Application for copies should be addressed to American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018).

(Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publication and Forms Center, 5801 Tabor Ave., Philadelphia, PA 19120).

ELECTRONIC INDUSTRIES ASSOCIATION

EIA STD RS359 - Standard Colors for Color Identification and Coding.

3. REQUIREMENTS

3.1 Specification sheets. The individual part requirements shall be as specified herein and in accordance with the applicable specification sheets listed in the supplement to this specification.

3.1.1 Precedence. In the event of conflict between the dimensional requirements of this specification and the associated specification sheets listed in the supplement, this specification shall govern. In the event of conflict between other requirements, of this specification and the associated specification sheets, the latter shall govern. In the event of conflict between this specification and the applicable documents (2.), this specification shall govern. The associated specification sheets take precedence over all documents referenced thereon.

3.2 Qualification. Connector assemblies and accessories furnished under this specification and the applicable specification sheet shall be products which are qualified for listing on the applicable Qualified Products List (QPL) at the time set for the opening of bids (see 4.4 and 6.3).

3.2.1 Use of military part numbers. Military part numbers shall not be applied to a product, except for qualification test samples (see 6.3), until notification has been received from the activity responsible for qualification that the product has been approved for listing on the Qualified Products List.

3.3 Materials. Materials shall be as specified herein; however, when a definite material is not specified, a material shall be used which will enable the connectors and accessories to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty for acceptance of the finished product.

3.3.1 Dissimilar metals. When dissimilar metals are employed in intimate contact with each other, suitable protection against electrolytic corrosion shall be provided as specified in Requirement 16 of MIL-STD-454.

3.3.2 Nonmagnetic materials (except class B, D, G, H and L). All parts shall be made from materials which are classed as nonmagnetic (see 3.5.20).

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3.3.3 Fungus resistant. Materials used in the construction of these connectors shall be fungus inert in accordance with requirement 4 of MIL-STD-454.

3.3.4 Contact materials. Contacts shall be made of suitably conductive copper alloys except that Class B, D, G, H and L contacts may be ferrous alloy. All contacts shall be protected against electrolytic corrosion.

3.3.4.1 Contact plating (solder type). Contacts shall be gold plated .00005 inch thick minimum in accordance with MIL-G-45204 over a suitable underplate, except silver underplate shall not be used. Gold plate shall be type I or II and hardness to meet the performance requirements specified herein. Accessory members of the socket contact need not be plated, but shall comply with the requirements for dissimilar metals specified in 3.3.1.

3.3.4.2 Contact plating (crimp type). Crimp contact shall be plated in accordance with MIL-C-23216 or the applicable MIL-C-39029 specification sheet.

3.3.5 Dielectric materials.

3.3.5.1 Insert and grommet. Insert and grommet materials shall be high grade dielectric, having hardness, electrical and mechanical characteristics suitable for the purpose intended.

3.3.5.1.1 Insert faces. The mating face of socket inserts shall be a hard non-resilient material. The mating face of pin inserts shall be a resilient material within a shore A durometer range of 40 to 50.

3.3.6 Connectors and accessories materials.

3.3.6.1 Connectors (class A, C, E, F, P and T) and accessories. Shells for connectors (except class B, D, G, H and L) and metal accessories (see 1.2.6) shall be fabricated of high grade aluminum such as Alloy 6061. Temper T6 in accordance with QQ-A-367.

3.3.6.2 Connectors (class B, D, G, H and L). Shells for class B, D, G, H, and L connectors shall be as specified (see 3.1).

3.3.6.3 Grounding spring members. Grounding springs shall be made from a spring temper copper alloy suitably protected to prevent corrosion.

3.3.6.4 Bayonet pins. Bayonet pins shall be made of corrosion resistant stainless steel.

3.3.6.5 Other component parts. Other metal component parts may be fabricated from aluminum conforming to QQ-A-591 or QQ-A-225; however, the rear retention ring may be made from another material and shall comply with the requirements for dissimilar metals (see 3.3.1).

3.3.7 Connectors and accessories finish. The finish of all connector shells and accessories shall be electrically conductive.

3.3.7.1 Connectors (class E and P) and accessories. Aluminum parts shall be cadmium plated in accordance with QQ-P-416, type II, class 3. A preliminary plating of another metal is permissible. The resulting finish shall be olive drab (light to dark).

3.3.7.2 Connectors (class A, C, F and T) and accessories. Aluminum parts shall be nickel plated in accordance with MIL-C-26074, class III, grade A.

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3.3.7.3 Connectors (class D, H and L). Ferrous alloy shells shall be tin plated in accordance wtih MIL-T-10727, type 1 or 2. Preliminary plating of another metal is permissible. The resultant finish shall be suitable for soldrring to a mounting surface.

3.3.7.4 Connectors (class B and G). Corrosion resistant steel shells shall be passivated in accordance with QQ-P-35.

3.3.7.5 Contact retention ring. The resultant finish of the contact retention ring shall be black (see figure 12).

3.4 Design and construction. Connectors shall be of the design, construction, and physical dimensions specified (see 3.1). Connectors shall be so designed that neither the pins nor the sockets will be damaged during normal mating of counterpart connectors. Dimensioning and tolerancing shall be in accordance with ANS Y14.5-1966. Connector interface dimensions shall conform to figures 4 thru 11. Connector back end configuration shall conform to figure 12A or 12B, as applicable.

3.4.1 Contact design. Contacts shall conform to MIL-C-39029, except as otherwise specified herein for solder contacts and through-bulkhead contacts.

3.4.1.1 Solder contacts. Solder contacts shall be as specified (see 3.1), and shall be nonremovable from the insert. Solder cups shall be so designed that during soldering no components will be damaged and no liquid solder shall escape.

3.4.1.2 Crimp contacts. Crimp contacts shall conform to the applicable military part number specified in table VI and shall be qualified to MIL-C-39029. The contact style shall be as designated in the connector part number, and the size and number of contacts shall be as shown in the applicable insert drawing. The contacts for use in each connector series are shown in table VI.

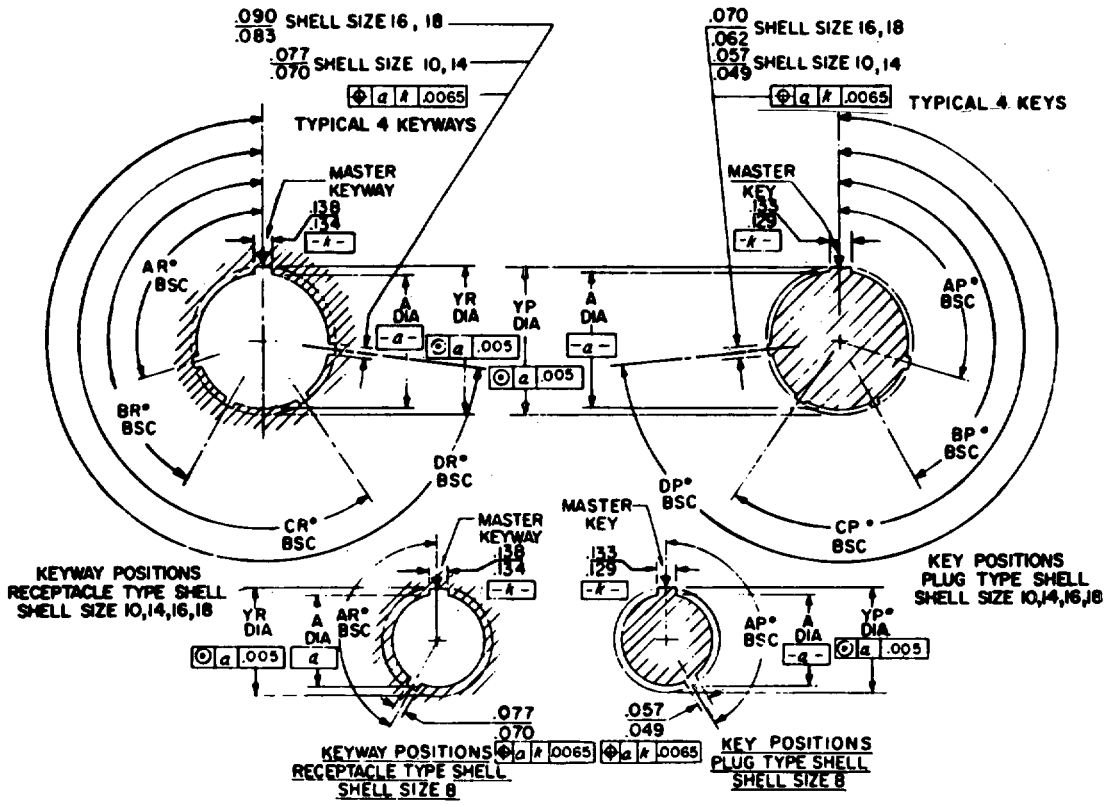
TABLE VI. Crimp contacts; insertion and removal tools.

<table>
<tr><th>Connector series</th><th>Crimp contact</th><th>Insertion tool</th><th>Removal tool</th></tr>
<tr><td>Series 1</td><td>Pin - MS90461
Soc - MS3343</td><td rowspan="2">MS3323</td><td>MS3342
MS3344</td></tr>
<tr><td>Series 2</td><td>Pin - MS90461
Soc - MS90460</td><td>MS3342
MS3344</td></tr>
<tr><td>Series 3</td><td>Pin - M39029/18
Soc - M39029/17</td><td colspan="2" rowspan="3">Insertion and removal tool
MS3160</td></tr>
<tr><td>Series 4</td><td>Pin - M39029/18
Soc - M39029/16</td></tr>
<tr><td>Series 3 & 4 class L</td><td>Soc - M39029/15</td></tr>
</table>

3.4.1.3 Contacts (class B, D, G, H, and L connectors). Nonremovable contacts for class B, D, G, H, and L connectors shall be as specified (see 3.1). (For class L crimp contacts, see 3.4.1.2).

3.4.2 Insert design and construction. Inserts shall be secured to prevent rotation or movement within the shell. The inserts shall be nonremovable from the shell and shall be installed in the position specified on the applicable specification sheet. The insert shall contain no metallic components. The contact retention mechanism shall consist entirely of dielectric material. The socket insert shall be designed to prevent a bent pin in a mating counterpart insert from penetrating the insert alongside the socket contact. Inserts for series 3 and 4 connectors shall be of voidless construction, eliminating all air paths between contacts.

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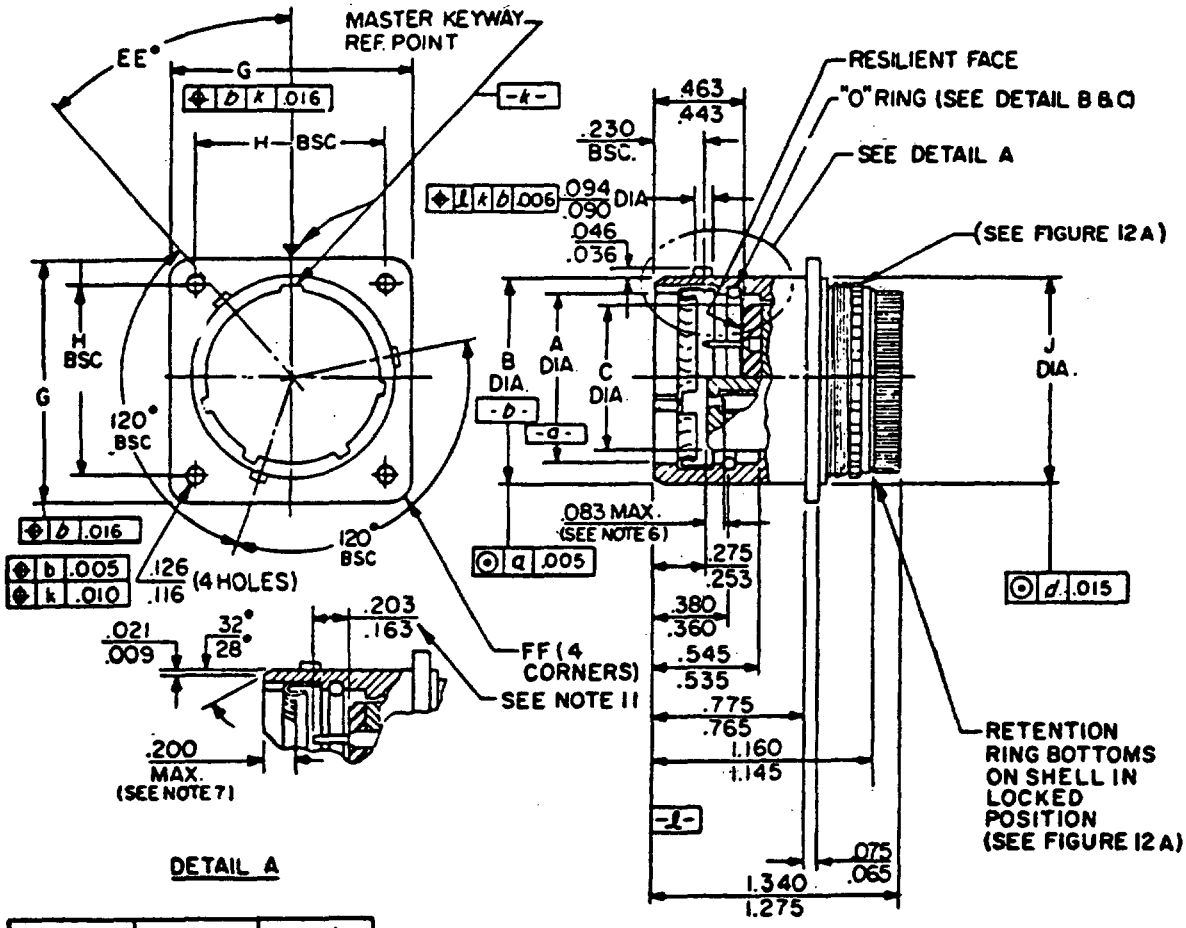
Shell size	Key & keyway arrangement	AR° or AP° BSC	BR° or BP° BSC	CR° or CP° BSC	DR° or DP° BSC	YR dia	YP dia
8	1	128	—	—	—	.462 (11.73) .450 (11.43)	.440 (11.18) .432 (10.97)
	2	190	—	—	—		
	3	205	—	—	—		
	4	220					
	5	235					
	6	250					
10 and 14	1	105	140	215	265	.587 (14.91) .575 (14.61)	.565 (14.35) .557 (14.15)
	2	102	132	248	320		
	3	80	118	230	312		
	4	35	140	205	275	.837 (21.26) .825 (20.96)	.815 (20.70) .807 (20.50)
	5	64	155	235	304		
	6	35	115	220	270		
16 and 18	1	105	140	215	265	.963 (24.46) .951 (24.16)	.941 (23.90) .933 (23.70)
	2	18	149	192	259		
	3	92	152	222	342		
	4	84	152	204	334	1.087 (27.61) 1.075 (27.31)	1.065 (27.05) 1.057 (26.85)
	5	24	135	199	240		
	6	98	152	268	338		

INCHES	MM
.005	.13
.008	.20
.049	1.24
.057	1.45
.062	1.57
.070	1.78
.077	1.96
.083	2.11
.090	2.29
.129	3.28
.133	3.38
.134	3.40
.138	3.51

NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Dimensions apply after plating.
5. For datum 'a' refer to figures 5, 6 and 7.

FIGURE 4. Connector, electrical, position key and keyways, mating (Series 2 & 4).

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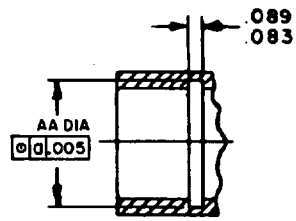


Contact size	X DIA	Y DIA
23 - 22	.035 (.89) .032 (.81)	.075 (1.91) .065 (1.65)
20 - 20	.053 (1.35) .050 (1.27)	.085 (2.16) .080 (2.03)
16 - 16	.077 (1.96) .074 (1.88)	.108 (2.74) .103 (2.62)
12 - 12	.130 (3.30) .126 (3.20)	.170 (4.32) .166 (4.22)

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.003	.08	.040	1.02	.094	2.39	.230	5.84	.535	13.59
.005	.13	.046	1.17	.116	2.95	.253	6.43	.545	13.84
.006	.15	.065	1.65	.126	3.20	.275	6.99	.765	19.43
.009	.23	.067	1.70	.157	3.99	.290	7.37	.775	19.69
.010	.25	.073	1.85	.163	4.14	.360	9.14	1.145	29.08
.015	.38	.075	1.91	.169	4.29	.380	9.65	1.160	29.46
.016	.41	.083	2.11	.200	5.08	.443	11.25	1.275	32.39
.021	.53	.090	2.29	.203	5.16	.463	11.76	1.340	34.04

FIGURE 5A. Connector, receptacle, electrical, flange mount (series 2).

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DETAIL C

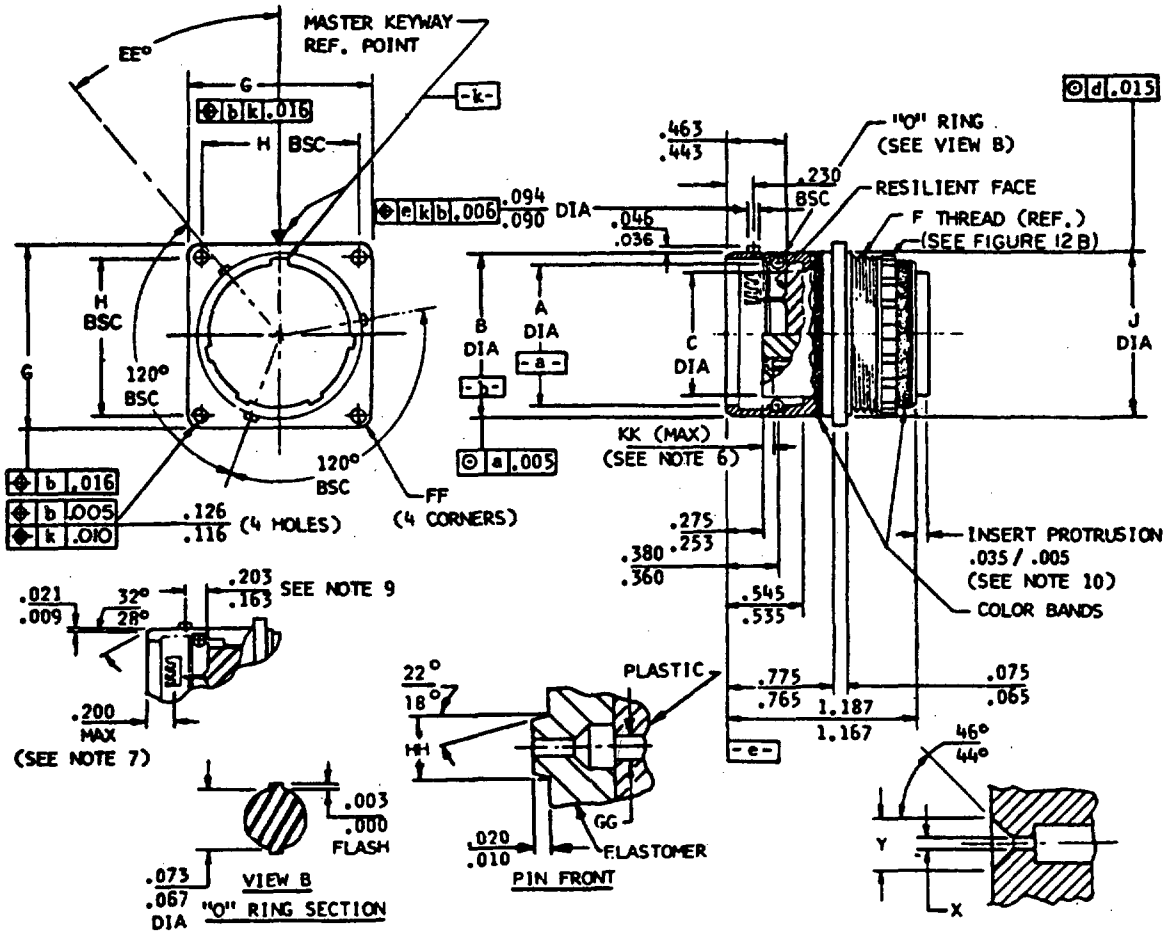
SIZE	AA DIA
8	.493 (12.52) .488 (12.40)
10	.618 (15.70) .613 (15.57)
14	.868 (22.05) .863 (21.92)
16	.994 (25.25) .989 (25.12)
18	1.118 (28.40) 1.113 (28.27)

Shell size	A dia	B dia	C dia	G	H BASIC	J dia	FF	EE° BSC Position 1	2	3	4	5	6
8	.3895 (9.89) .3845 (9.77)	.553 (14.05) .547 (13.89)	.290 (7.37) .280 (7.11)	.820 (20.83) .804 (20.42)	.594 (15.09)	.549 (13.94) .539 (13.69)	.120 (3.05) .100 (2.54)	40	40	40	40	40	40
10	.5145 (13.07) .5095 (12.94)	.678 (17.22) .672 (17.07)	.415 (10.54) .405 (10.29)	.945 (24.00) .929 (23.60)	.719 (18.26)	.674 (17.12) .664 (16.87)	.120 (3.05) .100 (2.54)	60	46	46	60	89	60
14	.7645 (19.42) .7595 (19.29)	.928 (23.57) .922 (23.42)	.665 (16.89) .655 (16.64)	1.133 (28.78) 1.117 (28.37)	.906 (23.01)	.924 (23.47) .914 (23.22)	.120 (3.05) .100 (2.54)	60	46	46	60	89	60
16	.8905 (22.62) .8855 (22.49)	1.054 (26.77) 1.048 (26.62)	.791 (20.09) .781 (19.84)	1.258 (31.95) 1.242 (31.55)	.969 (24.61)	1.050 (26.67) 1.040 (26.42)	.150 (3.81) .130 (3.30)	50	50	50	50	50	50
18	1.0145 (25.77) 1.0095 (25.64)	1.178 (29.92) 1.172 (29.77)	.916 (23.27) .906 (23.01)	1.351 (34.32) 1.335 (33.91)	1.062 (26.97)	1.173 (29.79) 1.163 (29.54)	.150 (3.81) .130 (3.30)	50	50	50	50	50	50

NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Dimensions apply after plating.
5. Mating keyway positions and dimensions are shown in figure 4.
6. Distance from face of insert to the point at which a gage pin, having the same basic diameter as the mating contact and a square face, engages the socket contact spring.
7. Distance from front of shell to the point at which a gage pin, having the same basic diameter as plug shell, engages the spring member.
8. This design information establishes connector intermating criteria and interchangeability of accessory hardware.
9. All dimensions are for connector with contact retention system in locked position.
10. Rear end connector design for attachment of non-rotatable accessory hardware is shown on figure 12A.
11. Dimension includes contact slack.

FIGURE 5A Connector, receptacle, electrical, flange mount (series 2)- Continued.

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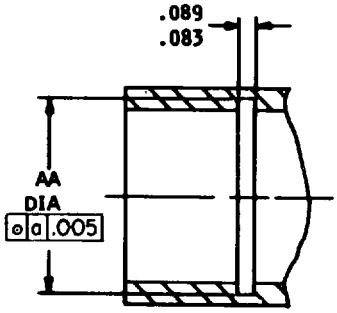


CONTACT SIZE	X DIA	Y DIA	GG DIA	HH DIA	KK(MAX)
23 - 22	.035 (.89) .032 (.81)	.075 (1.9°) .065 (1.65)	.0280(.71) .0300(.76)	.068 (1.73) .062 (1.57)	.083(2.11)
20 - 20	.053(1.35) .050(1.27)	.085 (2.16) .080 (2.03)	.0415(1.05) .0445(1.13)	.083 (2.11) .077 (1.96)	.090(2.29)
16 - 16	.077(1.96) .074(1.88)	.108 (2.74) .103 (2.62)	.0950(2.41) .0990(2.52)	.106 (2.69) .100 (2.54)	.105(2.67)
12 - 12	.130(3.30) .126(3.20)	.170 (4.32) .166 (4.22)	.1540(3.91) .1580(4.01)	.169 (4.29) .163 (4.14)	.105(2.67)

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.003	.08	.040	1.02	.094	2.39	.230	5.84	.535	13.59
.005	.13	.046	1.17	.100	2.54	.253	6.43	.545	13.84
.006	.15	.065	1.65	.116	2.95	.275	6.99	.765	19.43
.009	.23	.067	1.70	.126	3.20	.290	7.37	.775	19.69
.010	.25	.073	1.85	.157	3.99	.360	9.14	1.167	29.64
.015	.38	.075	1.91	.163	4.14	.380	9.65	1.187	30.15
.016	.41	.083	2.11	.169	4.29	.443	11.25		
.020	.51	.090	2.29	.200	5.08	.463	11.76		
.021	.53	.035	.89	.203	5.16				

FIGURE 5B. Connector, receptacle, electrical, flange mount (series 4).

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"O" RING GROOVE DIMENSIONS

SIZE	AA DIA
8	.493 (12.52) .488 (12.40)
10	.618 (15.70) .613 (15.57)
14	.868 (22.05) .863 (21.92)
16	.994 (25.25) .989 (25.12)
18	1.118 (28.40) 1.113 (28.27)

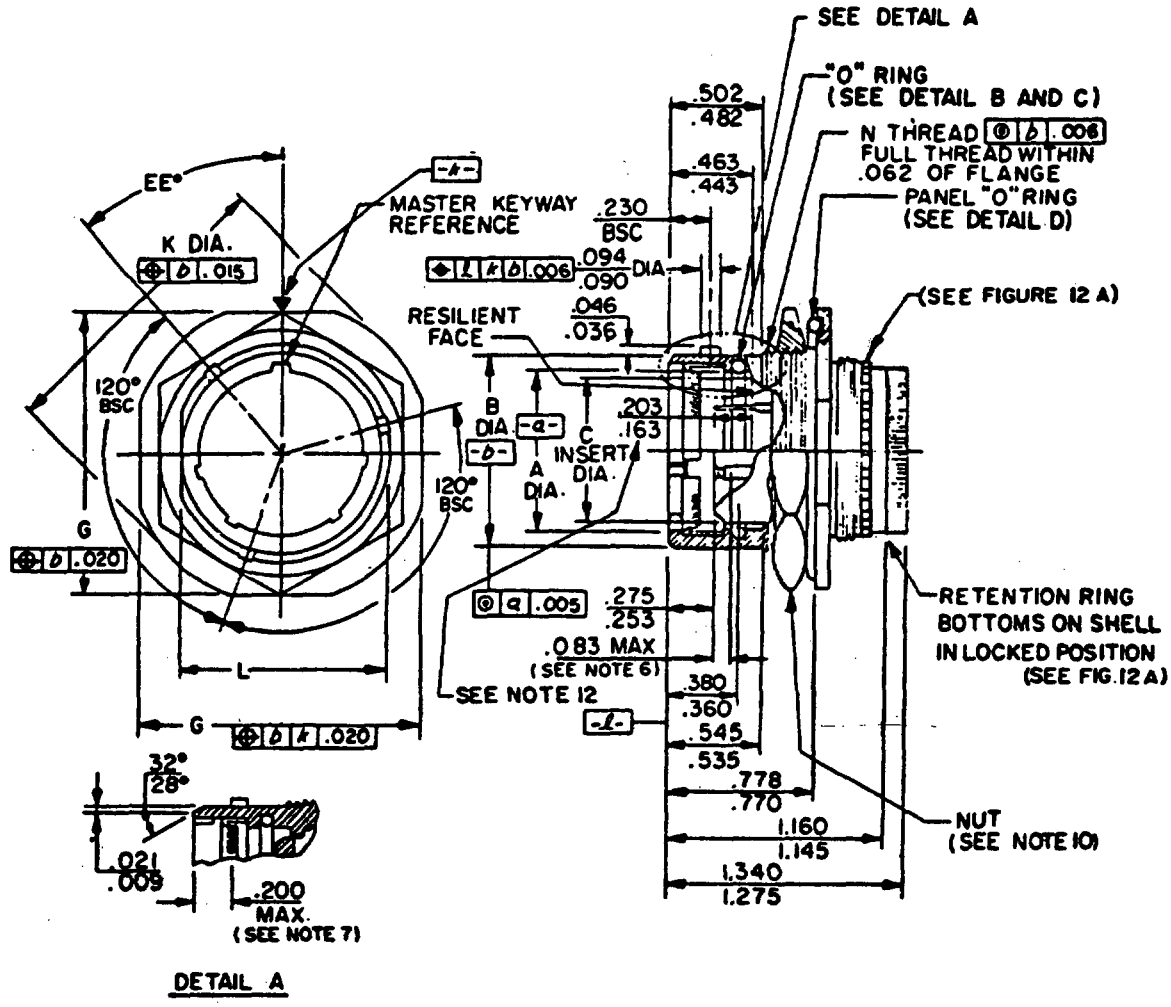
Shell size	A dia	B dia	C dia	G	H BASIC	J dia	FF	EE° BSC Position 1	2	3	4	5	6
8	.3895 (9.89) .3845 (9.77)	.553 (14.05) .547 (13.89)	.290 (7.37) .280 (7.11)	.820 (20.83) .804 (20.42)	.594 (15.09)	.549 (13.94) .539 (13.69)	.120 (3.05) .100 (2.54)	40	40	40	40	40	40
10	.5145 (13.07) .5095 (12.94)	.678 (17.22) .672 (17.07)	.415 (10.54) .405 (10.29)	.945 (24.00) .929 (23.60)	.719 (18.26)	.674 (17.12) .664 (16.87)	.120 (3.05) .100 (2.54)	60	46	46	60	89	60
14	.7645 (19.42) .7595 (19.29)	.928 (23.57) .922 (23.42)	.665 (16.89) .655 (16.64)	1.133 (28.78) 1.117 (28.37)	.906 (23.01)	.924 (23.47) .914 (23.22)	.120 (3.05) .100 (2.54)	60	46	46	60	89	60
16	.8905 (22.62) .8855 (22.49)	1.054 (26.77) 1.048 (26.62)	.791 (20.09) .781 (19.84)	1.258 (31.95) 1.242 (31.55)	.969 (24.61)	1.050 (26.67) 1.040 (26.42)	.150 (3.81) .130 (3.30)	50	50	50	50	50	50
18	1.0145 (25.77) 1.0095 (25.64)	1.178 (29.92) 1.172 (29.77)	.916 (23.27) .906 (23.01)	1.351 (34.32) 1.335 (33.91)	1.062 (26.97)	1.173 (29.79) 1.163 (29.54)	.150 (3.81) .130 (3.30)	50	50	50	50	50	50

NOTES:

1. DIMENSIONS ARE IN INCHES.
2. MILLIMETERS ARE IN PARENTHESES.
3. METRIC EQUIVALENTS (TO THE NEAREST .01 MM) ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25.4 MM.
4. DIMENSIONS APPLY AFTER PLATING.
5. MATING KEYWAY POSITIONS AND DIMENSIONS ARE SHOWN IN FIGURE 4.
6. DISTANCE FROM FACE OF INSERT TO THE POINT AT WHICH A GAGE PIN, HAVING THE SAME BASIC DIAMETER AS THE MATING CONTACT AND A SQUARE FACE, ENGAGES THE SOCKET CONTACT SPRING.
7. DISTANCE FROM FRONT OF SHELL TO THE POINT AT WHICH A GAGE PIN, HAVING THE SAME BASIC DIAMETER AS PLUG SHELL, ENGAGES THE SPRING MEMBER.
8. REAR END CONNECTOR DESIGN FOR ATTACHMENT OF NON-ROTATABLE ACCESSORY HARDWARE IS SHOWN ON FIGURE 12B.
9. DIMENSION INCLUDES CONTACT SLACK.
10. ARRANGEMENTS CONTAINING SIZE 12 CONTACTS, INSERT PROTRUSION. .020/.070

FIGURE 5B. CONNECTOR, RECEPTACLE, ELECTRICAL, FLANGE MOUNT (SERIES 4) - CONTINUED.

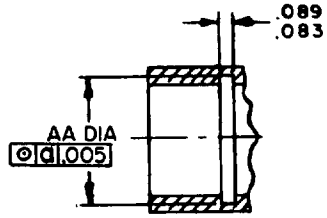
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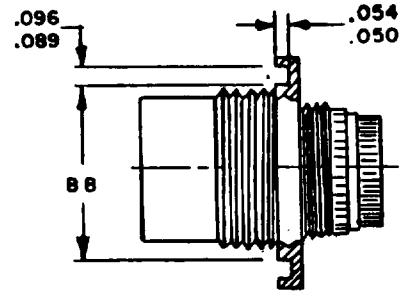
Contact size	X DIA	Y DIA
23 - 22	.035 (.89) .032 (.81)	.075 (1.91) .065 (1.65)
20 - 20	.053 (1.35) .050 (1.27)	.085 (2.16) .080 (2.03)
16 - 16	.077 (1.96) .074 (1.88)	.108 (2.74) .103 (2.62)
12 - 12	.130 (3.30) .126 (3.20)	.170 (4.32) .166 (4.22)

FIGURE 6A. Connector, receptacle, electrical, jam nut mount (series 2).

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DETAIL C



DETAIL D

SIZE	AA DIA
8	.493 (12.52) .488 (12.40)
10	.618 (15.70) .613 (15.57)
14	.868 (22.05) .863 (21.92)
16	.994 (25.25) .989 (25.12)
18	1.118 (28.40) 1.113 (28.27)

SIZE	BB DIA
8	.724 (18.39) .713 (18.11)
10	.849 (21.56) .838 (21.29)
14	1.099 (27.91) 1.088 (27.64)
16	1.221 (31.01) 1.210 (30.73)
18	1.356 (34.44) 1.345 (34.16)

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.003	.08	.040	1.02	.157	3.99	.290	7.37	.545	13.84
.005	.13	.046	1.17	.163	4.14	.360	9.14	.770	19.56
.006	.15	.062	1.57	.169	4.29	.380	9.65	.778	19.76
.009	.23	.067	1.70	.200	5.08	.443	11.25	1.145	29.08
.010	.25	.073	1.85	.203	5.16	.463	11.76	1.160	29.46
.015	.38	.083	2.11	.230	5.84	.482	12.24	1.275	32.39
.020	.51	.090	2.29	.253	6.43	.502	12.75	1.340	34.04
.021	.53	.094	2.39	.275	6.99	.535	13.59		

FIGURE 6A Connector, receptacle, electrical, jam nut mount (series 2) - continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
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Shell size	A dia	B dia	C dia	G across flats	K dia	L across flat	N thread	Panel "O" ring	EE BSC Position 1	2	3	4	5	6
8	.3895 (9.89)	.553 (14.05)	.290 (7.37)	.989 (25.12)	1.072 (27.23)	.598 (15.19)	5/8-20UN-2A	MS-9068-018	40	40	40	40	40	40
	.3845 (9.77)	.547 (13.89)	.280 (7.11)	.969 (24.61)	1.052 (26.72)	.590 (14.99)								
10	.5145 (13.07)	.678 (17.22)	.415 (10.54)	1.114 (28.30)	1.197 (30.40	.723 (18.36)	3/4-20UN-2A	MS-9068-020	60	46	46	60	89	60
	.5095 (12.94)	.672 (17.07)	.405 (10.29)	1.094 (27.79)	1.177 (29.90)	.715 (18.16)								
14	.7645 (19.42)	.928 (23.57)	.665 (16.89)	1.401 (35.59)	1.510 (38.35)	.973 (24.71)	1-20UNEF- 2A	MS-9068-024	60	46	46	60	89	60
	.7595 (19.29)	.922 (23.42)	.655 (16.64)	1.381 (35.08)	1.490 (37.85)	.965 (24.51)								
16	.8905 (22.62)	1.054 (26.77)	.791 (20.09)	1.526 (38.76)	1.635 (41.53)	1.098 (27.89)	1-1/8-20UN-2A	MS-9068-026	50	50	50	50	50	50
	.8855 (22.49)	1.048 (26.62)	.781 (19.84)	1.506 (38.25)	1.615 (41.02)	1.090 (27.69)								
18	1.0145 (25.77)	1.178 (29.92)	.916 (23.27)	1.651 (41.94)	1.745 (44.32)	1.222 (31.04)	1¼-20UN-2A	MS-9068-028	50	50	50	50	50	50
	1.0095 (25.64)	1.172 (29.77)	.906 (23.01)	1.631 (41.43)	1.725 (43.82)	1.214 (30.84)								

NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Dimensions apply after plating.
5. Mating keyway positions and dimensions are shown in figure 4.
6. Distance from face of insert to the point at which a gage pin, having the same basic diameter as the mating contact and a square face, engages the socket contact spring.
7. Distance from front of shell to the point at which a gage pin, having the same basic diameter as plug shell. engages the spring member.
8. This design information establishes connector intermating criteria and interchangeability of accessory hardware.
9. All dimensions are for connector with contact retention system in locked position.
10. Slotted nut shall conform to MIL-C-81511/20.
11. Rear end connector design for attachment of non-rotatable accessory hardwear is shown on figure 12A.
12. Dimension includes contact slack.

FIGURE 6A Connector, receptacle, electrical, jam nut mount (series 2) - continued.

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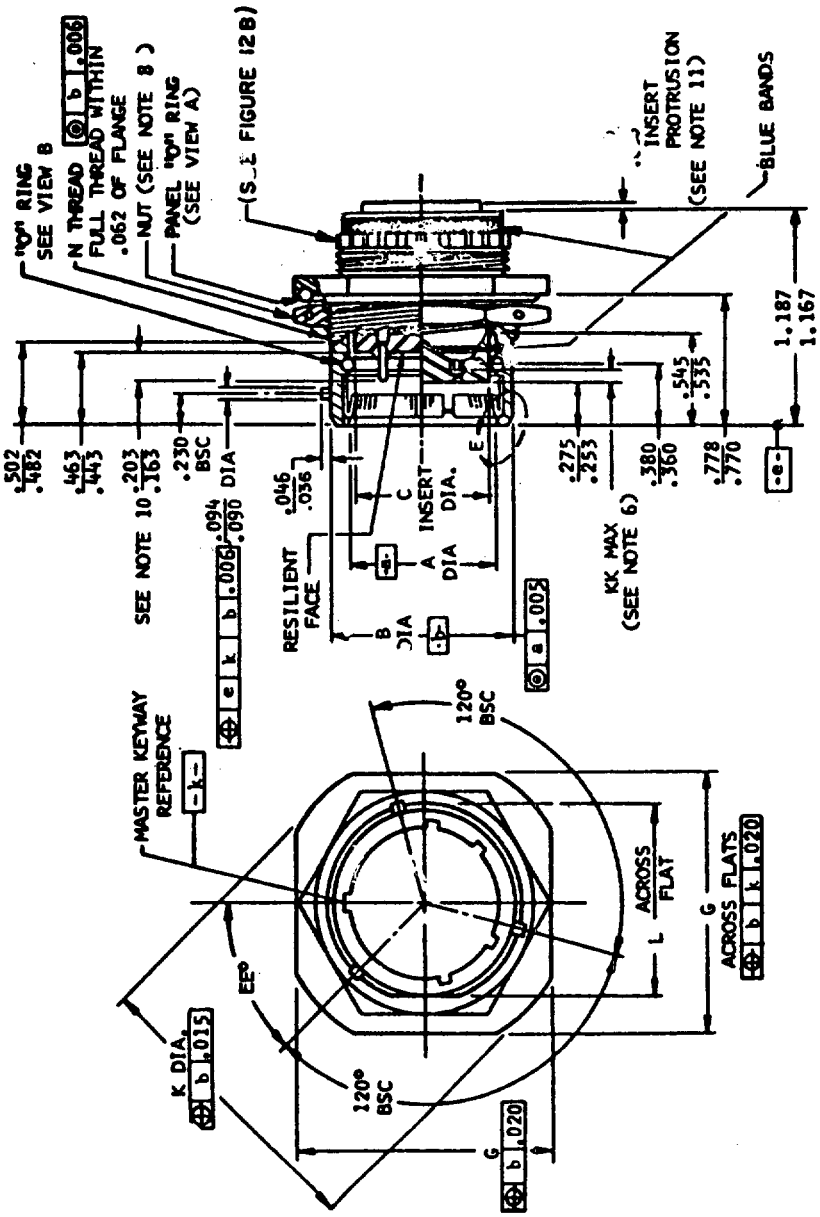
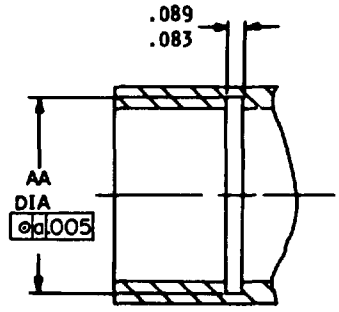
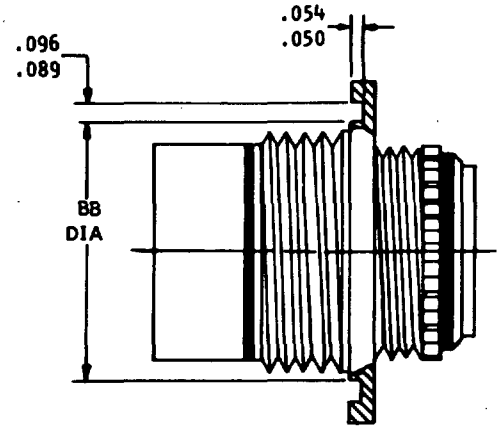


FIGURE 6B. Connector, receptacle, electrical, jam nut mount (series 4).

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"O" RING GROOVE
DIMENSIONS



PANEL 'O' RING
VIEW A

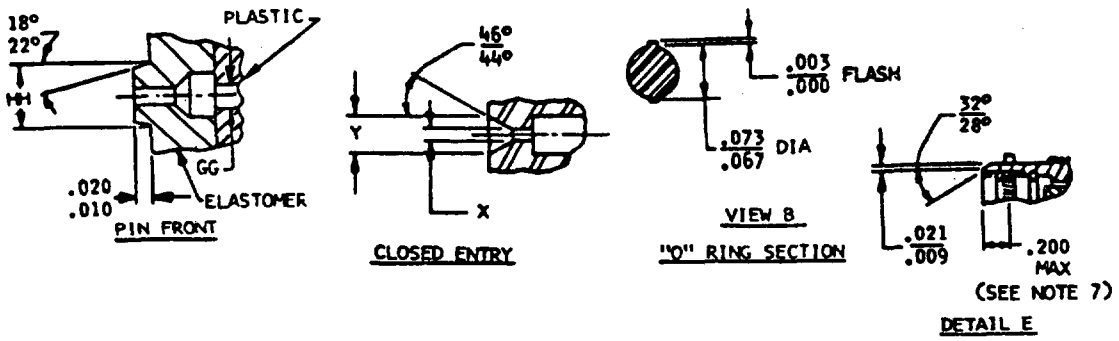
SIZE	AA DIA
8	.493 (12.52) .488 (12.40)
10	.618 (15.70) .613 (15.57)
14	.868 (22.05) .863 (21.92)
16	.994 (25.25) .989 (25.12)
18	1.118 (28.40) 1.113 (28.27)

SIZE	BB DIA
8	.724 (18.39) .713 (18.11)
10	.849 (21.56) .838 (21.29)
14	1.099 (27.91) 1.088 (27.64)
16	1.221 (31.01) 1.210 (30.73)
18	1.356 (34.44) 1.345 (34.16)

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.003	.08	.040	1.02	.157	3.99	.290	7.37	.545	13.84
.005	.13	.046	1.17	.163	4.14	.360	9.14	.770	19.56
.006	.15	.062	1.57	.169	4.29	.380	9.65	.778	19.76
.009	.23	.067	1.70	.200	5.08	.443	11.25	1.167	29.64
.010	.25	.073	1.85	.203	5.16	.463	11.76	1.187	30.15
.015	.38	.083	2.11	.230	5.84	.482	12.24		
.020	.51	.090	2.29	.253	6.43	.502	12.75		
.021	.53	.094	2.39	.275	6.99	.535	13.59		
.035	.89								

FIGURE 6B Connector, receptacle, electrical, jam nut mount (series 4) - continued.

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CONTACT SIZE	X DIA	Y DIA	GG DIA	HH DIA	KK(MAX)
23 - 22	.035 (.89) .032 (.81)	.075 (1.91) .065 (1.65)	.0280(.71) .0300(.76)	.068 (1.73) .062 (1.57)	.083(2.11)
20 - 20	.053(1.35) .050(1.27)	.085 (2.16) .080 (2.03)	.0415(1.05) .0445(1.13)	.083 (2.11) .077 (1.96)	.090(2.29)
16 - 16	.077(1.96) .074(1.88)	.108 (2.74) .103 (2.62)	.0950(2.41) .0990(2.52)	.106 (2.69) .100 (2.54)	.105(2.67)
12 - 12	.130(3.30) .126(3.20)	.170 (4.32) .166 (4.22)	.1540(3.91) .1580(4.01)	.169 (4.29) .163 (4.14)	.105(2.67)

FIGURE 6B. Connector, receptacle, electrical, jam nut mount (series 4) - Continued.

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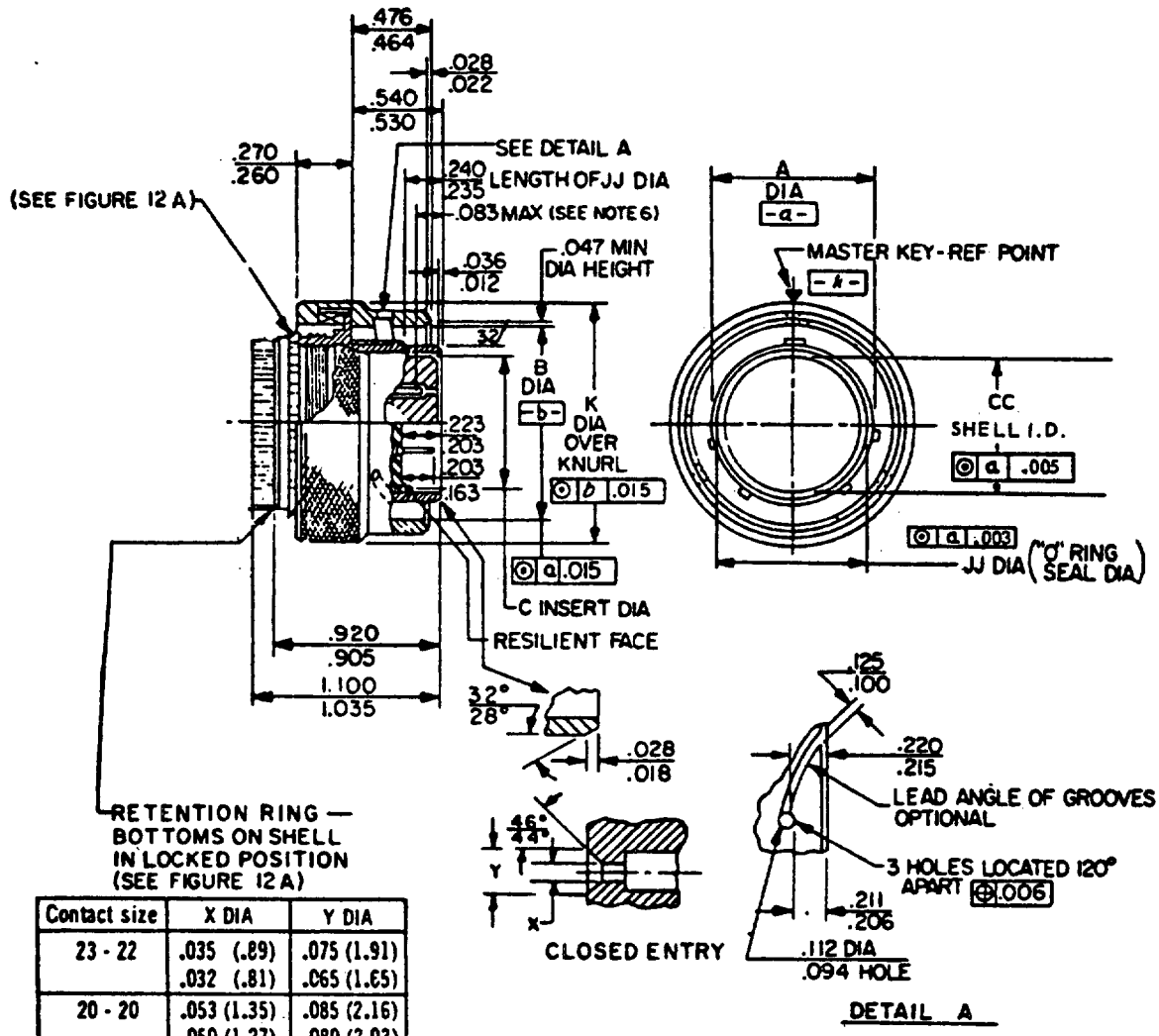
Shell size	A dia	B dia	C dia	G across flats	K dia	L across flat	N thread	Panel "O" ring	EE°BSC Position					
									1	2	3	4	5	6
8	.3895 (9.89)	.553 (14.05)	.290 (7.37)	.989 (25.12)	1.072 (27.23)	.598 (15.19)	5/8-20UN-2A	MS-9068-018	40	40	40	40	40	40
	.3845 (9.77)	547 (13.89)	.280 (7.11)	.969 (24. 61)	1.052 (26.72)	.590 (14.99)								
10	.5145 (13.07)	.678 (17.22)	.415 (10.54)	1.114 (28.30)	1.197 (30.40	.723 (18.36)	3/4-20UN-2A	MS-9068-020	60	46	46	60	89	60
	.5095 (12.94)	.672 (17.07)	.405 (10.29)	1.094 (27.79)	1.177 (29.90)	.715 (18.16)								
14	.7645 (19.42)	.928 (23.57)	.665 (16.89)	1.401 (35.59)	1.510 (38.35)	.973 (24.71)	1-20UNEF- 2A	MS-9068-024	60	46	46	60	89	60
	.7595 (19.29)	.922 (23.42)	.655 (16.64)	1.381 (35.08)	1.490 (37. 85)	.965 (24.51)								
16	.8905 (22.62)	1.054 (26.77)	.791 (20.09)	1.526 (38.76)	1.635 (41.53)	1.098 (27.89)	1-1/8-20UN-2A	MS-9068-026	50	50	50	50	50	50
	.8855 (22.49)	1.048 (26.62)	.781 (19.84)	1.506 (38.25)	1.615 (41.02)	1.090 (27.69)								
18	1.0145 (25.77)	1.178 (29.92)	.916 (23.27)	1.651 (41.94)	1.745 (44.32)	1.222 (31.04)	1⅛-20UN-2A	MS-9068-028	50	50	50	50	50	50
	1.0095 (25.64)	1.172 (29.77)	.906 (23.01)	1.631 (41.43)	1.725 (43.82)	1.214 (30.84)								

NOTES:

1. DIMENSIONS ARE IN INCHES.
2. MILLIMETERS ARE IN PARENTHESES.
3. METRIC EQUIVALENTS (TO THE NEAREST .01 MM) ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25.4 MM.
4. DIMENSIONS APPLY AFTER PLATING.
5. MATING KEYWAY POSITIONS AND DIMENSIONS ARE SHOWN IN FIGURE 4.
6. DISTANCE FROM FACE OF INSERT TO THE POINT AT WHICH A GAGE PIN, HAVING THE SAME BASIC DIAMETER AS THE MATING CONTACT AND A SQUARE FACE, ENGAGES THE SOCKET CONTACT SPRING.
7. DISTANCE FROM FRONT OF SHELL TO THE POINT AT WHICH A GAGE PIN, HAVING THE SAME BASIC DIAMETER AS PLUG SHELL, ENGAGES THE SPRING MEMBER.
8. SLOTTED NUT SHALL CONFORM TO MIL-C-81511/20.
9. REAR END CONNECTOR DESIGN FOR ATTACHMENT OF NON-ROTATABLE ACCESSORY HARDWARE IS SHOWN ON FIGURE 12B.
10. DIMENSION INCLUDES CONTACT SLACK.
11. ARRANGEMENTS CONTAINING SIZE 12 CONTACTS, INSERT PROTRUSION. .020/.070

FIGURE 6B. CONNECTOR, RECEPTACLE, ELECTRICAL, JAM NUT MOUNT (SERIES 4) - CONTINUED.

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Contact size	X DIA	Y DIA
23 - 22	.035 (.89) .032 (.81)	.075 (1.91) .065 (1.65)
20 - 20	.053 (1.35) .050 (1.27)	.085 (2.16) .080 (2.03)
16 - 16	.077 (1.96) .074 (1.88)	.108 (2.74) .103 (2.62)
12 - 12	.130 (3.30) .126 (3.20)	.170 (4.32) .166 (4.22)

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.005	.13	.036	.91	.163	4.14	.235	5.97	.474	12.04
.006	.15	.047	1.19	.169	4.29	.240	6.10	.530	13.46
.008	.20	.083	2.11	.203	5.16	.260	6.60	.540	13.72
.012	.30	.094	2.39	.206	5.23	.270	6.86	.905	22.99
.015	.38	.100	2.54	.211	5.36	.275	6.99	.920	23.37
.018	.46	.112	2.84	.215	5.46	.290	7.37	1.035	26.29
.022	.56	.125	3.18	.220	5.59	.464	11.79	1.100	27.94
.028	.71	.157	3.99						

FIGURE 7A. Connector, plug, electrical, (series 2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
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SHELL SIZE	A DIA	B DIA	C DIA	K DIA	CC DIA	JJ DIA
8	.3815 (9.69) .3755 (9.54)	.566 (14.38) .558 (14.17)	.290 (7.37) .280 (7.11)	.850 (21.59) .830 (21.08)	.301 (7.65) .293 (7.45)	.3815 (9.69) .373 (9.47)
10	.5065 (12.87) .5005 (12.71)	.691 (17.55) .683 (17.35)	.415 (10.54) .405 (10.29)	.975 (24.77) .955 (24.26)	.426 (10.82) .418 (10.62)	.5065 (12.87) .498 (12.65)
14	.7565 (19.22) .7505 (19.06)	.941 (23.90) .933 (23.70)	.665 (16.89) .655 (16.64)	1.225 (31.12) 1.205 (30.61)	.676 (17.17) .668 (16.97)	.7565 (19.22) .746 (18.95)
16	.8825 (22.42) .8765 (22.26)	1.068 (27.13) 1.060 (26.92)	.791 (20.09) .781 (19.84)	1.350 (34.29) 1.330 (33.78)	.802 (20.37) .794 (20.17)	8825 (22.42) .872 (22.15)
18	1.0065 (25.57) 1.0005 (25.41)	1.191 (30.25) 1.183 (30.05)	.916 (23.27) .906 (23.01)	1.475 (37.41) 1.455 (36.96)	.927 (23.55) .919 (23.35)	1.0065 (25.57) .996 (25.30)

NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Dimensions are after plating.
5. Mating key positions and dimensions are shown in figure 4.
6. Distance from front of insert and the point at which a gage pin, having the same basic diameter as the mating contact and a square face, engages the socket contact spring.
7. This design information establishes connector intermating criteria and interchangeability of accessory hardware.
8. All dimensions are for connector with contact retention system in locked position.
9. Rear end connector design for attachment of non-rotatable accessory hardware is shown on figure 12.

FIGURE 7A. Connector, plug, electrical, (series 2) - Continued.

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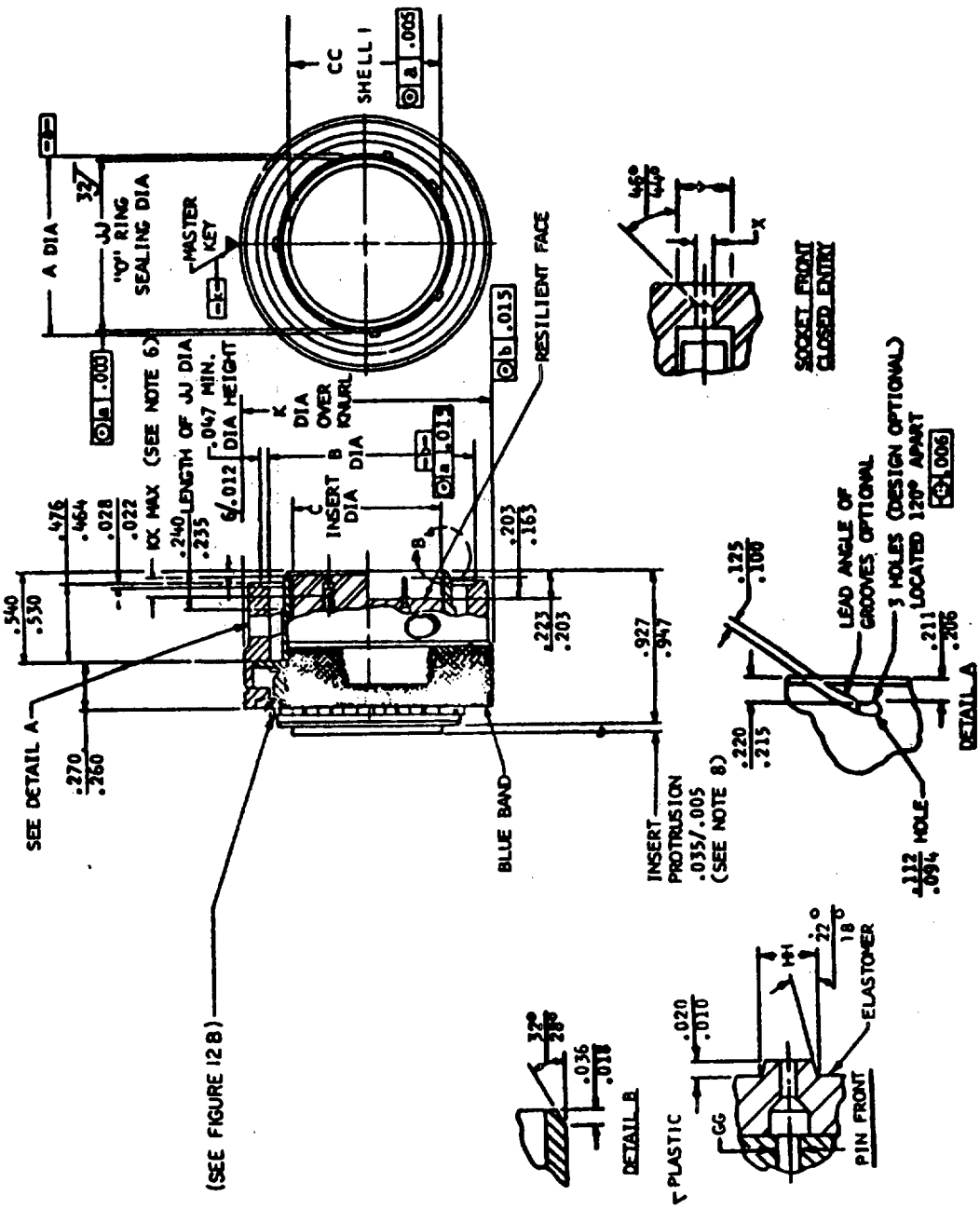


FIGURE 7B. Connector, plug, electrical, (series 4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
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SHELL SIZE	A DIA	B DIA	C DIA	K DIA	CC DIA	JJ DIA
8	.3815 (9.69) .3755 (9.54)	.566 (14.38) .558 (14.17)	.290 (7.37) .280 (7.11)	.850 (21.59) .830 (21.08)	.301 (7.65) .293 (7.45)	.3815 (9.69) .373 (9.47)
10	.5065 (12.87) .5005 (12.71)	.691 (17.55) .683 (17.35)	.415 (10.54) .405 (10.29)	.975 (24.77) .955 (24.26)	.426 (10.82) .418 (10.62)	.5065 (12.87) .498 (12.65)
14	.7565 (19.22) .7505 (19.06)	.941 (23.90) .933 (23.70)	.665 (16.89) .655 (16.64)	1.225 (31.12) 1.205 (30.61)	.676 (17.17) .668 (16.97)	.7565 (19.22) .746 (18.95)
16	.8825 (22.42) .8765 (22.26)	1.068 (27.13) 1.060 (26.92)	.791 (20.09) .781 (19.84)	1.350 (34.29) 1.330 (33.78)	.802 (20.37) .794 (20.17)	.8825 (22.42) .872 (22.15)
18	1.0065 (25.57) 1.0005 (25.41)	1.191 (30.25) 1.183 (30.05)	.916 (23.27) .906 (23.01)	1.475 (37.41) 1.455 (36.96)	.927 (23.55) .919 (23.35)	1.0065 (25.57) .996 (25.30)

NOTES:

1. DIMENSIONS ARE IN INCHES.
2. MILLIMETERS ARE IN PARENTHESES.
3. METRIC EQUIVALENTS (TO BE NEAREST .01 MM) ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25.4 MM.
4. DIMENSIONS ARE AFTER PLATING.
5. MATING KEY POSITIONS AND DIMENSIONS ARE SHOWN IN FIGURE 4.
6. DISTANCE FROM FRONT OF INSERT AND THE POINT AT WHICH A GAGE PIN, HAVING THE SAME BASIC DIAMETER AS THE MATING CONTACT AND A SQUARE FACE, ENGAGES THE SOCKET CONTACT SPRING.
7. REAR END CONNECTOR DESIGN FOR ATTACHMENT OF NON-ROTATABLE ACCESSORY HARDWARE IS SHOWN ON FIGURE 12B
8. ARRANGEMENTS CONTAINING SIZE 12 CONTACTS, INSERT PROTUSION .070/.020.

FIGURE 7B. Connector, plug electrical, (series 4) - Continued.

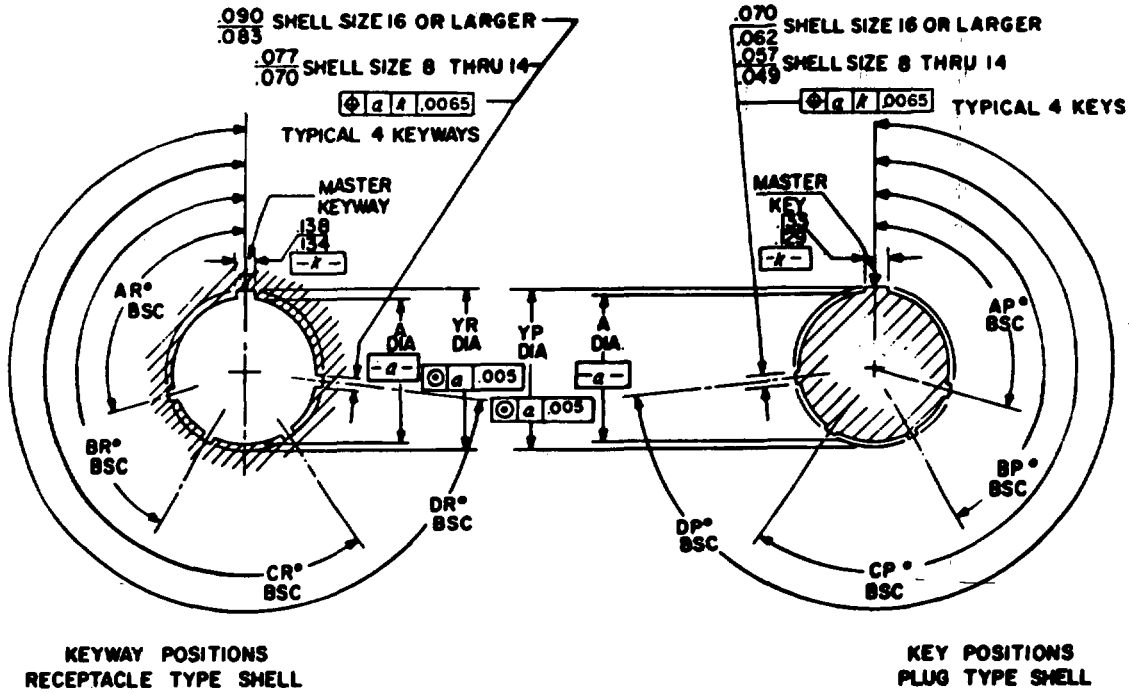
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

CONTACT SIZE	X DIA	Y DIA	GG DIA	HH DIA	KK (MAX)
23 - 22	.035 (.89) .032 (.81)	.075 (1.91) .065 (1.65)	.0280(.71) .0300(.76)	.068 (1.73) .062 (1.57)	.083(2.11)
20 - 20	.053(1.35) .050(1.27)	.085 (2.16) .080 (2.03)	.0415(1.05) .0445(1.13)	.083 (2.11) .077 (1.96)	.090(2.29)
16 - 16	.077(1.96) .074(1.88)	.108 (2.74) .103 (2.62)	.0950(2.41) .0990(2.52)	.106 (2.69) .100 (2.54)	.105(2.67)
12 - 12	.130(3.30) .126(3.20)	.170 (4.32) .166 (4.22)	.1540(3.91) .1580(4.01)	.169 (4.29) .163 (4.14)	.105(2.67)

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.005	.13	.036	.91	.163	4.14	.235	5.97	.474	12.04
.006	.15	.047	1.19	.169	4.29	.240	6.10	.530	13.46
.008	.20	.083	2.11	.203	5.16	.260	6.60	.540	13.72
.012	.30	.094	2.39	.206	5.23	.270	6.86	.927	23.55
.015	.38	.100	2.54	.211	5.36	.275	6.99	.947	34.05
.018	.46	.112	2.84	.215	5.46	.290	7.37		
.022	.56	.125	3.18	.220	5.59	.464	11.79		
.028	.71	.157	3.99						
.035	.89								

FIGURE 7B. Connector, plug, electrical, (series 4) - Continued.

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INCHES	MM	INCHES	MM
.005	.13	.083	2.11
.008	.20	.090	2.29
.049	1.24	.129	3.28
.057	1.45	.133	3.38
.062	1.57	.134	3.40
.070	1.78	.138	3.51
.077	1.96		

FIGURE 8. Connector, electrical, positions key and keyways, mating. (series 1 & 3)

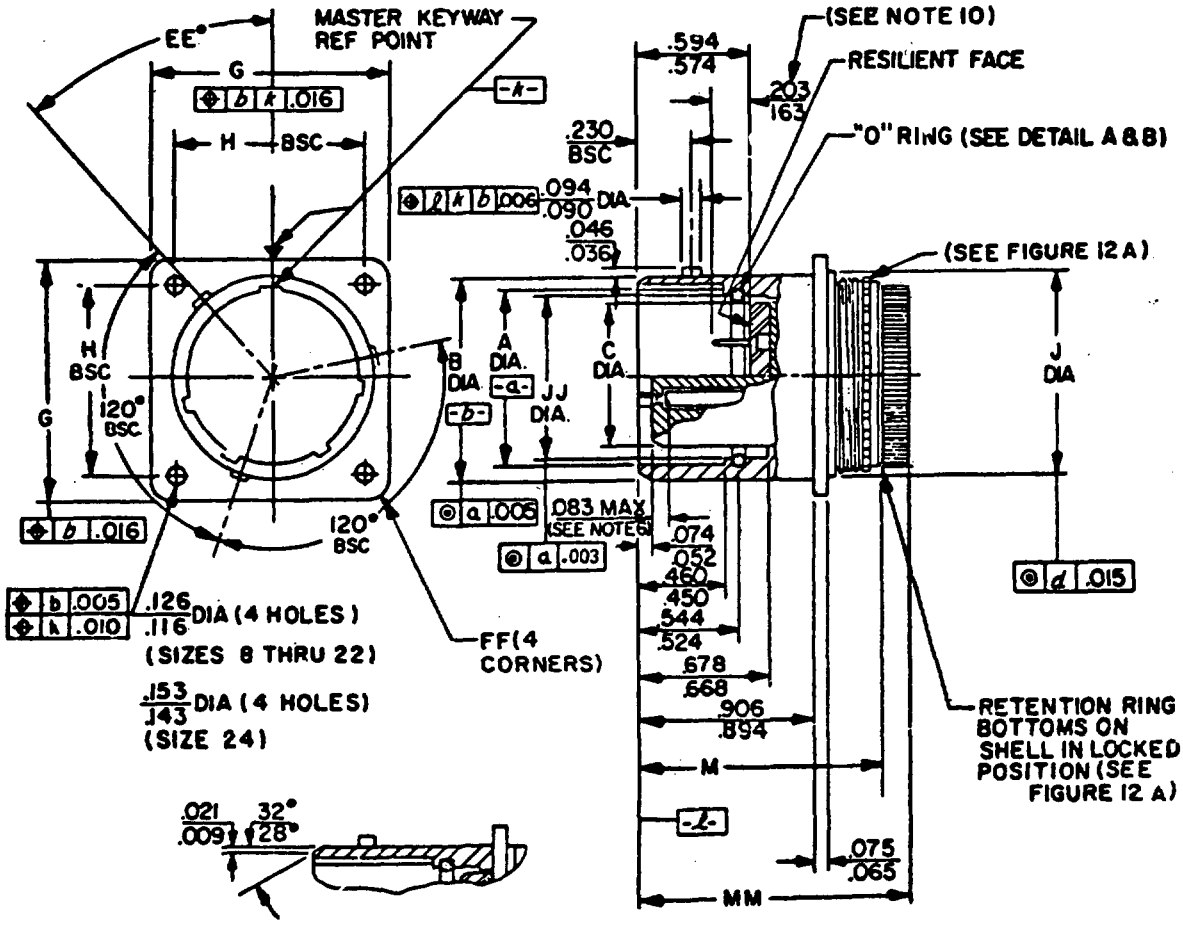
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Check the source to verify that this is the current version before use.

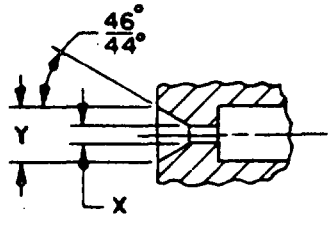
Shell size	Key & keyway arrangement	AR° or AP° BSC	BR° or BP° BSC	CR° or CP° BSC	DR° or DP° BSC	YR dia	YP dia
8	1	105	140	215	265	.482 (12.24) .470 (11.94)	.460 (11.68) .452 (11.48)
	2	102	132	248	320		
	3	80	118	230	312		
	4	35	140	205	275		
	5	64	155	234	304		
	6	91	131	197	240		
10 and 14	1	95	141	208	236	.607 (15.42) .595 (15.11) .857 (21.77) .845 (21.46)	.585 (14.86) .577 (14.66) .835 (21.21) .827 (21.01)
	2	113	156	182	292		
	3	90	145	195	252		
	4	53	156	220	255		
	5	119	146	176	298		
	6	51	141	184	242		
16 and 18	1	80	142	196	293	.983 (24.97) .971 (24.66) 1.107 (28.12) 1.095 (27.81)	.961 (24.41) .953 (24.21) 1.085 (27.56) 1.077 (27.36)
	2	135	170	200	310		
	3	49	169	200	244		
	4	66	140	200	257		
	5	62	145	180	280		
	6	79	153	197	272		
20, 22 and 24	1	80	142	196	293	1.232 (31.29) 1.220 (30.99) 1.357 (34.47) 1.345 (34.16) 1.482 (37.64) 1.470 (37.34)	1.210 (30.73) 1.202 (30.53) 1.335 (33.91) 1.327 (33.71) 1.460 (37.08) 1.452 (36.88)
	2	135	170	200	310		
	3	49	169	200	244		
	4	66	140	200	257		
	5	62	145	180	280		
	6	79	153	197	272		

NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Dimensions apply after plating.
5. For datum 'a' refer to figures 9, 10, and 11.

FIGURE 8. Connector, electrical, positions key and keyway, mating. (series 1 & 3) CONTINUED.

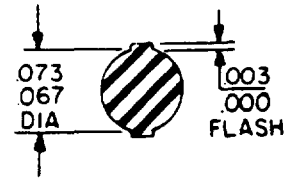
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CLOSED ENTRY

Contact size	X DIA	Y DIA
23 - 22	.035 (.89) .032 (.81)	.075 (1.91) .065 (1.65)
20 - 20	.053 (1.35) .050 (1.27)	.085 (2.16) .080 (2.03)
16 - 16	.077 (1.96) .074 (1.88)	.108 (2.74) .103 (2.62)
12 - 12	.130 (3.30) .126 (3.20)	.170 (4.32) .166 (4.22)

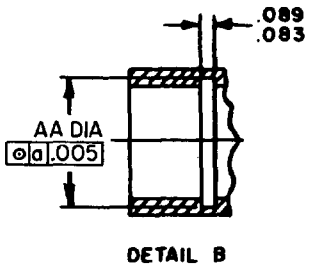


DETAIL A

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.003	.08	.040	1.02	.090	2.29	.275	6.99	.678	17.22
.004	.10	.046	1.17	.094	2.39	.290	7.37	.894	22.71
.005	.13	.052	1.32	.116	2.95	.450	11.43	.906	23.01
.006	.15	.065	1.65	.126	3.20	.460	11.68	1.276	32.41
.009	.23	.067	1.70	.157	3.99	.524	13.31	1.291	32.79
.010	.25	.073	1.85	.163	4.14	.544	13.82	1.406	35.71
.015	.38	.074	1.88	.169	4.29	.574	14.58	1.471	37.36
.016	.41	.075	1.91	.203	5.16	.594	15.09		
.021	.53	.083	2.11	.230	5.84	.668	16.97		

FIGURE 9A. Connector, receptacle, electrical, flange mount (series 1).

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DETAIL B

SIZE	AA DIA
8	.493 (12.52) .488 (12.40)
10	.618 (15.70) .613 (15.57)
14	.868 (22.05) .863 (21.92)
16	.994 (25.25) .989 (25.12)
18	1.118 (28.40) 1.113 (28.27)
20	1.243 (31.57) 1.238 (31.45)
22	1.368 (34.75) 1.363 (34.62)
24	1.493 (37.92) 1.488 (37.80)

NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Dimensions apply after plating.
5. Mating keyway positions and dimensions are shown in figure 8.
6. Distance from face of insert to the point at which a gage pin, having the same basic diameter as the mating contact and a square face, engages the socket contact spring.
7. This design information establishes connector intermating criteria and interchangeability of accessory hardware.
8. All dimensions are for connector with contact retention system in locked position.
9. Rear end connector design for attachment of non-rotatable accessory hardware is shown on figure 12A.
10. Dimension includes contact slack.

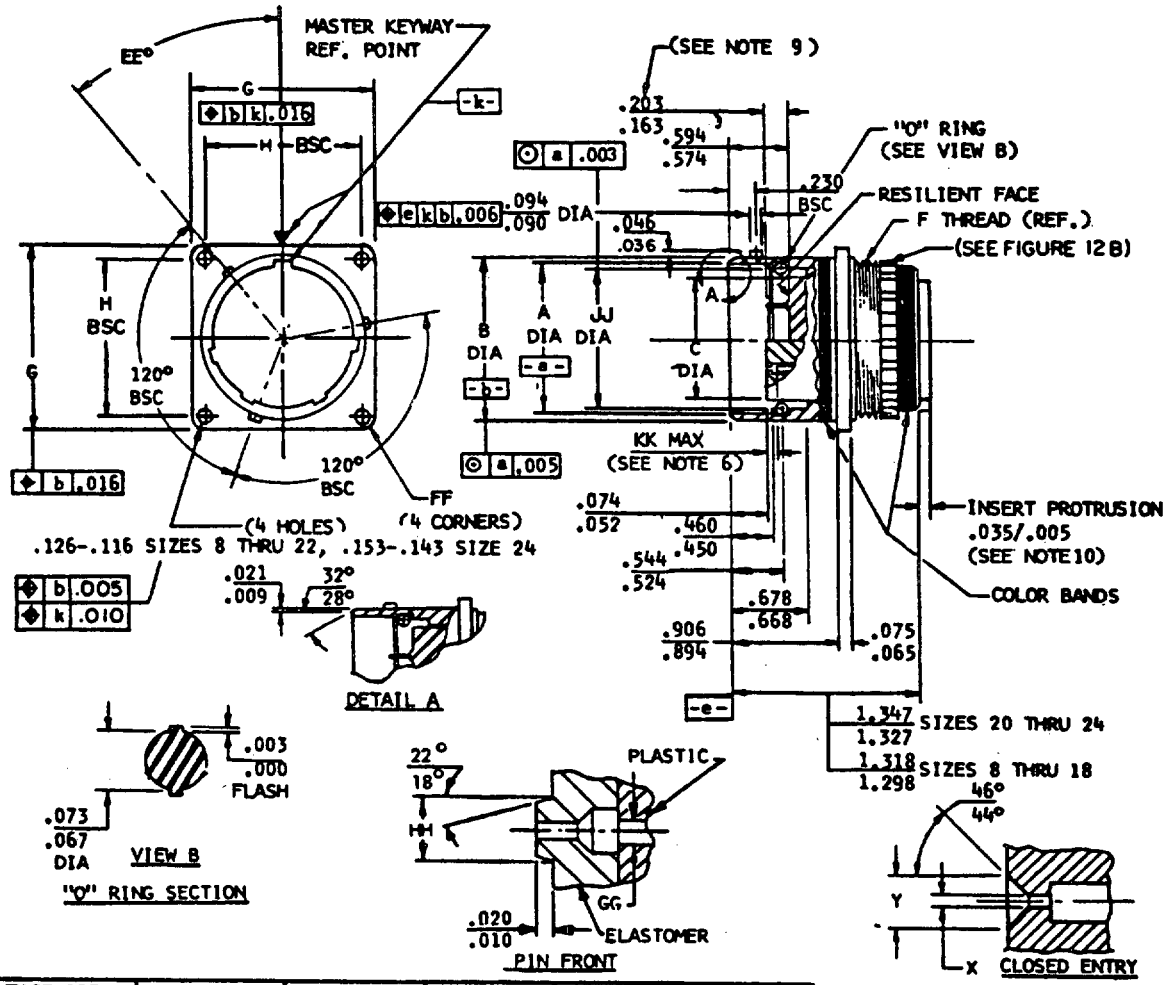
FIGURE 9A Connector, receptacle, electrical, flange mount. (series 1) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

Shell size	A dia	B dia	C dia	G	H BASIC	J dia	FF rad	JJ dia	M	MM	EE° BSC Position 1	2	3	4	5	6
8	.4095 (10.40) .4045 (10.27)	.553 (14.05) .547 (13.89)	.290 (7.37) .280 (7.11)	.820 (20.83) .804 (20.42)	.594 (15.09)	.549 (13.94) .539 (13.69)	.120 (3.05) .100 (2.54)	.3895 (9.89) .3845 (9.77)	1.291 (32.79) 1.276 (32.41)	1.471 (37.36) 1.406 (35.71)	60	46	43	60	89	43
10	.5345 (13.58) .5295 (13.45)	.678 (17.22) .672 (17.07)	.415 (10.54) .405 (10.29)	.945 (24.00) .929 (23.60)	.719 (18.26)	.674 (17.12) .664 (16.87)	.120 (3.05) .100 (2.54)	.5145 (13.07) .5095 (12.94)	1.291 (32.79) 1.276 (32.41)	1.471 (37.36) 1.406 (35.71)	53	87	51	77	87	87
14	.7845 (19.93) .7795 (19.80)	.928 (23.57) .922 (23.42)	.665 (16.89) .655 (16.64)	1.133 (28.78) 1.117 (28.37)	.906 (23.01)	.924 (23.47) .914 (23.22)	.120 (3.05) .100 (2.54)	.7645 (19.42) .7595 (19.29)	1.291 (32.79) 1.276 (32.41)	1.471 (37.36) 1.406 (35.71)	53	87	51	77	87	87
16	.9105 (23.13) .9055 (23.00)	1.054 (26.77) 1.048 (26.62)	.791 (20.09) .781 (19.84)	1.258 (31.95) 1.242 (31.55)	.969 (24.61)	1.050 (26.67) 1.040 (26.42)	.150 (3.81) .130 (3.30)	.8905 (22.62) .8855 (22.49)	1.291 (32.79) 1.276 (32.41)	1.471 (37.36) 1.406 (35.71)	98	98	98	98	98	98
18	1.0345 (26.28) 1.0295 (26.15)	1.178 (29.92) 1.172 (29.77)	.916 (23.27) .906 (23.01)	1.351 (34.32) 1.335 (33.91)	1.062 (26.97)	1.173 (29.79) 1.163 (29.54)	.150 (3.81) .130 (3.30)	1.0145 (25.77) 1.0095 (25.64)	1.291 (32.79) 1.276 (32.41)	1.471 (37.36) 1.406 (35.71)	98	98	98	98	98	98
20	1.1595 (29.45) 1.1545 (29.32)	1.303 (33.10) 1.297 (32.94)	1.041 (26.44) 1.031 (26.19)	1.475 (37.47) 1.459 (37.06)	1.156 (29.36)	1.298 (32.97) 1.288 (32.72)	.150 (3.81) .130 (3.30)	1.1395 (28.94) 1.1345 (28.82)	1.371 (34.82) 1.356 (34.44)	1.551 (39.40) 1.486 (37.74)	98	98	98	98	98	98
22	1.2835 (32.60) 1.2795 (32.50)	1.428 (36.27) 1.422 (36.12)	1.166 (29.62) 1.156 (29.36)	1.570 (39.88) 1.554 (39.47)	1.250 (31.75)	1.423 (36.14) 1.413 (35.89)	.150 (3.81) .130 (3.30)	1.2645 (32.12) 1.2595 (31.99)	1.371 (34.82) 1.356 (34.44)	1.551 (39.40) 1.486 (37.74)	98	98	98	98	98	98
24	1.4095 (35.80) 1.4045 (35.67)	1.553 (39.45) 1.547 (39.29)	1.291 (32.79) 1.281 (32.54)	1.711 (43.46) 1.695 (43.05)	1.375 (34.92)	1.548 (39.32) 1.538 (39.07)	.150 (3.81) .130 (3.30)	1.3895 (35.29) 1.3845 (35.17)	1.371 (34.82) 1.356 (34.44)	1.551 (39.40) 1.486 (37.74)	98	98	98	98	98	98

FIGURE 9A Connector, receptacle, electrical, flange mount. (series 1) - Continued.

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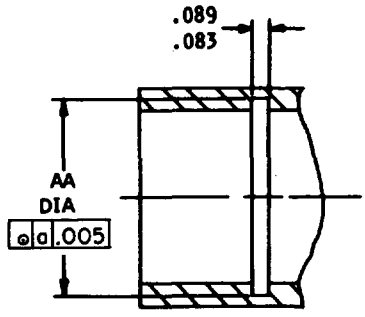


CONTACT SIZE	X DIA	Y DIA	GG DIA	HH DIA	KK (MAX)
23 - 22	.035 (.89) .032 (.81)	.075 (1.91) .065 (1.65)	.0280(.71) .0300(.76)	.068 (1.73) .062 (1.57)	.083(2.11)
20 - 20	.053(1.35) .050(1.27)	.085 (2.16) .080 (2.03)	.0415(1.05) .0445(1.13)	.083 (2.11) .077 (1.96)	.090(2.29)
16 - 16	.077(1.96) .074(1.88)	.108 (2.74) .103 (2.62)	.0950(2.41) .0990(2.52)	.106 (2.69) .100 (2.54)	.105(2.67)
12 - 12	.130(3.30) .126(3.20)	.170 (4.32) .166 (4.22)	.1540(3.91) .1580(4.01)	.169 (4.29) .163 (4.14)	.105(2.67)

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.003	.08	.040	1.02	.090	2.29	.275	6.99	.678	17.22
.004	.10	.046	1.17	.094	2.39	.290	7.37	.894	22.71
.005	.13	.052	1.32	.100	2.54	.450	11.43	.906	23.01
.006	.15	.065	1.65	.116	2.95	.460	11.68	1.298	32.97
.009	.23	.067	1.70	.126	3.20	.524	13.31	1.318	33.48
.010	.25	.073	1.85	.143	3.63	.544	13.82	1.327	33.71
.015	.38	.074	1.88	.153	3.89	.574	14.58	1.347	34.21
.016	.41	.075	1.91	.163	4.14	.594	15.09		
.020	.51	.083	2.11	.203	5.16	.668	16.97		
.021	.53	.035	.89	.230	5.84				

FIGURE 9B. Connector, receptacle, electrical, flange mount (series 3).

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"O" RING GROOVE
DIMENSIONS

SIZE	AA DIA
8	.493 (12.52) .488 (12.40)
10	.618 (15.70) .613 (15.57)
14	.868 (22.05) .863 (21.92)
16	.994 (25.25) .989 (25.12)
18	1.118 (28.40) 1.113 (28.27)
20	1.243 (31.57) 1.238 (31.45)
22	1.368 (34.75) 1.363 (34.62)
24	1.493 (37.92) 1.488 (37.80)

NOTES:

1. DIMENSIONS ARE IN INCHES.
2. MILLIMETERS ARE IN PARENTHESES.
3. METRIC EQUIVALENTS (TO THE NEAREST .01 MM) ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25.4 MM.
4. DIMENSIONS APPLY AFTER PLATING.
5. MATING KEYWAY POSITIONS AND DIMENSIONS ARE SHOWN IN FIGURE 8.
6. DISTANCE FROM FACE OF INSERT TO THE POINT AT WHICH A GAGE PIN, HAVING THE SAME BASIC DIAMETER AS THE MATING CONTACT AND A SQUARE FACE, ENGAGES THE SOCKET CONTACT SPRING.
7. REAR END CONNECTOR DESIGN FOR ATTACHMENT OF NON-ROTATABLE ACCESSORY HARDWARE IS SHOWN ON FIGURE 12B.
8. DIMENSION INCLUDES CONTACT SLACK.
9. ARRANGEMENTS CONTAINING SIZE 12 CONTACTS, INSERT PROTRUSION. .020/.070

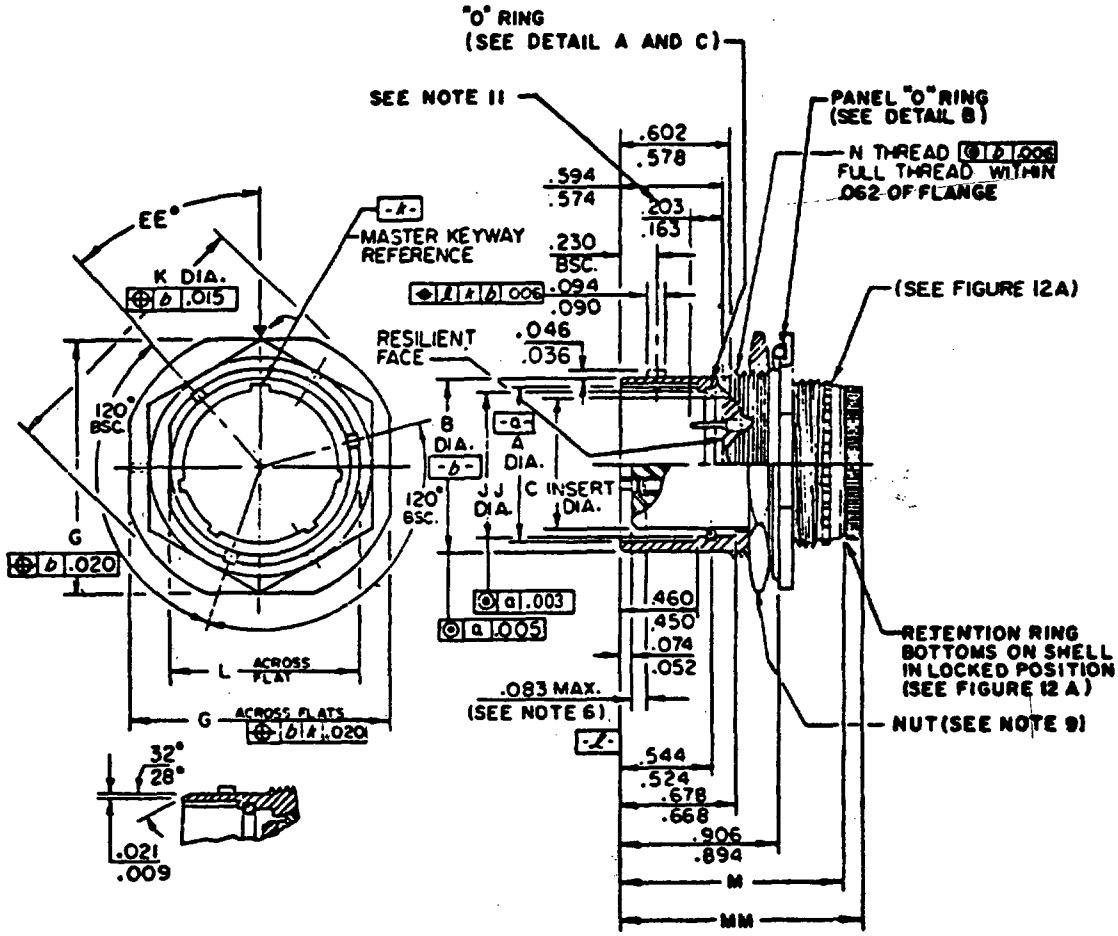
FIGURE 9B Connector, receptacle, electrical, flange mount. (series 3) - Continued.

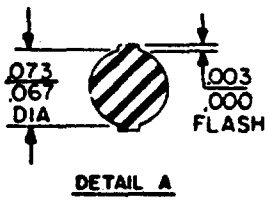
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Shell size	A dia	B dia	C dia	G	H BASIC	J dia	FF rad	JJ dia	EE° BSC Position 1	2	3	4	5	6
8	.4095 (10.40)	.553 (14.05)	.290 (7.37)	.820 (20.83)	.594 (15.09)	.549 (13.94)	.120 (3.05)	.3895 (9.89)	60	46	43	60	89	43
	.4045 (10.27)	.547 (13.89)	.280 (7.11)	.804 (20.42)		.539 (13.69)	.100 (2.54)	.3845 (9.77)						
10	.5345 (13.58)	.678 (17.22)	.415 (10.54)	.945 (24.00)	.719 (18.26)	.674 (17.12)	.120 (3.05)	.5145 (13.07)	53	87	51	77	87	87
	.5295 (13.45)	.672 (17.07)	.405 (10.29)	.929 (23.60)		.664 (16.87)	.100 (2.54)	.5095 (12.94)						
14	.7845 (19.93)	.928 (23.57)	.665 (16.89)	1.133 (28.78)	.906 (23.01)	.924 (23.47)	.120 (3.05)	.7645 (19.42)	53	87	51	77	87	87
	.7795 (19.80)	.922 (23.42)	.655 (16.64)	1.117 (28.37)		.914 (23.22)	.100 (2.54)	.7595 (19.29)						
16	.9105 (23.13)	1.054 (26.77)	.791 (20.09)	1.258 (31.95)	.969 (24.61)	1.050 (26.67)	.150 (3.81)	.8905 (22.62)	98	98	98	98	98	98
	.9055 (23.00)	1.048 (26.62)	.781 (19.84)	1.242 (31.55)		1.040 (26.42)	.130 (3.30)	.8855 (22.49)						
18	1.0345 (26.28)	1.178 (29.92)	.916 (23.27)	1.351 (34.32)	1.062 (26.97)	1.173 (29.79)	.150 (3.81)	1.0145 (25.77)	98	98	98	98	98	98
	1.0295 (26.15)	1.172 (29.77)	.906 (23.01)	1.335 (33.91)		1.163 (29.54)	.130 (3.30)	1.0095 (25.64)						
20	1.1595 (29.45)	1.303 (33.10)	1.041 (26.44)	1.475 (37.47)	1.156 (29.36)	1.298 (32.97)	.160 (4.06)	1.1395 (28.94)	98	98	98	98	98	98
	1.1545 (29.32)	1.297 (32.94)	1.031 (26.19)	1.459 (37.06)		1.288 (32.72)	.130 (3.30)	1.1345 (28.82)						
22	1.2835 (32.60)	1.428 (36.27)	1.166 (29.62)	1.570 (39.88)	1.250 (31.75)	1.423 (36.14)	.160 (4.06)	1.2645 (32.12)	98	98	98	98	98	98
	1.2795 (32.50)	1.422 (36.12)	1.156 (29.36)	1.554 (39.47)		1.413 (35.89)	.130 (3.30)	1.2595 (31.99)						
24	1.4095 (35.80)	1.553 (39.45)	1.291 (32.79)	1.711 (43.46)	1.375 (34.92)	1.548 (39.32)	.160 (4.06)	1.3895 (35.29)	98	98	98	98	98	98
	1.4045 (35.67)	1.547 (39.29)	1.281 (32.54)	1.695 (43.05)		1.538 (39.07)	.130 (3.30)	1.3845 (35.17)						

FIGURE 9B Connector, receptacle, electrical, flange mount. (series 3) - Continued.

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Contact size	X DIA	Y DIA
23 - 22	.035 (.89) .032 (.81)	.075 (1.91) .065 (1.65)
20 - 20	.053 (1.35) .050 (1.27)	.085 (2.16) .080 (2.03)
16 - 16	.077 (1.96) .074 (1.88)	.108 (2.74) .103 (2.62)
12 - 12	.130 (3.30) .126 (3.20)	.170 (4.32) .166 (4.22)

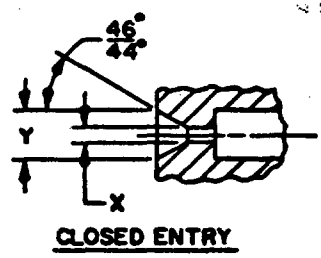
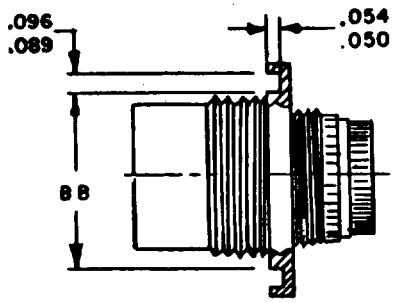
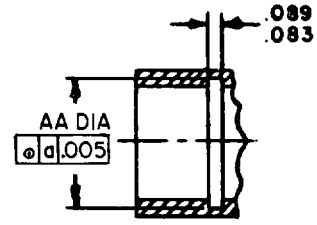


FIGURE 10A. Connector, receptacle, electrical, jam nut mount (series 1).

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DETAIL B



DETAIL C

Size	BB DIA
8	.724 (18.39) .713 (18.11)
10	.849 (21.56) .838 (21.29)
14	1.099 (27.91) 1.088 (27.64)
16	1.221 (31.01) 1.210 (30.73)
18	1.356 (34.44) 1.345 (34.16)
20	1.481 (37.62) 1.470 (37.34)
22	1.606 (40.79) 1.595 (40.51)
24	1.731 (43.97) 1.720 (43.69)

SIZE	AA DIA
8	.493 (12.52) .488 (12.40)
10	.618 (15.70) .613 (15.57)
14	.868 (22.05) .863 (21.92)
16	.994 (25.25) .989 (25.12)
18	1.118 (28.40) 1.113 (28.27)
20	1.243 (31.57) 1.238 (31.45)
22	1.368 (34.75) 1.363 (34.62)
24	1.493 (37.92) 1.488 (37.80)

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.003	.08	.040	1.02	.094	2.39	.460	11.68	.894	22.71
.004	.10	.046	1.17	.157	3.99	.524	13.31	.906	23.01
.005	.13	.052	1.32	.163	4.14	.544	13.82	1.276	32.41
.006	.15	.062	1.57	.169	4.29	.574	14.58	1.291	32.79
.009	.23	.067	1.70	.203	5.16	.578	14.68	1.406	35.71
.010	.25	.073	1.85	.230	5.84	.594	15.09	1.471	37.36
.015	.38	.074	1.88	.450	11.43	.602	15.29		
.020	.51	.083	2.11			.668	16.97		
.021	.53	.090	2.29			.678	17.22		

FIGURE 10A Connector, receptacle, electrical, jam nut mount (series 1).

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SHELL SIZE	A DIA	B DIA	C DIA	G ACROSS FLAT	K DIA	L ACROSS FLAT	H THREAD	JJ DIA	M	CC DIA	PANEL "O" RING	EE° BSC POSITION						MM
												1	2	3	4	5	6	
8	.4095 (10.40) .4045 (10.27)	.553 (14.05) .547 (13.89)	.290 (7.37) .280 (7.11)	.989 (25.12) .969 (24.61)	1.072 (27.23) 1.052 (26.72)	.598 (15.19) .590 (14.99)	5/8-20UN-2A	.3895 (9.89) .3845 (9.77)	1.291 (32.79) 1.276 (32.41)	.301 (7.65) .293 (7.45)	MS9068-018	60	46	43	60	89	43	1.471 (37.36) 1.406 (35.71)
10	.5345 (13.58) .5295 (13.45)	.678 (17.22) .672 (17.07)	.415 (10.54) .405 (10.29)	1.114 (28.30) 1.094 (27.79)	1.197 (30.40) 1.177 (29.90)	.723 (18.36) .715 (18.16)	3/4-20UNEF-2A	.5145 (13.07) .5095 (12.94)	1.291 (32.79) 1.276 (32.41)	.426 (10.82) .418 (10.62)	MS9068-020	53	87	51	77	87	87	1.471 (37.36) 1.406 (35.71)
14	.7845 (19.93) .7795 (19.80)	.928 (23.57) .922 (23.42)	.665 (16.89) .655 (16.64)	1.401 (35.59) 1.381 (35.08)	1.510 (38.35) 1.490 (37.85)	.973 (24.71) .965 (24.51)	1-20UNEF-2A	.7645 (19.42) .7595 (19.29)	1.291 (19.42) 1.276 (32.41)	.676 (17.17) .668 (16.97)	MS9068-024	53	87	51	77	87	87	1.471 (37.36) 1.406 (35.71)
16	.9105 (23.13) .9055 (23.00)	1.054 (26.77) 1.048 (26.62)	.791 (20.09) .781 (19.84)	1.526 (38.76) 1.506 (38.25)	1.635 (41.53) 1.615 (41.02)	1.098 (27.89) 1.090 (27.69)	1-1/8-20UN-2A	.8905 (22.62) .8855 (22.49)	1.291 (32.79) 1.276 (32.41)	.802 (20.37) .794 (20.17)	MS9068-026	98	98	98	98	98	98	1.471 (37.36) 1.406 (35.71)
18	1.0345 (26.28) 1.0295 (26.15)	1.178 (29.92) 1.172 (29.77)	.916 (23.27) .906 (23.01)	1.651 (41.94) 1.631 (41.43)	1.745 (44.32) 1.725 (43.82)	1.222 (31.04) 1.214 (30.84)	1-1/4-20UN-2A	1.0145 (25.77) 1.0095 (25.64)	1.291 (32.79) 1.276 (32.41)	.927 (23.55) .919 (23.35)	MS9068-028	98	98	98	98	98	98	1.471 (37.36) 1.406 (35.71)
20	1.1595 (29.45) 1.1545 (29.32)	1.303 (33.10) 1.297 (32.94)	1.041 (26.44) 1.031 (26.19)	1.776 (45.11) 1.756 (44.60)	1.870 (47.50) 1.850 (46.99)	1.347 (28.94) 1.339 (34.01)	1-3/8-20UN-2A	1.1395 (28.94) 1.1345 (28.82)	1.371 (34.82) 1.356 (34.44)	1.052 (26.72) 1.044 (26.52)	MS9068-029	98	98	98	98	98	98	1.551 (39.40) 1.486 (37.74)
22	1.2845 (32.63) 1.2795 (32.50)	1.428 (36.27) 1.422 (36.12)	1.166 (29.62) 1.156 (29.36)	1.901 (48.29) 1.881 (47.78)	1.995 (50.67) 1.975 (50.17)	1.472 (37.39) 1.464 (37.19)	1-1/2-20UN-2A	1.2645 (32.12) 1.2595 (31.99)	1.371 (34.82) 1.356 (34.44)	1.177 (29.90) 1.169 (29.70)	MS9068-030	98	98	98	98	98	98	1.551 (39.40) 1.486 (37.74)
24	1.4095 (35.80) 1.4045 (35.67)	1.553 (39.45) 1.547 (39.29)	1.291 (32.79) 1.281 (32.54)	2.026 (51.46) 2.006 (50.95)	2.120 (53.85) 2.100 (53.34)	1.597 (40.56) 1.589 (40.36)	1-5/8-20UN-2A	1.3895 (35.29) 1.3845 (35.17)	1.371 (34.82) 1.356 (34.44)	1.302 (33.07) 1.295 (32.90)	MS9068-031	98	98	98	98	98	98	1.551 (39.40) 1.486 (37.74)

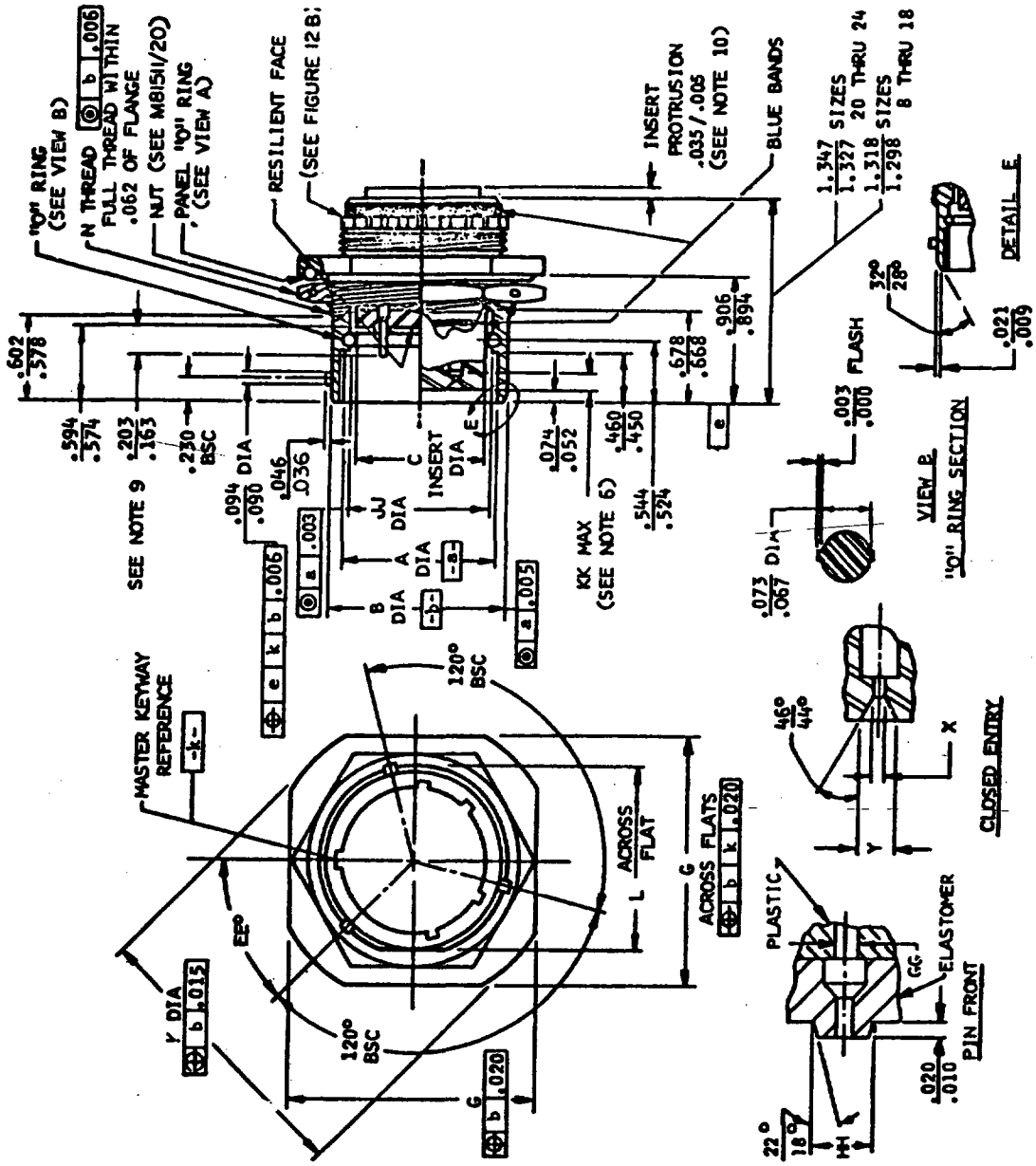
FIGURE 10A. Connector, receptacle, electrical, jam nut mount (series 1) - Continued.

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NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Dimensions apply after plating.
5. Mating keyway positions and dimensions are shown in figure 8.
6. Distance from face of insert to the point at which a gage pin, having the same basic diameter as the mating contact and a square face, engages the socket contact spring.
7. This design information establishes connector intermating criteria and interchangeability of accessory hardware.
8. All dimensions are for connector with contact retention system in locked position.
9. Slotted nut shall conform to MIL-C-81511/20.
10. Rear end connector design for attachment of non-rotatable accessory hardware is shown on figure 12A.
11. Dimension includes contact slack.

FIGURE 10A. Connector, receptacle, electrical, jam nut mount (series 1) - Continued.

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CONTACT SIZE	X DIA	Y DIA	GG DIA	HH DIA	KK(MAX)
23 - 22	.035 (.89) .032 (.81)	.075 (1.91) .065 (1.65)	.0280(.71) .0300(.76)	.068 (1.73) .062 (1.57)	.083(2.11)
20 - 20	.053(1.35) .050(1.27)	.085 (2.16) .080 (2.03)	.0415(1.05) .0445(1.13)	.083 (2.11) .077 (1.96)	.090(2.29)
16 - 16	.077(1.96) .074(1.88)	.108 (2.74) .103 (2.62)	.0950(2.41) .0990(2.52)	.106 (2.69) .100 (2.54)	.105(2.67)
12 - 12	.130(3.30) .126(3.20)	.170 (4.32) .166 (4.22)	.1540(3.91) .1580(4.01)	.169 (4.29) .163 (4.14)	.105(2.67)

FIGURE 10B. Connector, receptacle, electrical, jam nut mount (series 3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
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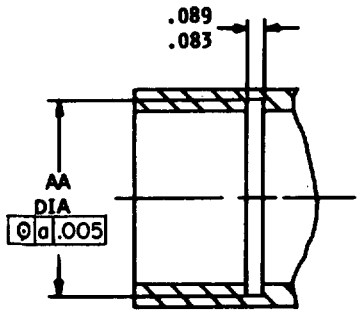
Shell size	A dia	B dia	C dia	G across flat	K dia	L across flat	N thread	JJ dia	Panel 'O' ring	EE° BSC Position 1	2	3	4	5	6
8	.4095 (10.40) .4045 (10.27)	.553 (14.05) .547 (13.89)	.290 (7.37) .280 (7.11)	.989 (25.12) .969 (24.61)	1.072 (27.23) 1.052 (26.72)	.598 (15.19) .590 (14.99)	5/8-20UN-2A	.3895 (9.89) .3845 (9.77)	MS9068-018	60	46	43	60	89	43
10	.5345 (13.58) .5295 (13.45)	.678 (17.22) .672 (17.07)	.415 (10.54) .405 (10.29)	1.114 (28.30) 1.094 (27.79)	1.197 (30.40) 1.177 (29.90)	.723 (18.36) .715 (18.16)	¾-20UNEF-2A	.5145 (13.07) .5095 (12.94)	MS9068-020	53	87	51	77	87	87
14	.7845 (19.93) .7795 (19.80)	.928 (23.57) .922 (23.42)	.665 (16.89) .655 (16.64)	1.401 (35.59) 1.381 (35.08)	1.510 (38.35) 1.490 (37.85)	.973 (24.71) .965 (24.51)	1-20UNEF-2A	.7645 (19.42) .7595 (19.29)	MS9068-024	53	87	51	77	87	87
16	.9105 (23.13) .9055 (23.00)	1.054 (26.77) 1.048 (26.62)	.791 (20.09) .781 (19.84)	1.526 (38.76) 1.506 (38.25)	1.635 (41.53) 1.615 (41.02)	1.098 (27.89) 1.090 (27.69)	1-1/8-20UN-2A	.8905 (22.62) .8855 (22.49)	MS9068-026	98	98	98	98	98	98
18	1.0345 (26.28) 1.0295 (26.15)	1.178 (29.92) 1.172 (29.77)	.916 (23.27) .906 (23.01)	1.651 (41.94) 1.631 (41.43)	1.745 (44.32) 1.725 (43.82)	1.222 (31.04) 1.214 (30.84)	1-¼-20UN-2A	1.0145 (25.77) 1.0095 (25.64)	MS9068-028	98	98	98	98	98	98
20	1.1595 (29.45) 1.1545 (29.32)	1.303 (33.10) 1.297 (32.94)	1.041 (26.44) 1.031 (26.19)	1.776 (45.11) 1.756 (44.60)	1.870 (47.50) 1.850 (46.99)	1.347 (34.21) 1.339 (34.01)	1-3/8-20UN-2A	1.1395 (28.94) 1.1345 (28.82)	MS9068-029	98	98	98	98	98	98
22	1.2845 (32.63) 1.2795 (32.50)	1.428 (36.27) 1.422 (36.12)	1.166 (29.62) 1.156 (29.36)	1.901 (48.29) 1.881 (47.78)	1.995 (50.67) 1.975 (50.17)	1.472 (37.39) 1.464 (37.19)	1-½-20UN-2A	1.2645 (32.12) 1.2595 (31.99)	MS9068-030	98	98	98	98	98	98
24	1.4095 (35.80) 1.4045 (35.67)	1.553 (39.45) 1.547 (39.29)	1.291 (32.79) 1.281 (32.54)	2.026 (51.46) 2.006 (50.95)	2.120 (53.85) 2.100 (53.34)	1.597 (40.56) 1.589 (40.36)	1-5/8-20UN-2A	1.3895 (35.29) 1.3845 (35.17)	MS9068-031	98	98	98	98	98	98

NOTES:

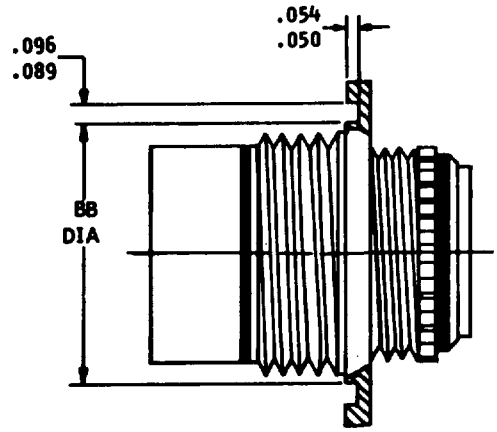
1. DIMENSIONS ARE IN INCHES.
2. MILLIMETERS ARE IN PARENTHESES.
3. METRIC EQUIVALENTS (TO THE NEAREST .01 MM) ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25.4 MM.
4. DIMENSIONS APPLY AFTER PLATING.
5. MATING KEYWAY POSITIONS AND DIMENSIONS ARE SHOWN IN FIGURE 8.
6. DISTANCE FROM FACE OF INSERT TO THE POINT AT WHICH A GAGE PIN, HAVING THE SAME BASIC DIAMETER AS THE MATING CONTACT AND A SQUARE FACE, ENGAGES THE SOCKET CONTACT SPRING.
7. REAR END CONNECTOR DESIGN FOR ATTACHMENT OF NON-ROTATABLE ACCESSORY HARDWARE IS SHOWN ON FIGURE 12B.
8. DIMENSION INCLUDES CONTACT SLACK.
9. ARRANGEMENTS CONTAINING SIZE 12 CONTACTS, INSERT PROTRUSION. .020/.070

FIGURE 10 B. CONNECTOR, RECEPTACLE, ELECTRICAL, JAM-NUT MOUNT (SERIES 3) - CONTINUED.

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"O" RING GROOVE
DIMENSIONS



PANEL "O" RING
VIEW A

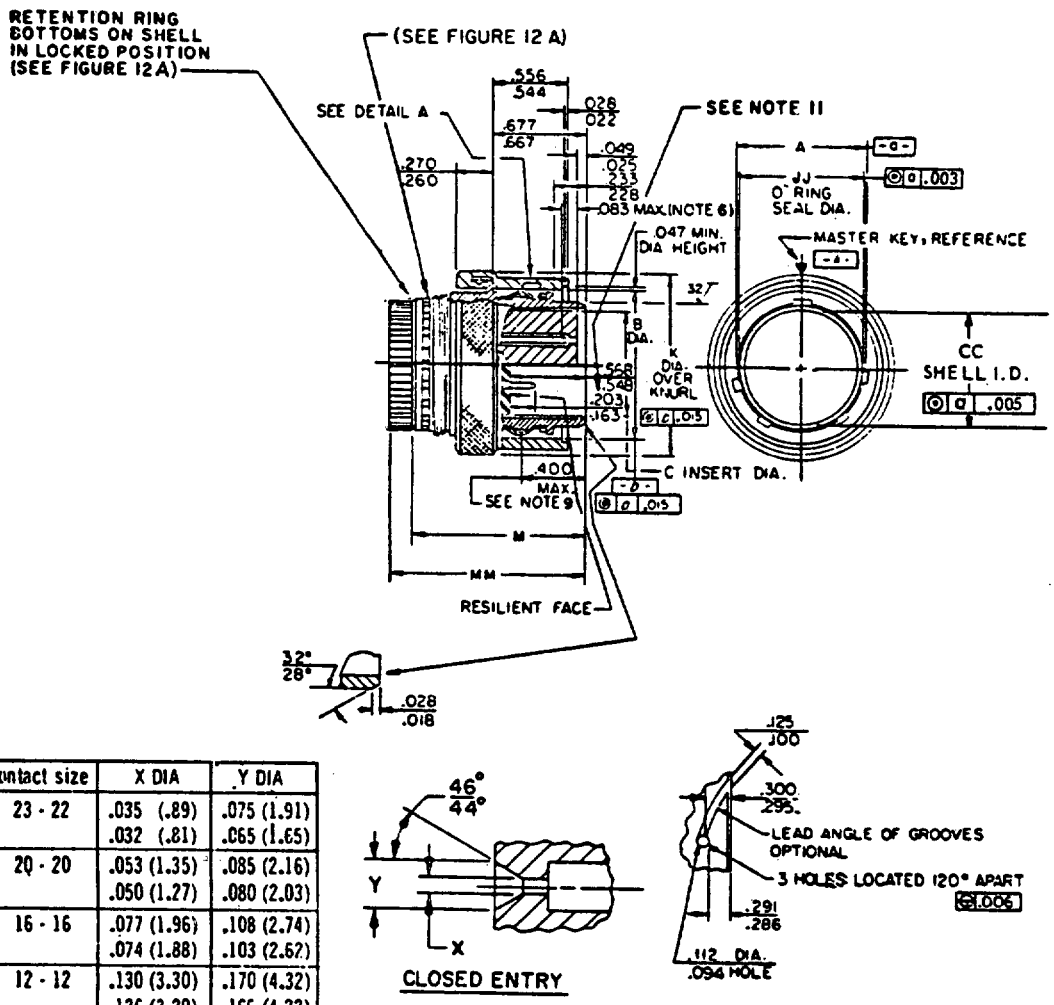
SIZE	AA DIA
8	.493 (12.52) .488 (12.40)
10	.618 (15.70) .613 (15.57)
14	.868 (22.05) .863 (21.92)
16	.994 (25.25) .989 (25.12)
18	1.118 (28.40) 1.113 (28.27)
20	1.243 (31.57) 1.238 (31.45)
22	1.368 (34.75) 1.363 (34.62)
24	1.493 (37.92) 1.488 (37.80)

Size	BB DIA
8	.724 (18.39) .713 (18.11)
10	.849 (21.56) .838 (21.29)
14	1.099 (27.91) 1.088 (27.64)
16	1.221 (31.01) 1.210 (30.73)
18	1.356 (34.44) 1.345 (34.16)
20	1.481 (37.62) 1.470 (37.34)
22	1.606 (40.79) 1.595 (40.51)
24	1.731 (43.97) 1.720 (43.69)

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.003	.08	.040	1.02	.094	2.39	.460	11.68	.894	22.71
.004	.10	.046	1.17	.157	3.99	.524	13.31	.906	23.01
.005	.13	.052	1.32	.163	4.14	.544	13.82	1.298	32.97
.006	.15	.062	1.57	.169	4.29	.574	14.58	1.318	33.48
.009	.23	.067	1.70	.203	5.16	.578	14.68	1.327	33.71
.010	.25	.073	1.85	.230	5.84	.594	15.09	1.347	34.21
.015	.38	.074	1.88	.450	11.43	.602	15.29		
.020	.51	.083	2.11			.668	16.97		
.021	.53	.090	2.29			.678	17.22		
.035	.89								

FIGURE 10B. CONNECTOR, RECEPTACLE, ELECTRICAL, JAM NUT MOUNT (SERIES 3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.



Contact size	X DIA	Y DIA
23 - 22	.035 (.89) .032 (.81)	.075 (1.91) .065 (1.65)
20 - 20	.053 (1.35) .050 (1.27)	.085 (2.16) .080 (2.03)
16 - 16	.077 (1.96) .074 (1.88)	.108 (2.74) .103 (2.62)
12 - 12	.130 (3.30) .126 (3.20)	.170 (4.32) .166 (4.22)

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.005	.13	.047	1.19	.163	4.14	.286	7.26	.556	14.12
.006	.15	.049	1.24	.169	4.29	.290	7.37	.568	14.43
.008	.20	.083	2.11	.203	5.16	.291	7.39	.667	16.94
.015	.38	.094	2.39	.228	5.79	.295	7.49	.677	17.20
.018	.46	.100	2.54	.233	5.92	.300	7.62	1.250	31.75
.022	.56	.112	2.84	.260	6.60	.400	10.16	1.265	32.13
.025	.64	.125	3.18	.270	6.86	.544	13.82		
.028	.71	.157	3.99	.275	6.99	.548	13.92		

FIGURE 11A. Connector, plug, electrical (series 1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
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SHELL SIZE	A DIA	B DIA	C DIA	K DIA	M	CC DIA	JJ DIA	MM
8	.4015 (10.20) .3955 (10.05)	.566 (14.38) .558 (14.17)	.290 (7.37) .280 (7.11)	.850 (21.59) .830 (21.08)	1.265 (32.13) 1.250 (31.75)	.301 (7.65) .293 (7.45)	.3815 (9.69) .373 (9.47)	1.445 (36.70) 1.380 (35.05)
10	.5265 (13.37 .5205 (13.22)	.691 (17.55) .683 (17.35)	.415 (10.54) .405 (10.29)	.975 (24.77) .955 (24.26)	1.265 (32.13) 1.250 (31.75)	.426 (10.82) .418 (10.62)	.5065 (12.87) .498 (12.65)	1.445 (36.70) 1.380 (35.05)
14	.7765 (19.72) .7705 (19.57)	.941 (23.90) .933 (23.70)	.665 (16.89) .655 (16.64)	1.225 (31.12) 1.205 (30.61)	1.265 (32.13) 1.250 (31.75)	.676 (17.17) .668 (16.97)	.7565 (19.22) .746 (18.95)	1.445 (36.70) 1.380 (35.05)
16	.9025 (22.92) .8965 (22.77)	1.068 (27.13) 1.060 (26.92)	.791 (20.09) .781 (19.84)	1.350 (34.29) 1.330 (33.78)	1.265 (32.13) 1.250 (31.75)	.802 (20.37) .794 (20.17)	.8825 (22.42) .872 (22.15)	1.445 (36.70) 1.380 (35.05)
18	1.0265 (26.07) 1.0205 (25.92)	1.191 (30.25) 1.183 (30.05)	.916 (23.27) .906 (23.01)	1.475 (37.47) 1.455 (36.96)	1.265 (32.13) 1.250 (31.75)	.927 (23.55) .919 (23.35)	1.0065 (25.57) .996 (25.30)	1.445 (36.70) 1.380 (35.05)
20	1.1515 (29.25) 1.1455 (29.10)	1.316 (33.43) 1.308 (33.22)	1.041 (26.44) 1.031 (26.19)	1.624 (41.25) 1.604 (40.74)	1.345 (34.16) 1.330 (33.78)	1.052 (26.72) 1.044 (26.52)	1.1315 (28.74) 1.121 (28.47)	1.525 (38.74) 1.460 (37.08)
22	1.2765 (32.42) 1.2705 (32.27)	1.441 (36.60) 1.433 (36.40)	1.166 (29.62) 1.156 (29.36)	1.749 (44.42) 1.729 (43.92)	1.345 (34.16) 1.330 (33.78)	1.177 (29.90) 1.169 (29.70)	1.2565 (31.92) 1.246 (31.65)	1.525 (38.74) 1.460 (37.08)
24	1.4015 (35.60) 1.3955 (35.45)	1.566 (39.78) 1.558 (39.57)	1.291 (32.79) 1.281 (32.54)	1.874 (47.60) 1.854 (47.09)	1.345 (34.16) 1.330 (33.78)	1.302 (33.07) 1.295 (32.90)	1.3815 (35.09) 1.371 (34.82)	1.525 (38.74) 1.460 (37.08)

NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Dimensions apply after plating.
5. Mating key positions and dimensions are shown in figure 8.
6. Distance from front of insert and to point at which a gage pin, having the same basic diameter as the mating contact and a square face, engages the socket contact spring.
7. This design information establishes connector intermating criteria and interchangeability of accessory hardware.
8. All dimensions are for connector with contact retention system in locked position.
9. Distance from front of shell to the point at which a gage ring, having the same basic diameter as the receptacle shell engages the spring member.
10. Rear end connector design for attachment of non-rotatable accessory hardware is shown on figure 12A.
11. Dimension includes contact slack.

FIGURE 11A. Connector, plug, electrical (series 1) - Continued.

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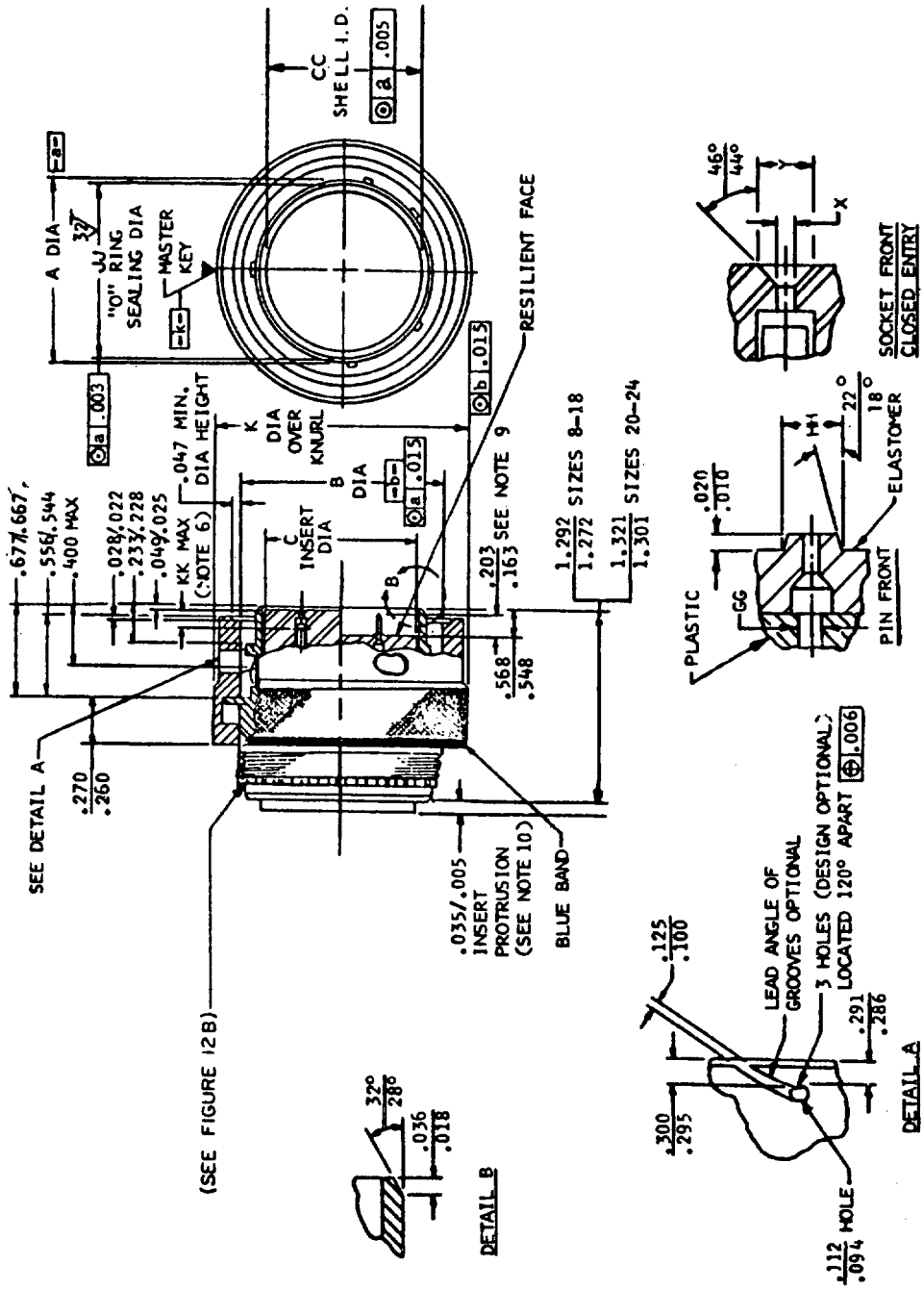


FIGURE 11B. Connector, plug, electrical (series 3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
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SHELL SIZE	A DIA	B DIA	C DIA	K DIA	C_C DIA	JJ DIA
8	.4015 (10.20) .3955 (10.05)	.566 (14.38) .558 (14.17)	.290 (7.37) .280 (7.11)	.850 (21.59) .830 (21.08)	.301 (7.65) .293 (7.45)	.3815 (9.69) .373 (9.47)
10	.5265 (13.37) .5205 (13.22)	.691 (17.55) .683 (17.35)	.415 (10.54) .405 (10.29)	.975 (24.77) .955 (24.26)	.426 (10.82) .418 (10.62)	.5065 (12.87) .498 (12.65)
14	.7765 (19.72) .7705 (19.57)	.941 (23.90) .933 (23.70)	.665 (16.89) .655 (16.64)	1.225 (31.12) 1.205 (30.61)	.676 (17.17) .668 (16.97)	.7565 (19.22) .746 (18.95)
16	.9025 (22.92) .8965 (22.77)	1.068 (27.13) 1.060 (26.92)	.791 (20.09) .781 (19.84)	1.350 (34.29) 1.330 (33.78)	.802 (20.37) .794 (20.17)	.8825 (22.42) .872 (22.15)
18	1.0265 (26.07) 1.0205 (25.92)	1.191 (30.25) 1.183 (30.05)	.916 (23.27) .906 (23.01)	1.475 (37.47) 1.455 (36.96)	.927 (23.55) .919 (23.35)	1.0065 (25.57) .996 (25.30)
20	1.1515 (29.25) 1.1455 (29.10)	1.316 (33.43) 1.308 (33.22)	1.041 (26.44) 1.031 (26.19)	1.624 (41.25) 1.604 (40.74)	1.052 (26.72) 1.044 (26.52)	1.1315 (28.74) 1.121 (28.47)
22	1.2765 (32.42) 1.2705 (32.27)	1.441 (36.60) 1.433 (36.40)	1.166 (29.62) 1.156 (29.36)	1.749 (44.42) 1.729 (43.92)	1.177 (29.90) 1.169 (29.70)	1.2565 (31.92) 1.246 (31.65)
24	1.4015 (35.60) 1.3955 (35.45)	1.566 (39.78) 1.558 (39.57)	1.291 (32.79) 1.281 (32.54)	1.874 (47.60) 1.854 (47.09)	1.302 (33.07) 1.295 (32.90)	1.3815 (35.09) 1.371 (34.82)

NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Dimensions apply after plating.
5. Mating keyway positions and dimensions are shown in figure 8.
6. Distance from face of insert to the point at which a gage pin, having the same basic diameter as the mating contact and a square face, engages the socket contact spring.
7. Distance from front of shell to the point at which a gage pin, having the same basic diameter as plug shell, engages the spring member.
8. Rear end connector design for attachment of non-rotatable accessory hardware is shown on figure 12B.
9. Dimension includes contact slack.
10. Arrangements containing size 12 contacts, insert protrusion. .020/.070

FIGURE 11B. Connector, plug, electrical (series 3) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
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CONTACT SIZE	X DIA	Y DIA	GG DIA	HH DIA	KK(MAX)
23 - 22	.035 (.89) .032 (.81)	.075 (1.91) .065 (1.85)	.0280(.71) .0300(.76)	.068 (1.73) .062 (1.57)	.083(2.11)
20 - 20	.053(1.35) .050(1.27)	.085 (2.16) .080 (2.03)	.0415(1.05) .0445(1.13)	.083 (2.11) .077 (1.96)	.090(2.29)
16 - 16	.077(1.96) .074(1.88)	.108 (2.74) .103 (2.62)	.0950(2.41) .0990(2.52)	.106 (2.69) .100 (2.54)	.105(2.67)
12 - 12	.130(3.30) .126(3.20)	.170 (4.32) .166 (4.22)	.1540(3.91) .1580(4.01)	.169 (4.29) .163 (4.14)	.105(2.67)

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.005	.13	.047	1.19	.163	4.14	.286	7.26	.556	14.12
.006	.15	.049	1.24	.169	4.29	.290	7.37	.568	14.43
.008	.20	.083	2.11	.203	5.16	.291	7.39	.667	16.94
.015	.38	.094	2.39	.228	5.79	.295	7.49	.677	17.20
.018	.46	.100	2.54	.233	5.92	.300	7.62	1.272	32.31
.022	.56	.112	2.84	.260	6.60	.400	10.16	1.292	32.82
.025	.64	.125	3.18	.270	6.86	.544	13.82	1.301	33.05
.028	.71	.157	3.99	.275	6.99	.548	13.92	1.321	33.55
.035	.89								

FIGURE 11B. Connector, plug, electrical (series 3) - Continued.

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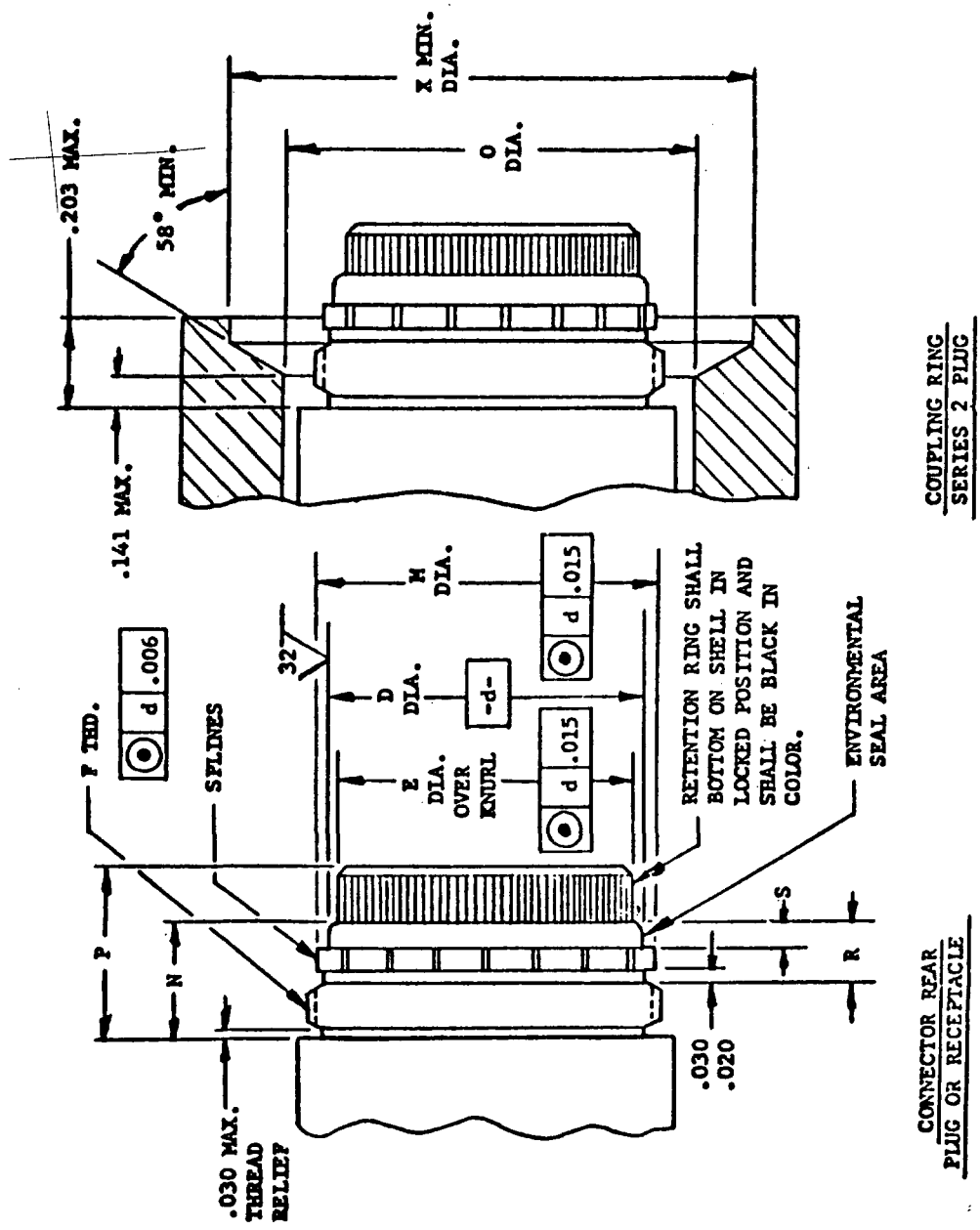
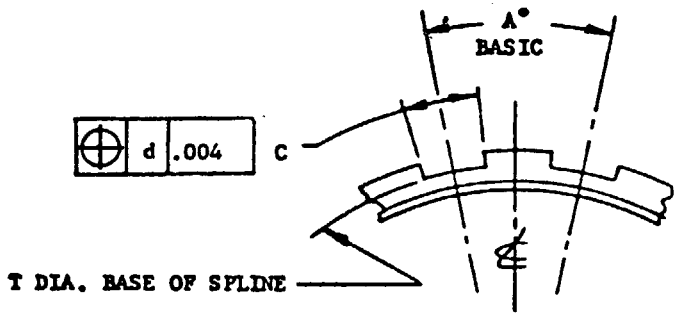


FIGURE 12A. Series 1 & 2 connector back end configuration.

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INCHES	M.M.
.020	.51
.030	.76
.141	3.58
.203	5.16

SPLINE DETAIL

NOTES:
1. DIMENSIONS ARE IN INCHES.
2. DIMENSIONS APPLY AFTER PLATING, AND WITH CONTACT RETENTION SYSTEM IN LOCKED POSITION.
3. METRIC EQUIVALENTS (TO THE NEAREST .01mm) ARE GIVEN FOR GENERAL INFORMATION ONLY.

FIGURE 12A. Series 1 & 2 connector back end configuration - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

SHELL SIZE	A (DEGREES)	C	D DIA.	E DIA.	F THREAD	N DIA.	N
8	20	.051 (1.30) .046 (1.17)	.430 (10.92) .424 (10.77)	.422 (10.72) .407 (10.34)	1/2-28UNEF-2A	.463 (11.76) .457 (11.61)	.290 (7.37) .280 (7.11)
10	15	.049 (1.24) .044 (1.12)	.555 (14.10) .549 (13.94)	.548 (13.92) .533 (13.54)	5/8-28UN-2A	.588 (14.94) .582 (14.78)	.290 (7.37) .280 (7.11)
14	12	.054 (1.37) .049 (1.24)	.805 (20.45) .799 (20.29)	.798 (20.27) .783 (19.89)	7/8-28UN-2A	.838 (21.29) .832 (21.13)	.290 (7.37) .280 (7.11)
16	10	.053 (1.35) .048 (1.22)	.930 (23.62) .924 (23.47)	.924 (23.47) .909 (23.09)	1-28UN-2A	.963 (24.46) .957 (24.31)	.290 (7.37) .280 (7.11)
18	9	.053 (1.35) .048 (1.22)	1.055 (26.80) 1.049 (26.64)	1.048 (26.62) 1.033 (26.24)	1-1/8-28UN-2A	1.088 (27.64) 1.082 (27.48)	.290 (7.37) .280 (7.11)
20	9	.059 (1.50) .054 (1.37)	1.180 (29.97) 1.174 (29.82)	1.173 (29.79) 1.158 (29.41)	1-1/4-28UN-2A	1.213 (30.81) 1.207 (30.66)	.364 (9.25) .349 (8.86)
22	9	.062 (1.57) .057 (1.45)	1.305 (33.15) 1.299 (32.99)	1.298 (32.97) 1.283 (32.59)	1-3/8-28UN-2A	1.338 (33.99) 1.332 (33.83)	.364 (9.25) .349 (8.86)
24	9	.069 (1.75) .064 (1.63)	1.430 (36.32) 1.424 (36.17)	1.423 (36.14) 1.408 (35.76)	1-1/2-28UN-2A	1.463 (37.16) 1.457 (37.01)	.364 (9.25) .349 (8.86)

SHELL	P MAX.	R	S	T DIA.	O DIA.	X DIA. MIN.
8	.454 (11.53)	.169 (4.29) .157 (3.99)	.076 (1.93) .063 (1.60)	.439 (11.15) .433 (11.00)	.579 (14.71) .559 (14.20)	.645 (16.38)
10	.454 (11.53)	.169 (4.29) .157 (3.99)	.076 (1.93) .063 (1.60)	.564 (14.33) .558 (14.17)	.704 (17.88) .684 (17.37)	.770 (19.56)
14	.454 (11.53)	.169 (4.29) .157 (3.99)	.076 (1.93) .063 (1.60)	.814 (20.68) .808 (20.52)	.954 (24.23) .934 (23.72)	1.020 (25.91)
16	.454 (11.53)	.169 (4.29) .157 (3.99)	.076 (1.93) .063 (1.60)	.939 (23.85) .933 (23.70)	1.082 (27.48) 1.060 (26.92)	1.145 (29.08)
18	.454 (11.53)	.169 (4.29) .157 (3.99)	.076 (1.93) .063 (1.60)	1.064 (27.03) 1.058 (26.87)	1.204 (30.58) 1.184 (30.07)	1.279 (32.49)
20	.530 (13.46)	.217 (5.51) .205 (5.21)	.092 (2.34) .079 (2.01)	1.189 (30.20) 1.183 (30.05)	1.315 (33.40) 1.305 (33.15)	
22	.530 (13.46)	.217 (5.51) .205 (5.21)	.092 (2.34) .079 (2.01)	1.314 (33.38) 1.308 (33.22)	1.440 (36.58) 1.430 (36.32)	
24	.530 (13.46)	.217 (5.51) .205 (5.21)	.092 (2.34) .079 (2.01)	1.439 (36.55) 1.433 (36.40)	1.565 (39.75) 1.555 (39.50)	

FIGURE 12A. Series 1 & 2 connector back end configuration - Continued.

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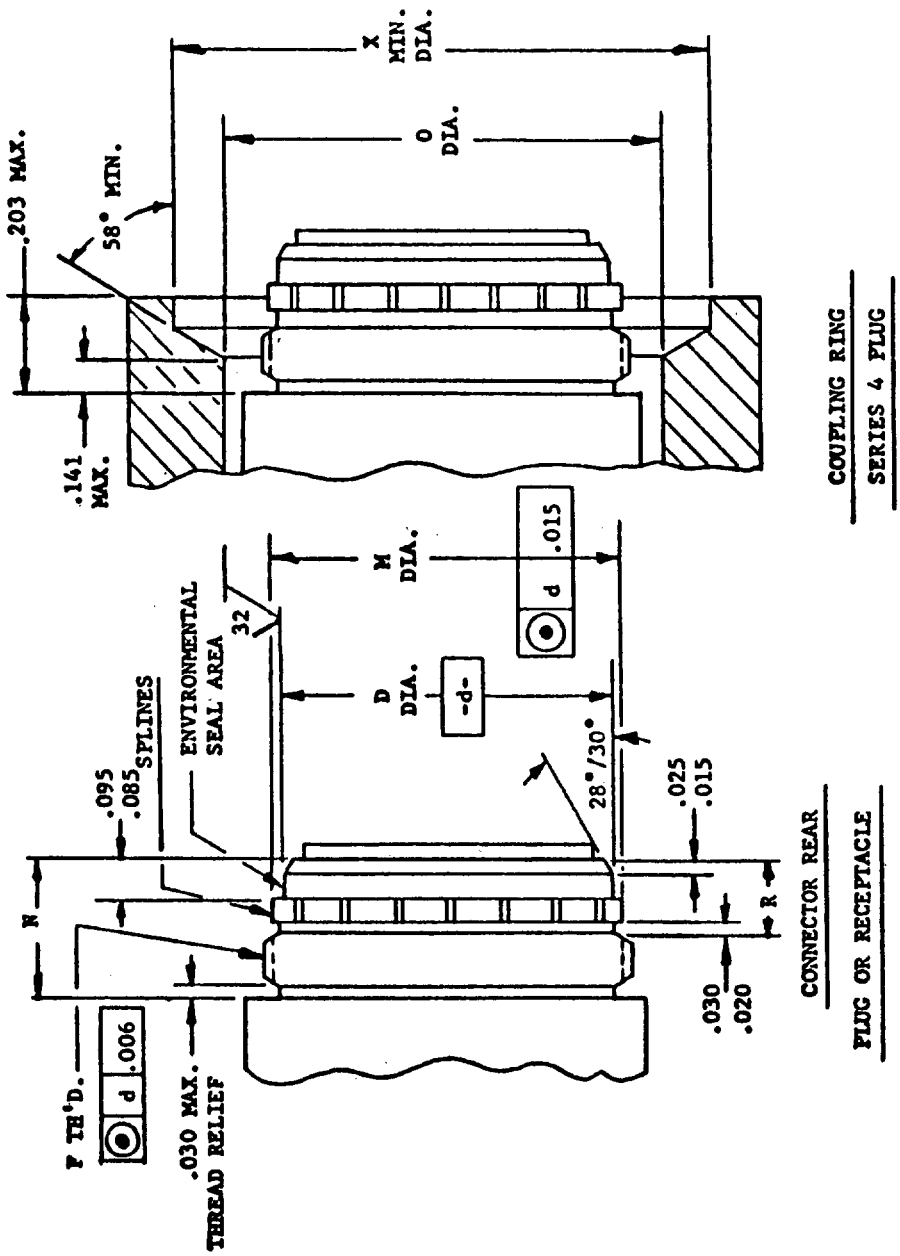


FIGURE 12B. Series 3 & 4 connector back end configuration.

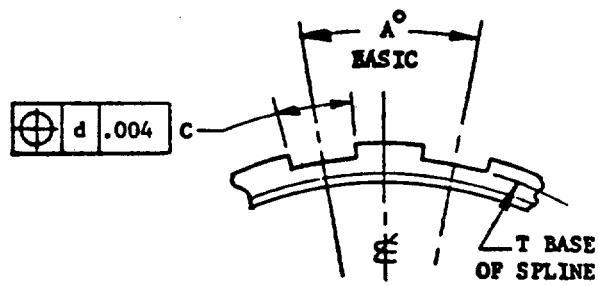
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SHELL SIZE	A (DEGREES)	C	D DIA.	F THREAD	H DIA.	O DIA.
8	20	.051 (1.30) .046 (1.17)	.430 (10.92) .424 (10.77)	1/2-28UNEF-2A	.463 (11.76) .457 (11.61)	.579 (14.71) .559 (14.20)
10	15	.049 (1.24) .044 (1.12)	.555 (14.10) .549 (13.94)	5/8-28UN-2A	.588 (14.94) .582 (14.78)	.704 (17.88) .684 (17.37)
14	12	.054 (1.37) .049 (1.24)	.805 (20.45) .799 (20.29)	7/8-28UN-2A	.838 (21.29) .832 (21.13)	.954 (24.23) .934 (23.72)
16	10	.053 (1.35) .048 (1.22)	.930 (23.62) .924 (23.47)	1-28UN-2A	.963 (24.46) .957 (24.31)	1.082 (27.48) 1.069 (26.92)
18	9	.053 (1.35) .048 (1.22)	1.055 (26.80) 1.049 (26.64)	1-1/8-28UN-2A	1.088 (27.64) 1.082 (27.48)	1.264 (30.58) 1.184 (30.07)
20	9	.059 (1.50) .054 (1.37)	1.180 (29.97) 1.174 (29.82)	1-1/4-28UN-2A	1.213 (30.81) 1.207 (30.66)	
22	9	.062 (1.57) .057 (1.45)	1.305 (33.15) 1.299 (32.99)	1-3/8-28UN-2A	1.338 (33.99) 1.332 (33.83)	
24	9	.069 (1.75) .064 (1.63)	1.430 (36.32) 1.424 (36.17)	1-1/2-28UN-2A	1.463 (37.16) 1.457 (37.01)	

SHELL SIZE	N	T DIA.	R	X DIA.
8	.320 (8.13) .300 (7.62)	.439 (11.15) .433 (11.00)	.190 (4.83) .160 (4.06)	.645 (16.38)
10	.320 (8.13) .300 (7.62)	.564 (14.33) .558 (14.17)	.190 (4.83) .160 (4.06)	.770 (19.56)
14	.320 (8.13) .300 (7.62)	.814 (20.68) .808 (20.52)	.190 (4.83) .160 (4.06)	1.020 (25.91)
16	.320 (8.13) .300 (7.62)	.939 (23.85) .933 (23.70)	.190 (4.83) .160 (4.06)	1.145 (29.08)
18	.320 (8.13) .300 (7.62)	1.064 (27.03) 1.058 (26.87)	.190 (4.83) .160 (4.06)	1.279 (32.49)
20	.358 (9.09) .338 (8.59)	1.189 (31.20) 1.183 (30.05)	.220 (5.59) .190 (4.83)	
22	.358 (9.09) .338 (8.59)	1.314 (33.38) 1.308 (33.22)	.220 (5.59) .190 (4.83)	
24	.358 (9.09) .338 (8.59)	1.439 (36.55) 1.433 (36.40)	.220 (5.59) .190 (4.83)	

FIGURE 12B. Series 3 & 4 connector back end configuration - Continued.

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SPLINE DETAIL

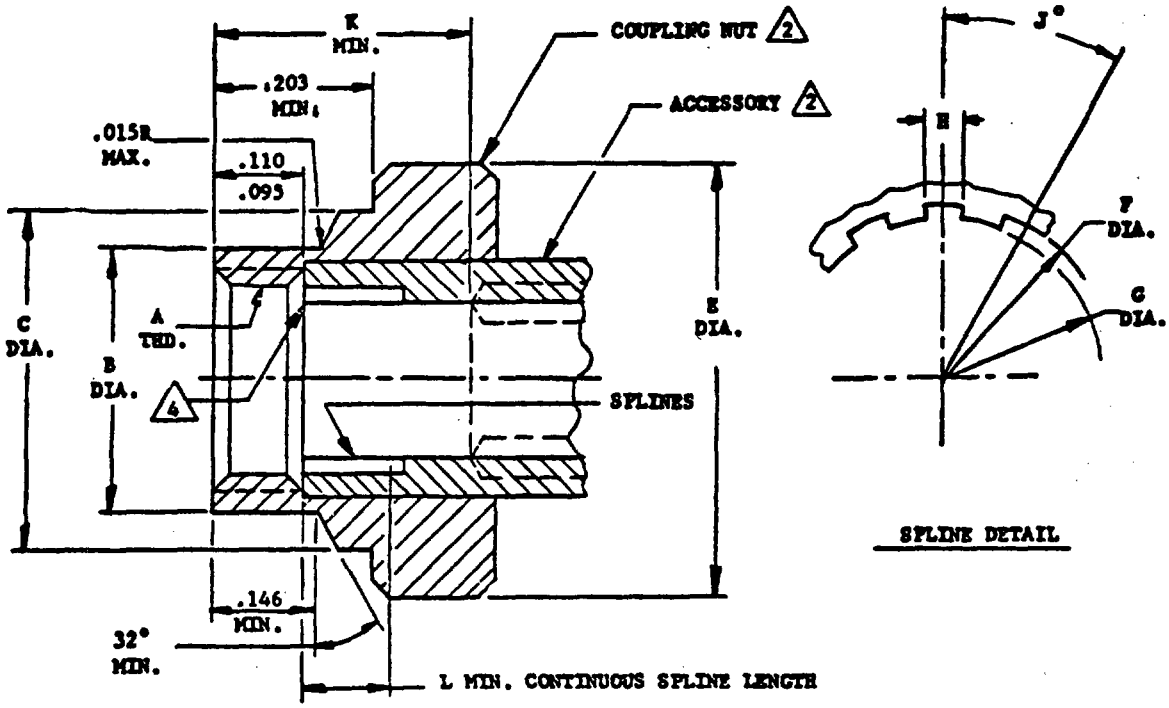
INCHES	M.M.
.015	.38
.020	.51
.025	.64
.030	.76
.085	2.16
.095	2.41
.141	3.58
.203	5.16

NOTES:

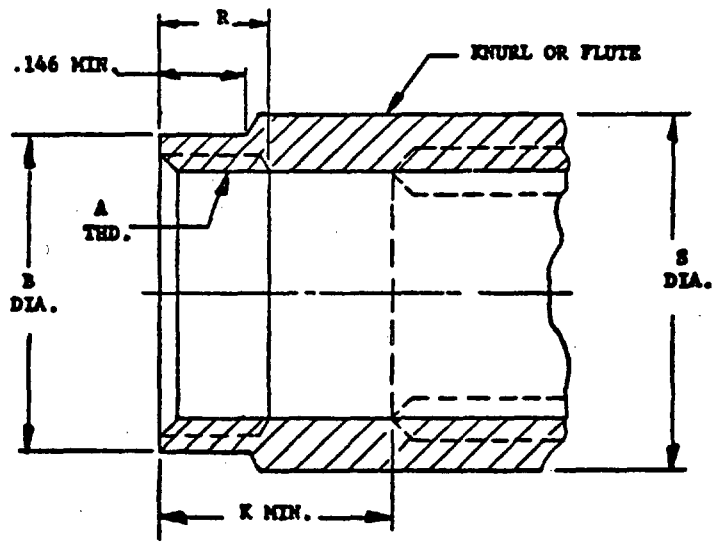
1. DIMENSIONS ARE IN INCHES.
2. DIMENSIONS APPLY AFTER PLATING.
3. METRIC EQUIVALENTS (TO THE NEAREST .01mm) ARE GIVEN FOR GENERAL INFORMATION ONLY.

FIGURE 12B. Series 3 & 4 connector back end configuration - Continued.

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DESIGN STANDARD FOR NON-ROTATABLE ACCESSORY HARDWARE



INCHES	M.M.
.015	.38
.095	2.41
.110	2.79
.146	3.71
.203	5.16

DESIGN STANDARD FOR ROTATABLE ACCESSORY HARDWARE

NOTES:
1. DIMENSIONS IN INCHES.
2. COUPLING NUT SHALL BE CAPTIVATED TO AND FREE TO ROTATE ON ACCESSORY.
3. DIMENSIONS APPLY AFTER PLATING.
4. NON-ROTATABLE HARDWARE SHALL BOTTOM ON LEAD THREAD (F THD.) OF CONNECTORS (FIGURES 12A AND 12B).
5. METRIC EQUIVALENTS (TO THE NEAREST .01mm) ARE GIVEN FOR GENERAL INFORMATION ONLY.

FIGURE 12C. Design standard for rotatable and non-rotatable connector accessories (For use on series 1, 2, 3 & 4).

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SHELL SIZE	A THREAD	MOD. MINOR DIA.	B DIA.	C DIA.	E DIA. MAX.	F DIA.	G DIA.
8	1/2-28UNEF-2B	.470 (11.94) .464 (11.79)	.558 (14.17) .546 (13.87)	.640 (16.26) .635 (16.13)	.850 (21.59)	.473 (12.01) .469 (11.91)	.446 (11.34) .443 (11.25)
10	5/8-28UN-2B	.595 (15.11) .589 (14.96)	.683 (17.35) .671 (17.04)	.765 (19.43) .760 (19.30)	.975 (24.77)	.598 (15.19) .594 (15.09)	.571 (14.50) .568 (14.43)
14	7/8-28UN-2B	.845 (21.46) .839 (21.31)	.933 (23.70) .921 (23.39)	1.015 (25.78) 1.010 (25.65)	1.225 (31.12)	.848 (21.54) .844 (21.44)	.821 (20.85) .818 (20.78)
16	1-28UN-2B	.970 (24.64) .964 (24.49)	1.059 (26.90) 1.047 (26.59)	1.140 (28.96) 1.135 (28.83)	1.350 (34.29)	.973 (24.71) .969 (24.61)	.946 (24.03) .943 (23.95)
18	1-1/8-28UN-2B	1.095 (27.81) 1.089 (27.66)	1.183 (30.05) 1.171 (29.74)	1.274 (32.36) 1.269 (32.23)	1.475 (37.47)	1.098 (27.89) 1.094 (27.79)	1.071 (27.20) 1.068 (27.13)
20	1-1/4-28UN-2B	1.220 (30.99) 1.214 (30.84)	1.308 (33.22) 1.296 (32.92)	1.410 (35.81) 1.405 (35.69)	1.624 (41.25)	1.223 (31.06) 1.219 (30.96)	1.196 (30.38) 1.193 (30.30)
22	1-3/8-28UN-2B	1.345 (34.16) 1.339 (34.01)	1.433 (36.40) 1.421 (36.09)	1.535 (38.99) 1.530 (38.86)	1.749 (44.42)	1.348 (34.24) 1.344 (34.14)	1.321 (33.55) 1.318 (33.48)
24	1-1/2-28UN-2B	1.470 (37.34) 1.464 (37.19)	1.558 (39.57) 1.546 (39.27)	1.649 (41.88) 1.644 (41.76)	1.874 (47.60)	1.473 (37.41) 1.469 (37.31)	1.446 (36.73) 1.443 (36.65)

SHELL SIZE	H	J	K MIN.	L MIN.	R	S DIA.
8	.050 (1.27) .048 (1.22)	20°	.500 (12.7)	.078 (1.98)	.300 (7.62) .190 (4.83)	.640 (16.26) .590 (14.99)
10	.048 (1.22) .046 (1.17)	15°	.500 (12.7)	.078 (1.98)	.300 (7.62) .190 (4.83)	.765 (19.43) .715 (18.16)
14	.053 (1.35) .051 (1.30)	12°	.500 (12.7)	.078 (1.98)	.300 (7.62) .190 (4.83)	1.015 (25.78) .967 (24.56)
16	.052 (1.32 .050 (1.27)	10°	.500 (12.7)	.078 (1.98)	.300 (7.62) .190 (4.83)	1.140 (28.96) 1.090 (27.69)
18	.052 (1.32) .050 (1.27)	9°	.500 (12.7)	.078 (1.98)	.300 (7.62) .190 (4.83)	1.274 (32.36) 1.220 (30.99)
20	.059 (1.50) .056 (1.42)	9°	.576 (14.63)	.109 (2.77)	.328 (8.33) .218 (5.54)	1.410 (35.81) 1.340 (34.04)
22	.062 (1.57) .059 (1.50)	9°	.576 (14.63)	.109 (2.77)	.328 (8.33) .218 (5.54)	1.535 (38.99) 1.465 (37.21)
24	.069 (1.75) .066 (1.68)	9°	.576 (14.63)	.109 (2.77)	.328 (8.33) .218 (5.54)	1.649 (41.88) 1.590 (40.39)

FIGURE 12C. Design standard for rotatable and non-rotatable connector accessories - Continued. (For use on series 1, 2, 3 & 4).

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3.4.2.1 Contact retention and release.

3.4.2.1.1 Removable contacts (series 1 and 2). The inserts shall be designed so that positive locking of the contacts in the inserts is provided. The contact retaining system shall be free of foreign material, adhesive, or any obstruction that would prevent smooth contact insertion and positive retention. Contacts for class A, F, and E removable contact connectors shall be retained by an internal insert construction and a ganged contact mechanism. To lock contacts, the retention system is operated by rotating the rear retention ring clockwise until the yellow color band is no longer exposed and metal to metal contact occurs between the shell and retention ring. Unlocking is accomplished by rotating the rear retention ring counterclockwise from the locked position until the yellow color band is fully exposed. The color shall be within the limits described in MIL-STD-104.

3.4.2.1.1.1 Insertion and removal tools. The insertion and removal tools used to install and remove contacts from the connectors are shown in table VI.

3.4.2.1.2 Removable contacts (series 3 and 4). The inserts shall be designed to provide positive individual contact retention (ICR), and individual contact release from the wire side of the connector, using the applicable tools. The insert design shall permit individual insertion and extraction of contacts without use of a retention ring, or removal of the insert or sealing members. The individual contact retention members shall be surrounded by rigid dielectric material, configured so as to limit their flexure during contact insertion and removal as required to prevent overstressing. Insertion/extraction shall be possible with a full complement of maximum O.D. wire (see table V-B).

3.4.2.1.3 Nonremovable contacts. Class A, F, E, C, T, and P nonremoveable contact connectors shall have the contacts permanently retained in place. Contacts for class B, D, G, H, and L receptacles shall be fused in vitreous material.

3.4.2.2 Grommet construction. The design of the grommet shall accommodate and seal insulated wire conductors having outside diameters within the ranges specified in table V. Connectors with nonremovable contacts shall have a removable grommet if specified (see 3.1).

3.4.2.3 Interfacial seal. Plugs and receptacles with pin inserts shall have a resilient face seal permanently bonded in place to provide an interfacial seal with the hard face of the socket insert in the mated condition. The interfacial sealing member for class B, D, G, H, and L connectors shall be permanently bonded to the insert. The interfacial sealing members of series 3 and 4 connectors shall provide raised seal barriers around each pin contact, as shown in figures 5B, 6B, 7B, 9B, 10B, and 11B. There shall be .005 inch minimum displacement of resilient material when the connectors are fully mated, as tested in 4.6.32.

3.4.2.4 Insert arrangement. The arrangement of contacts in the insert shall be as specified by the applicable MS standard.

3.4.2.5 Contact alignment and stability. With all contacts in place, the alignment of pin and socket contacts shall permit engagement irrespective of buildup of allowable tolerances on hole locations, distortion of contacts due to crimping, and insert location in the shell.

3.4.3 Shell design. The connector shall be of single-piece shell design, constructed to positively retain the insert interface in its specified position in the shell. The shell shall be designed to accept and retain a cable support or other accessory (see figures 4 through 12). The shells of series 3 and 4 connectors shall have a blue color band in accordance with EIA STD. RS359 indicating individual contact release from the rear of the connector.

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3.4.3.1 Grounding spring member. Series 1 and 3 plugs shall have a minimum of six fingers per inch. Series 2 and 4 receptacles shall have the following minimum fingers per shell size:

Shell size	8	6	fingers
"	" 10	7	"
"	" 14	9	"
"	" 16	14	"
"	" 18	17	"

Spring fingers shall be convex shaped in a manner which will ensure proper engagement of the mating shell. The grounding spring shall be permanently fixed to the shell periphery by riveting, welding, brazing, or soldering. Attachment of the grounding spring to the shell shall not impede spring flexing. The spring fingers shall engage the mating shell at a minimum distance of .090 inch prior to pin-socket engagement.

3.4.3.2 Jam nut mounting receptacle shell. The jam nut mounting receptacle shell shall be provided with a mounting nut with provisions for locking (see figures 6 and 10).

3.4.3.3 Screw threads. Screw threads shall be in accordance with MIL-S-7742. Out-of-roundness is not objectionable if the threads can be checked without forcing the thread gages.

3.4.3.4 Lubrication. Bayonet coupling slots, including inside diameter in the bayonet coupling slot area and threads shall be coated with a suitable lubricant to aid in mating connectors. Features which are intended to provide potting compound anchorage shall be free of lubricant.

3.4.4 Connector engagement. The coupling type shall be a 3-pin bayonet (see figures 4 through 11). Keys and keyways shall be designed to prevent engagement of the contacts with the mating contacts or with the mating surface of the counterpart connector until the keyways are properly aligned for engagement (see figure 4 or figure 8). The coupling ring shall be nonremovable and of a fail-safe design. The coupling ring shall be knurled or fluted and designed to assist in engaging and disengaging contacts as the coupling ring is tightened or loosened. Counterpart connectors of any arrangement and size shall be capable of being fully engaged without the use of tools. Full engagement shall be indicated by an audible sound at the completion of the coupling cycle, and a positive detent shall be included in the coupling mechanism to lock the connectors in the engaged position. Connectors shall be provided with spring fingers that are permanently attached to the shell. The spring fingers shall make contact with the shell before pin-socket engagement is made. (See figures 5 and 6 (series 2 and 4) or figure 11 (series 1 and 3).)

3.4.4.1 Shell polarization. The polarization of the mating plugs and receptacles shall be accomplished by means of mating integral keys on the plug shell and keyways on the receptacle shell, as specified (see 3.1). The key and keyway positions shall be in accordance with figure 4 (series 2 and 4) or figure 8 (series 1 and 3).

3.4.4.2 Bayonet pins. Bayonet pins on receptacle shells shall have end surfaces white in color. The color shall be within the limits described in MIL-STD-104. The ends of the pins shall be visible through suitable holes in the coupling ring when the connectors are fully engaged and the coupling ring is in the locked position (see figures 5, 6, 7, 9, 10, and 11).

3.4.4.3 Engagement seal. Connectors shall contain sealing means so that engaged connectors comply with the requirement specified herein. A seal-ring shall be provided to achieve a peripheral seal around the mated shells.

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3.4.4.4 Protective covers and stowage receptacles. When mated to counterpart connectors, the protective covers and stowage receptacles shall maintain the connector free of moisture, prevent air leakage, and comply with the applicable requirements of table XXIV. Protective covers and stowage receptacles shall be provided with an electromagnetic shielding feature if required.

3.4.5 Panel seal. Jam nut mounting receptacles shall be provided with an MS9068 nonconductive "O"-ring panel seal. Flange mounted receptacles shall be provided with a qualified MS90484 flange gasket panel seal only if specified. Jam nut mounting receptacles shall be provided with a qualified MS3322 conductive "O"-ring panel seal only if specified (see 3.1).

3.4.6 Interchangeability. All connectors and accessories having the same military part number shall be completely interchangeable with each other with respect to installation (physical) and performance (function) as specified herein. Accessories shall be interchangeable between series and shall conform to figure 12C.

3.4.7 Intermateability. All series 1 and series 3 connectors having the same shell size, keying, contact arrangement, and contact style shall be intermateable. All series 2 and series 4 connectors having the same shell size, keying, contact arrangement, and contact style shall be intermateable. (See 6.1 and figures 4 through 11.)

3.5 Performance. Connectors and accessories shall be designed to meet the performance requirements stated herein when tested in accordance with the specified method of section 4.

3.5.1 Contact protection.

3.5.1.1 Protection, receptacles or plugs designed for pin contacts (series 1 and 3). Connectors designed to provide protection in receptacles or plugs equipped with pin contacts shall not permit the edge of the shell of the mating plug or receptacle to touch the pin contacts. (See figures 9, 10, and 11.) The procedure described in 4.6.2 shall be used.

3.5.1.2 Protection, receptacles designed for pin contacts (series 2 and 4). Connectors designed to provide protection only in receptacles equipped with pin contacts shall not permit the edge of the shell of the mating plug to touch the pin contacts. (See figures 5, 6, and 7.) The procedure described in 4.6.2 shall be used.

3.5.2 Resistance to probe damage (size 12 and smaller socket contacts). When tested as specified in 4.6.3, socket contacts shall withstand the bending moment and depth of test probe insertion without evidence of visible damage, and shall meet the requirements of subsequent tests.

3.5.3 Contact insertion and removal forces (crimp removable contact connectors only). When tested as specified in 4.6.4, contacts shall meet the insertion and removal forces specified in table VII, during maintenance aging.

TABLE VII-A. Contact insertion and removal forces with locking mechanism disengaged. (Series 1 and 2 only.)

Contact size	Insertion force pounds (max)	Removal force, pounds						
		During 1st cycle		During 5th cycle		During 10 cycle		Maximum
		Minimum average	Minimum	Minimum average	Minimum	Minimum average	Minimum	
23-28	12	7	6	6	5	5	4	15
23-22	12	7	6	6	5	5	4	15
20-20	20	10	8	8	7	7	6	15
16-16	25	11	10	8	7	7	6	15
12-12	30	11	10	8	7	7	6	15

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TABLE VII-B. Contact insertion and removal forces (Series 3 and 4 only).

Contact size	Insertion or removal force, pounds (max.) (1st or 10th cycles)
23-28	3
23-22	3
20-20	5
16-16	7
12-12	10

3.5.4 Contact engaging and separating force. When tested as specified in 4.6.5, the contact engaging and separating force shall not exceed the applicable values specified in table VIII for non-removable contacts, and the values specified on the applicable MIL-C-39029 specification sheet for removable crimp contacts.

TABLE VIII. Contact engagement and separation forces (non-removable contacts only).

Mating end size	Minimum Separation force (ounces) Minimum diameter MS3197 pin	Maximum average engagement force (ounces) Maximum diameter MS3197 pin	Maximum engagement force (ounces) Maximum diameter MS3197 pin
23	0.5	6.8	8
20	0.7	12	18
16	2	24	30
12	3	24	30

3.5.5 Operating force. When tested as specified in 4.6.6, the coupling torque for mating and unmating of counterpart connectors shall meet the torque requirements of table IX. The connectors used in this test shall have the complete complement of contacts. The dummy storage receptacles and protective covers, when mated with their applicable connectors, shall also comply with the torque requirements of table IX.

TABLE IX. Coupling - torque.

Shell size	Torque (pounds-inches)	
	Maximum engagement and disengagement	Minimum disengagement
8	8	1
10	12	1
14	20	4
16	24	4
18	28	4
20	30	6
22	32	7
24	34	7

3.5.6 Contact retention. Contacts shall be retained in their inserts when the axial loads of table X are applied. The axial displacement of the contact shall not exceed 0.012 inch while under the specified axial load on the mating end of the contact (see 4.6.7).

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TABLE X. Axial loads for contact retention.

Mating end size	Axial load (pounds-Minimum)	
	Series 1 and 2	Series 3 and 4
23	10	12
20	15	15
16	25	25
12	25	25

3.5.7 Thermal shock. When tested as specified in 4.6.8, there shall be no damage detrimental to the operation of the connector. The temperature extremes of table XI shall be used.

TABLE XI. Temperature extremes.

Low temperature	High temperature		
All classes	Class E, H, and P	Class D, F, G, L and T	Class A, B and C
$-65\ ^{+0°}_{-3°}$ C	$+150\ ^{+3°}_{-0°}$ C	$+175\ ^{+3°}_{-0°}$ C	$+200\ ^{+3°}_{-0°}$ C

3.5.8 Dielectric withstanding voltage. When tested as specified in 4.6.9, connectors shall show no evidence of breakdown or flashover when subjected to the test voltages and altitude of table XII; corona shall not be considered as breakdown. Maximum recommended working voltages are as shown in table XXX.

TABLE XII. Test voltage (ac, rms, not working voltage).

Simulated altitude condition	Service rating I		Service rating II	
	Mated	Unmated	Mated	Unmated
Sea level	1800	1500	2300	2300
50,000 feet	1000	700	1500	800
70,000 feet	1000	375	1500	500
110,000 feet	1000	200	1500	200

3.5.9 EMI/RFI shielding effectiveness.

3.5.9.1 Shielding effectiveness. When tested as specified in 4.6.10.1, the shells shall exhibit a radio frequency leakage attenuation equal to or greater than that specified in table XIII.

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TABLE XIII. EMI/RFI shielding effectiveness.

Frequency MHz	Leakage attenuation (dB)	Frequency MHz	Leakage attenuation (dB)
100	65	400	55
150	60	600	50
200	60	800	45
300	55	1,000	45

3.5.9.2 Shell to shell continuity. When tested as specified in 4.6.10.2, the dc resistance shall not exceed 0.005 ohms.

3.5.9.3 Shell spring finger forces. When tested as specified in 4.6.10.3, the axial force necessary to overcome the shell spring finger interference shall be no less than .25 pounds nor more than 15 pounds.

3.5.10 Fluid immersion. Connectors shall be immersed in the fluids specified in table XIV and shall then meet the requirements of operating force (3.5.5), dielectric withstanding voltage (3.5.8), and insulation resistance (3.5.15). The test method shall be as specified in 4.6.11.

TABLE XIV. Fluids for fluid immersion.

Class	Fluids
E, H, and P	MIL-H-5606 MIL-L-23699
A, B, C, D, F, G, L, T, and W	MIL-L-7808 MIL-L-23699 MIL-H-5606 Hydraulic fluid MIL-A-8243 MIL-C-25769 MIL-T-5624 (JP-5) Coolant dielectric fluid, synthetic silicate ester base MIL-G-3056 (type 1) Solvent (a), Solvent (b), Solvent (c) } MIL-STD-202, method 215

3.5.10.1 Retention system fluid exposure (class A, C, E, F, L, P, and T). When tested as specified in 4.6.11.1, the connectors shall meet the requirements of subsequent tests as specified herein. (Effects of fluids on resilient sealing members shall not be a consideration of this test).

3.5.11 Durability. When tested as specified in 4.6.12, counterpart connectors shall show no damage detrimental to the operation of the connector.

3.5.12 Vibration.

3.5.12.1 Vibration (conditioning). Mated connectors shall not be damaged, there shall be no loosening of parts due to vibration, and counterpart connectors shall be retained in full engagement (see 4.6.13).

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3.5.12.2 Vibration (resistance change). Mated connectors shall not be damaged and there shall be no loosening of parts due to vibration. Counterpart connectors shall be retained in full engagement and the increase in contact resistance due to vibration shall not exceed 5 ohms in 1 microsecond (see 4.6.13).

3.5.13 Shock.

3.5.13.1 Shock (conditioning). Mated connectors shall not be damaged and there shall be no loosening of parts during the exposure to mechanical shock (see 4.6.14.1).

3.5.13.2 Shock (resistance change). Mated connectors shall not be damaged and there shall be no loosening of parts, nor shall the increase in contact resistance exceed 5 ohms in 1 microsecond, during the exposure to mechanical shock (see 4.6.14.2).

3.5.14 Moisture resistance. When tested as specified in 4.6.15, mated connectors shall show no deterioration and the insulation resistance shall be 500 megohms or greater at 25°C.

3.5.15 Insulation resistance.

3.5.15.1 Insulation resistance at ambient temperature. When tested as specified in 4.6.16.1, the insulation resistance at ambient temperature (25°C) shall be greater than 5,000 megohms.

3.5.15.2 Insulation resistance at elevated temperature. After exposure for 1000 hours to the applicable high temperature specified in table I, the insulation resistance shall be as specified in table I for each class (see 4.6.16.2).

3.5.16 External bending moment. When tested as specified in 4.6.17 using the applicable bending moment shown in table XV, connectors shall show no evidence of damage nor shall there be any interruption of electrical continuity.

TABLE XV. External bending moment.

Plug shell size	Bending moment (inch-pounds)
8	75
10	230
14	350
16	419
18	433
20	460
22	500
24	525

3.5.17 Salt spray (corrosion). When tested as specified in 4.6.18, unmated connectors, stowage receptacles, protective covers, mating shells and accessories shall show no exposure of basic metal due to corrosion which will affect performance in subsequent tests.

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3.5.18 Temperature life. All connectors shall perform satisfactorily after exposure for 1000 hours to the applicable high temperature specified in table XI when tested per 4.6.19.

3.5.18.1 Temperature life with contact retention. During exposure for 1000 hours to the applicable high temperature specified in table XI, the contacts of removable contact connectors shall support an axial load equal to 50% of the values specified in table X. The contacts shall maintain their specified locations (see figures 4 through 11). There shall be no electrical discontinuity in excess of one microsecond. The test shall be performed per 4.6.19.1.

3.5.19 Ozone exposure. When tested as specified in 4.6.20, the connectors shall show no evidence of cracking of materials or other damage due to ozone exposure that will adversely affect subsequent performance in the qualification test sequence.

3.5.20 Magnetic permeability. When tested as specified in 4.6.21, the relative permeability of connectors and accessories except class B, D, G, L, and H, shall be less than 2.

3.5.21 Insert retention.

3.5.21.1 Insert retention (classes A, C, E, F, P, and T). When tested as specified in 4.6.22.1, inserts shall not be dislocated from their original positions when an axial load of 75 pounds psig is applied.

3.5.21.2 Insert retention (classes B, D, G, L, and H). When tested as specified in 4.6.22.2, inserts shall not be dislocated from their original positions or damaged when an effective pressure differential of 2000 pounds psig is applied.

3.5.22 Air leakage.

3.5.22.1 Connectors, with solder contacts and environmental sealing adapters. When tested as specified in 4.6.23.1, the air leakage rate shall not be greater than 1 cubic inch-per-hour (4.55×10^{-3} cc/sec).

3.5.22.2 Class B, C, G, L, and H connectors. When tested as specified in 4.6.23.2, the air leakage rate shall not be greater than 10^{-8} ft^3/hour ($7.86 \times 10^{-8} cm^3$/sec). The specified leakage rate shall apply only through the connector and not through the flange and mounting surface area, unless solder mounted.

3.5.22.3 Stowage receptacle and protective covers. When tested as specified in 4.6.23.3, the air leakage rate shall not be greater than 1 cubic inch-per-hour ($4.55 \times 10^{-3} cm^3$/sec).

3.5.23 Water immersion (applicable to nonremovable contact connectors; class A, F, and E with J adapter). When tested as specified in 4.6.24, receptacle inserts and panel seals shall show no leakage and there shall be no evidence of moisture at the interface and cable housing. Insulation resistance of mated connectors at the end of 48 hours while still immersed and the insulation resistance of unmated connectors after the drying period shall meet a minimum requirement of 100 megohms.

3.5.24 Altitude immersion. Connectors shall have an insulation resistance of at least 5,000 megohms after being subjected to altitude immersion as specified in 4.6.25, and shall maintain a dielectric withstanding voltage at sea level as specified in table XII.

3.5.25 Contact resistance. When tested as specified in 4.6.26, nonremovable contacts in the mated condition shall meet the contact resistance requirements of table XVI. Removable crimp contacts shall meet the requirements of the applicable MS standard or military specification sheet. Both solder and crimp-type contacts for all connector classes shall meet the requirements for low-level contact resistance as described in MIL-C-39029.

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TABLE XVI. Contact resistance limits (non-removable contacts).

Connector class	Mating end size	Wire size	Test current (amps)	Millivolt drop (Max)			
				Initial		After corrosion	
A,F,and E	23	28	1.5	60		70	
		32	1.0	55		65	
	23	22	5.0	70		80	
		26	2.0	60		70	
	20	20	7.5	55		65	
		22	5.0	45		55	
		24	3.0	45		55	
	16	16	13.0	50		60	
		18	10.0	45		55	
		20	7.5	45		55	
	12	12	23.0	50		60	
		14	17.0	45		55	
				Avg	Indiv	Avg	Indiv
B,D,G,L, and H	23	22	4.0	90	115	105	180
	20	20	5.0	70	105	90	165
	16	16	10.0	65	95	80	165
	12	12	17.0	65	95	80	165

3.5.26 Cover chains tensile strength. When tested in accordance with 4.6.27, protective covers with chains or wire rope as specified shall withstand a 50-pound tensile test without damage.

3.5.27 Insertion/removal tool abuse. (Series 3 and 4 only) When tested as specified in 4.6.28, there shall be no damage to the contacts, the connector insert, or the contact retaining mechanism. The connectors shall meet the requirements of subsequent testing.

3.5.28 Gage location and retention. When tested as specified in 4.6.29, the connectors shall meet the contact retention requirements of 3.5.6. Contact location shall be within the limits specified in figures 4 thru 11.

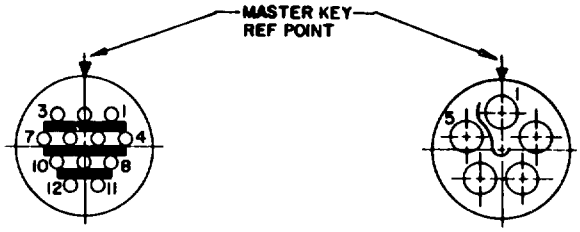
3.5.29 Pin contact stability. (Series 3 and 4 only) When tested as specified in 4.6.30 the total displacement of the tip of the pin contact probe shall not exceed the applicable value specified in table XXIX with the specified load applied.

3.5.30 Contact walk-out. When tested as specified in 4.6.31, contacts shall not become dislodged from their normal position.

3.6 Marking. Each connector and accessory shall be legibly and permanently marked on the shell or coupling ring in accordance with MIL-STD-130 and MIL-STD-1285 (see 3.1). The part number shall be as shown in 1.2, and shall be separate from other required markings. Series 3 and 4 connectors shall be marked with a blue color band denoting individual contact insertion and removal from the wire side of the connector (see 3.4.3).

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3.6.1 Insert marking. Inserts shall be marked as illustrated on figure 13. Raised or depressed characters shall not be used on mating faces. For series 3 and 4 connectors, a bracket [( )] shall be added to the rear grommet marking, identifying each 10th cavity. The pin front of series 3 and 4 connectors may be marked with an interrupted line, applied to the raised seal barriers.



Markings are shown for the pin insert front and socket grommet. The socket face and pin grommet are the opposite.
All characters on the insert grommets shall be white in color.

FIGURE 13 Insert arrangement marking (typical).

3.6.1.1 Contact designations. Contact locations shall be designated by identifiable characters of contrasting color on the front and rear faces of the insert-grommet assembly. Positioning and arrangement of the characters shall be such that the appropriate contact cavities are identifiable and shall remain identifiable after completion of the tests specified in tables XVIII through XXIV.

3.7 Workmanship. Connectors and accessories shall meet all design dimensions and intermateability requirements of this specification. Loose contacts, poor molding fabrication, loose materials, defective bonding, damaged or improperly assembled contacts, peeling or chipping of plating or finish, galling of mating parts, nicks and burrs of metal parts, and post molding warpage will be considered adequate basis for rejection of items of quality inferior for the purpose intended. Emphasis shall be on the quality of the molded dielectric retention system parts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-C-45662.

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4.2. Classification of inspection. The inspections specified herein are classified as follows:

a. Qualification inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions and preparation of sample.

4.3.1 Inspection conditions. Unless otherwise specified, all inspections shall be made in accordance with the following conditions:

a. Temperature - 15° to 35°C.
b. Relative humidity - 45 to 75 percent.
c. Barometric pressure - 650 to 800 mm of mercury.

4.3.2 Preparation of sample. Connectors shall be wired with approximately 3 feet of wire except for those connectors being monitored for resistance change under shock and vibration, which shall have six inch lead lengths. Termination of wires to contacts shall be accomplished as follows: A crimping tool in accordance with specification MIL-C-22520 as indicated on the applicable crimp contact specification sheet, shall be used for removable contacts. Soldering shall be in accordance with requirement 5 of MIL-STD-454 for nonremovable contacts.

4.4 Qualification inspection. Qualification inspection shall consist of the examinations and tests performed in the sequence specified in either tables XVIII, XIX, XX, XXI, XXII, XXIII, or XXIV, on the qualification test samples specified in 4.4.1. After receipt of the letter of authorization from the activity responsible for qualification (see 6.3), the applicant shall submit three copies of his test reports (certified by the Government inspector indicating the extent to which the tests were witnessed), together with required samples specified in 4.4.1, to the activity responsible for qualification.

4.4.1 Samples. All samples shall be taken from a production run and shall be produced with equipment and procedures normally used in production (see 6.3.2). One-half of all classes, accessories or shells, selected shall be submitted to the applicable qualification inspections specified in paragraphs 4.4.1.2 thru 4.4.1.6. The remaining untested half shall be forwarded to the qualifying activity with the qualification test reports. Where counterpart connectors are required for tests requiring mating assemblies, the counterpart connectors provided for this purpose shall be new, previously qualified connectors or new connectors submitted for qualification testing. Supplies not producing mating connectors shall submit substantiating certification data that tests were performed with approved counterpart connectors.

4.4.1.1 Connectors (series 1 or 2, class E, H, and P). A minimum of 14 connectors, class E, containing a minimum of contact cavities per contact size in accordance with table XVII shall be selected. The connectors for test samples 1, 4, and 5 shall be assembled with wire approaching the applicable minimum wire diameter shown on table V. Connectors for test samples 2, 3, 6, and 7 shall be assembled with wire approaching the applicable maximum wire diameter shown in table V. Class H connectors shall be assembled with table V wire having nominal diameters as applicable. Periodic testing of class E connectors shall be deemed sufficient to demonstrate control of class P. Samples 1, 2, and 4 shall be assembled with pin contacts in the receptacles. A minimum of 2 connectors, class H, containing a minimum of contacts per contact size in accordance with table XVII shall be selected (as sample group "H"). Additional samples may be required to provide the contacts as specified in table XVII. The sample subjected to qualification testing shall be provided with a counterpart connector for those tests requiring mating assemblies. One-half of all classes selected shall be submitted to the applicable qualification inspections of tables XVII and XIX.

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4.4.1.2 Connectors (series 1, 2, 3, and 4, class A, B, C, D, E, F, G, L, T or W). A minimum of 14 connectors in the series for which qualification is desired as described (see 4.4.6) shall be selected. These connectors shall be class A or F, containing a minimum of contact cavities per size in accordance with table XVII. The connectors for test samples 1, 4, 5, and 7 shall be assembled with wire approaching the applicable minimum wire diameter shown in table V. Class C or T connectors shall be assembled with table V wire having nominal diameters as applicable. Periodic testing of class A connectors shall be deemed sufficient to demonstrate control of class C. Periodic testing of class F connectors shall be deemed sufficient to demonstrate control of class T. Samples 1, 2, and 4 shall be assembled with pin contacts in the receptacles. A minimum of 2 connectors, class B, D, G, L or W, containing a minimum of contacts per contact size in accordance with table XVII shall be selected (as sample group "H"). Additional samples may be required to provide the contacts as specified in table XVII. The sample subjected to qualification testing shall be provided with a counterpart connector for those tests requiring mating assemblies. One-half of all classes selected shall be submitted to the applicable qualification inspections of tables XX and XXI.

4.4.1.3 Connectors for tests of dielectric retention systems (removable contact types, series 1, 2, 3, and 4, class A, C, E, F, L, P, T and W). A minimum of 18 connectors in the series for which qualification is desired (see 4.4.6) shall be selected. These connectors shall be class A, E, F, L or W containing a minimum of contact cavities per size in accordance with table XVIIA. Each contact cavity size to be qualified shall be represented in each test group (see table XXII). Class L samples shall consist of receptacles only. Contacts shall be wired with nominal diameter wire in accordance with table V unless otherwise specified in the test paragraph. Periodic testing of a given class shall be deemed sufficient to demonstrate control of all other classes which use identical retention systems. One-half of all classes selected shall be submitted to the applicable qualification inspections of table XXII.

TABLE XVII. Samples.

Contact size	Class A, F, and E contact cavities (minimum)	Class B, D, G, H, and L contacts (minimum)
23-22	220	55
20-20	128	32
16-16	72	18
12-12	40	10

4.4.1.4 Connectors for intermateability tests (series 3 and 4, class A or F). A minimum of 8 connector halves in the series for which qualification is desired (see 4.4.6) shall be selected. These connectors shall be class A or F. Sample 1 shall be a plug with sockets. Sample 2 shall be a plug with pins. Sample 3 shall be a receptacle with sockets, and sample 4 shall be a receptacle with pins. An equal number of series 1 or 2 mating counterpart connectors shall be provided. Counterpart connectors shall be class A, E, or F, but need not be of the same class as the series 3 or 4 connectors. All samples shall be assembled with wire approaching the applicable maximum wire diameter shown in table V. One-half of all classes selected shall be submitted to the qualification inspections of table XXIII.

4.4.1.5 Accessories. Four J adapters shall be selected for qualification. They shall be tested with a qualified connector or an accessory device having identical mating features. One-half of the adapters selected shall be submitted to the qualification inspection specified in table XXIV. The remaining untested half shall be forwarded to the qualifying activity with the qualification test reports. The samples shall be produced with equipment and procedures normally used in production.

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4.4.1.6 Shells for EMI testing. Four pairs of mating shells shall be selected for qualification and shall be tested without inserts or contacts installed. Two pairs of mating shells shall be submitted to the applicable qualification sequence specified in table XVIII, XIX, XX, or XXI. The remaining two pairs shall be forwarded to the qualifying activity with the qualification test reports. These samples shall be produced with equipment and procedures normally used in production.

4.4.1.7 Connectors for fluid immersion testing (series 1, 2, 3, and 4, class A, B, C, D, F, G, L, T or W). A minimum of 24 connectors in the series for which qualification is desired (see 4.4.6) shall be selected. These connectors shall be class A, B, C, D, F, G, L, T or W containing a minimum of contact cavities per size in accordance with table XVII. The connectors for test samples 5 and 6 of table XXIIB shall be assembled with wire approaching the applicable minimum wire diameter shown in table V. Class C or T connectors shall be assembled with table V wire having nominal diameters as applicable. Periodic testing of class A connectors shall be deemed sufficient to demonstrate control of class C. Periodic testing of class F connectors shall be deemed sufficient to demonstrate control of class T. Additional samples may be required to provide the contacts as specified in table XVII. The samples subjected to qualification testing shall be provided with counterpart connectors. One-half of all classes selected shall be submitted to the applicable qualification inspections of table XXIIB.

4.4.2 Test routine. Sample units shall be subjected to the applicable qualification inspection specified in tables XVIII, XIX, XX, XXI, XXII, or XXIII, in the order shown. Applicable connector accessories shall be subjected to the inspection test sequence of table XXIV, in the order shown.

4.4.3 Qualification rejection. There shall be no failures during any examination or tests of the connectors or accessories detrimental to the operation of the connector submitted for qualification tests. After notification of any failure, detrimental to the operation of the connector, the activity responsible for qualification testing shall receive details of corrective action from the supplier before initiating any further tests deemed necessary to assure compliance with specification requirements.

4.4.4 Retention of qualification. In order to retain qualification, a summary of group B test results shall be furnished at one or two year intervals (see 4.5.1.4.1) and shall cover the results of group B tests performed during that period. The summary shall also include the number and type of any part failures and shall be forwarded via the Government inspector to the activity responsible for qualification. If the summary of results indicates nonconformance with specification requirements, 6.4 (noncompliance) shall apply. Failure to furnish the summary shall result in loss of qualification for that period.

4.4.5 Assembly plants. Assembly plants must be listed on or approved for listing on the applicable qualified products list. The qualified connector supplier shall certify that the assembly plant is approved for the distribution of the supplier's parts. The assembly plant shall use only piece parts supplied by the qualified connector supplier. No testing other than visual examination is required of certified piece parts obtained from the qualified connector supplier, except when there is cause for rejection. All assemblies produced at the assembly plant shall be subjected to the quality assurance provisions specified herein. Quality control requirements, including Government inspection surveillance, shall be the same as required for the qualified connector supplier.

4.4.6 Qualification of additional connectors. For connectors of related series and similar construction which differ only in shell size and configuration from those which have been qualified, the supplier's test report need only provide test data necessary to validate the difference. The connectors shall be subjected to the tests specified in table XXV, and the results forwarded to the qualifying activity (see 6.3.1). However, qualification of series 1 or 2 connectors shall not admit qualification of series 3 or 4 connectors under the provisions of this paragraph, nor shall qualification of series 3 or 4 connectors admit qualification of series 1 or 2 connectors. Each class for which qualification is requested shall be represented in each applicable test group, unless otherwise authorized by the qualifying activity.

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TABLE XVIII. Qualification and group B inspection for removable contact type connectors (series 1 and 2, class E and P).

Title	Requirement paragraph	Test paragraph	Test sample group							Mating shell 1 and 2
			1	2	3	4	5	6	7	
Examination of product	3.1, 3.3 thru 3.4.6, 3.6 and 3.7	4.6.1	X	X	X	X	X	X	X	X
Interfacial seal compression	3.4.2.3	4.6.32	X	X	X	X	X	X	X	
Contact insertion and removal force	3.5.3									
Series 1 and 2		4.6.4.1	X	X	X	X	X	X	X	
Series 3 and 4		4.6.4.2	X	X	X	X	X	X	X	
Shell to shell continuity	3.5.9.2	4.6.10.2	X		X			X		
Shell spring finger forces	3.5.9.3	4.6.10.3								X 1/
EMI/RFI shielding effectiveness	3.5.9.1	4.6.10.1								X
Contact protection, connector mating										
Series 1 and 3	3.5.1.1	4.6.2	X	X		X				
Series 2 and 4	3.5.1.2	4.6.2			X		X	X		
Probe damage	3.5.2	4.6.3	X			X	X			
Operating force	3.5.5	4.6.6			X	X				
Contact engaging and separating force	3.5.4	4.6.5	X	X	X	X	X	X	X	
Contact retention	3.5.6									
Series 1 and 2		4.6.7.2	X	X	X	X	X	X	X	
Series 3 and 4		4.6.7.3	X	X	X	X	X	X	X	
Thermal shock	3.5.7	4.6.8	X	X	X	X	X	X	X	
Dielectric withstanding voltage (unmated at sea level)	3.5.8	4.6.9.1	X	X	X	X	X	X	X	
Altitude immersion	3.5.24	4.6.25	X			X	X			
Fluid immersion	3.5.10	4.6.11				X	X			
Durability	3.5.11	4.6.12			X				X	X
Vibration (conditioning)	3.5.12.1	4.6.13.1	X				X			
Vibration (resistance change)	3.5.12.2	4.6.13.2		X						
Shock (conditioning)	3.5.13.1	4.6.14.1	X							
Shock (resistance change)	3.5.13.2	4.6.14.2		X						
Moisture resistance	3.5.14	4.6.15	X							
Insulation resistance (ambient temp.)	3.5.15.1	4.6.16.1	X						X	
External bending moment	3.5.16	4.6.17							X	
Salt spray (corrosion)	3.5.17	4.6.18	X							X
Temperature life with cont. reten.	3.5.18.1	4.6.19.1		X				X		
Insulation resistance (elevated temperature)	3.5.15.2	4.6.16.2		X				X		
Ozone exposure (unmated)	3.5.19	4.6.20			X					
Magnetic permeability	3.5.20	4.6.21			X					
Insulation resistance (ambient temp.)	3.5.14.1	4.6.16.1			X				X	
Insert retention (class A, C, E, and P)	3.5.21.1	4.6.22.1	X							
Operating force	3.5.5	4.6.6	X	X	X	X	X	X		
Dielectric withstanding voltage (unmated at sea level)	3.5.8	4.6.9.1	X	X	X			X	X	
Contact retention	3.5.6									
Series 1 and 2		4.6.7.2	X	X	X			X		
Series 3 and 4		4.6.7.3	X	X	X			X		
Insulation resistance (ambient temp.)	3.5.15.1	4.6.16.1		X		X	X	X		
Dielectric withstanding voltage (mated at altitude)	3.5.8	4.6.9.2		X	X	X		X		
Dielectric withstanding voltage (unmated at altitude)	3.5.8	4.6.9.2	X	X				X		
Contact insertion and removal force	3.5.3									
Series 1 and 2		4.6.4.1		X	X	X		X		
Series 3 and 4		4.6.4.2		X	X	X		X		
Contact resistance	3.5.25	4.6.26	X	X						
Shell to shell continuity	3.5.9.2	4.6.10.2	X		X			X		
EMI/RFI shielding effectiveness	3.5.9.1	4.6.10.1								X
Examination of product	3.1, 3.3 thru 3.4.6, 3.6, and 3.7	4.6.1	X	X	X	X	X	X	X	

1/ See 4.6.10.3

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TABLE XIX. Qualification and group B inspection for nonremovable contact type connectors (series 1 and 2, class E, H and P).

Title	Requirement paragraph	Test paragraph	Test sample group								Mating shell 1 and 2
			1	2	3	4	5	6	7	H	
Examination of product	3.1, 3.3 thru 3.4.6, 3.6, 3.7	4.6.1	X	X	X	X	X	X	X	X	X
Interfacial seal compression	3.4.2.3	4.6.32	X	X	X	X	X	X	X		
Shell to shell continuity	3.5.9.2	4.6.10.2	X		X			X			
Shell spring finger forces	3.5.9.3	4.6.10.3									X 1/
EMI/RFI shielding effectiveness	3.5.9.1	4.6.10.1									X
Contact protection, connector mating											
Series 1	3.5.1.1	4.6.2	X	X		X					
Series 2	3.5.1.2	4.6.2			X		X	X			
Probe damage	3.5.2	4.6.3	X			X	X				
Operating force	3.5.5	4.6.6			X	X					
Contact retention	3.5.6	4.6.7	X	X	X	X	X	X			
Contact engaging and separating force	3.5.4	4.6.5	X	X	X	X	X	X	X	X	
Thermal shock	3.5.7	4.6.8	X	X	X	X	X	X		X	
Contact retention	3.5.6	4.6.7	X	X	X	X	X	X			
Dielectric withstanding voltage (unmated at sea level)	3.5.8	4.6.9.1	X	X	X	X	X	X		X	
Altitude immersion	3.5.24	4.6.25	X			X	X				
Fluid immersion	3.5.10	4.6.11				X	X			X	
Durability	3.5.11	4.6.12			X				X	X	X
Vibration (conditioning)	3.5.12.1	4.6.13.1	X				X			X	
Vibration (resistance change)	3.5.12.2	4.6.11.2		X	X					X	
Shock (conditioning)	3.5.13.1	4.6.14.1	X								
Shock (resistance change)	3.5.13.2	4.6.14.2		X							
Moisture resistance	3.5.14	4.6.15	X							X	
Insulation resistance at ambient temperature	3.5.15.1	4.6.16.1	X	X			X		X	X	
External bending moment	3.5.16	4.6.17							X		
Salt spray (corrosion)	3.5.17	4.6.18	X							X	
Temperature life	3.5.18	4.6.19		X				X		X	
Insulation resistance (elevated temperature)	3.5.15.2	4.6.16.2		X				X		X	
Ozone exposure (unmated)	3.5.19	4.6.20			X						
Magnetic permeability	3.5.20	4.6.21			X						
Insulation resistance (ambient temp.)	3.5.15.1	4.6.16.1			X				X		
Insert retention (class A, C, E, and P)	3.5.21.1	4.6.22.1	X								
Insert retention (class B and H)	3.5.21.2	4.6.22.2								X	
Operating force	3.5.5	4.6.6	X	X	X	X	X	X		X	
Dielectric withstanding voltage (unmated at sea level)	3.5.8	4.6.9.1	X	X	X			X	X	X	
Air leakage	3.5.22.1	4.6.23.1		X	X		X				
Air leakage (class B and H)	3.5.22.2	4.6.23.2								X	
Water immersion	3.5.23	4.6.24		X	X		X		X	X	
Contact retention	3.5.6	4.6.7	X	X	X			X			
Insulation resistance (ambient temp.)	3.5.15.1	4.6.16.1		X		X	X	X		X	
Dielectric withstanding voltage (mated at altitude)	3.5.8	4.6.9.2		X	X	X		X		X	
Dielectric withstanding voltage (unmated at altitude)	3.5.8	4.6.9.2	X	X				X		X	
Contact resistance	3.5.25	4.6.26		X							
Shell to shell continuity	3.5.9.2	4.6.10.2	X		X			X			
EMI/RFI shielding effectiveness	3.5.9.1	4.6.10.1									X
Examination of product	3.1, 3.3 thru 3.4.6, 3.6, 3.7	4.6.1	X	X	X	X	X	X	X	X	

1/ See 4.6.10.3.

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TABLE XX. Qualification and group B inspection for removable contact type connectors (series 1, 2, 3, and 4, class A, C, F, T and W.

Title	Requirement paragraph	Test paragraph	Test sample group							Mating shell
			1	2	3	4	5	6	7	1 and 2
Examination of product	3.1, 3.3 thru 3.4.6, 3.6, 3.7	4.6.1	X	X	X	X	X	X	X	X
Interfacial seal compression	3.4.2.3	4.6.32	X	X	X	X	X	X	X	
Contact insertion and removal force										
Series 1 and 2, removable contacts		4.6.4.1	X	X	X	X	X	X	X	
Series 3 and 4, removable contacts		4.6.4.2	X	X	X	X	X	X	X	
Shell to shell continuity	3.5.9.2	4.6.10.2	X		X			X		
Shell spring finger forces	3.5.9.3	4.6.10.3								X 1/
EMI/RFI shielding effectiveness	3.5.9.1	4.6.10.1								X
Contact protection, connector mating										
Series 1 and 3	3.5.1.1	4.6.2	X	X		X				
Series 2 and 4	3.5.1.2	4.6.2			X		X		X	
Probe damage	3.5.2	4.6.3	X			X	X			
Operating force	3.5.5	4.6.6			X	X				
Contact engaging and separating force	3.5.4	4.6.5	X	X	X	X	X	X		
Contact retention										
Series 1 and 2, removable contacts		4.6.7.2	X	X	X	X	X	X		
Series 3 and 4, removable contacts		4.6.7.3	X	X	X	X	X	X		
Thermal shock	3.5.7	4.6.8	X	X	X	X	X	X		
Dielectric withstanding voltage (unmated at sea level)	3.5.8	4.6.9.1	X	X	X	X	X	X		
Altitude immersion	3.5.24	4.6.25	X			X	X			
Durability	3.5.11	4.6.12			X				X	X
Vibration (conditioning)	3.5.12.1	4.6.13.1	X				X			
Vibration (resistance change)	3.5.12.2	4.6.13.2		X						
Chock (conditioning)	3.5.13.1	4.6.14.1	X							
Shock (resistance change)	3.5.13.2	4.6.14.2		X						
Moisture resistance	3.5.14	4.6.15	X							
Insulation resistance (ambient temp.)	3.5.15.1	4.6.16.1	X						X	
External bending moment	3.5.16	4.6.17							X	
Durability (first 50 cycles, class W only)	3.5.11	4.6.12	X							
Salt spray (corrosion)	3.5.17	4.6.18	X							X
Durability (remaining 450 cycles, class W only)	3.5.11	4.6.12	X							
Temperature life with cont. reten.	3.5.18.1	4.6.19.1		X					X	
Insulation resistance (elevated temperature)	3.5.15.2	4.6.16.2		X					X	
Ozone exposure (unmated)	3.5.19	4.6.20			X					
Magnetic permeability	3.5.20	4.6.21			X					
Insulation resistance (ambient temp.)	3.5.15.1	4.6.16.1			X				X	
Insert retention	3.5.21.1	4.6.22.1	X							
Operating force	3.5.5	4.6.6	X	X	X	X	X	X		
Dielectric withstanding voltage (unmated at sea level)	3.5.8	4.6.9.1	X	X	X			X	X	
Contact retention										
Series 1 and 2, removable contacts		4.6.7.2	X	X	X			X		
Series 3 and 4, removable contacts		4.6.7.3	X	X	X			X		
Insulation resistance (ambient temp.)	3.5.15.1	4.6.16.1		X		X	X	X		
Dielectric withstanding voltage (mated at altitude)	3.5.8	4.6.9.2		X	X	X	X	X		
Dielectric withstanding voltage (unmated at altitude)	3.5.8	4.6.9.2	X	X		X	X	X		
Contact insertion and removal force										
Series 1 and 2, removable contacts		4.6.4.1			X	X	X			
Series 3 and 4, removable contacts		4.6.4.2			X	X	X			
Contact resistance	3.5.25	4.6.26	X	X				X		
Shell to shell continuity	3.5.9.2	4.6.10.2	X		X					
EMI/RFI shielding effectiveness	3.5.9.1	4.6.10.1								X
Examination of product	3.1, 3.3 thru 3.4.6, 3.6, 3.7	4.6.1	X	X	X	X	X	X	X	X

1/ See 4.6.10.3.

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TABLE XXI. Qualification and group B inspection for nonremovable contact type connectors (series 1, 2, 3, 4 - class A, B, C, D, F, G, and T) (and for removable contact type connectors (series 3 and 4 - class L).

Title	Requirement paragraph	Test paragraph	Test sample group								Mating shell 1 and 2
			1	2	3	4	5	6	7	H	
Examination of product	3.1, 3.3 thru 3.4.6, 3.6, 3.7	4.6.1	X	X	X	X	X	X	X	X	X
Interfacial seal compression	3.4.2.3	4.6.32	X	X	X	X	X	X	X	X	
Contact insertion and removal force											
Series 3 and 4, class L	3.5.3	4.6.4.2								X	
Shell to shell continuity	3.5.9.2	4.6.10.2	X		X			X			
Shell spring finger forces	3.5.9.3	4.6.10.3									X 1/
EMI/RFI shielding effectiveness	3.5.9.1	4.6.10.1									X
Contact protection, connector mating											
Series 1 and 3	3.5.1.1	4.6.2	X	X		X					
Series 2 and 4	3.5.1.2	4.6.2			X		X	X			
Probe damage	3.5.2	4.6.3	X			X	X				
Operating force	3.5.5	4.6.6			X	X					
Contact retention	3.5.6										
(Non-removable contacts)		4.6.7.1	X	X	X	X	X	X			
(class L only)		4.6.7.3.2								X	
Contact engaging and separating force	3.5.4	4.6.5	X	X	X	X	X	X	X	X	
Thermal shock	3.5.7	4.6.8	X	X	X	X	X	X		X	
Contact retention	3.5.6										
(Non-removable contacts)		4.6.7.1	X	X	X	X	X	X			
(class L only)		4.6.7.3.2								X	
Dielectric withstanding voltage (unmated at sea level)	3.5.8	4.6.9.1	X	X	X	X	X	X		X	
Altitude immersion	3.5.24	4.6.25	X			X	X				
Durability	3.5.11	4.6.12			X				X	X	X
Vibration (conditioning)	3.5.12.1	4.6.13.1	X				X			X	
Vibration (resistance change)	3.5.12.2	4.6.13.2		X	X					X	
Shock (conditioning)	3.5.13.1	4.6.14.1	X								
Shock (resistance change)	3.5.13.2	4.6.14.2		X							
Moisture resistance	3.5.14	4.6.15	X							X	
Insulation resistance (ambient temp.)	3.5.15.1	4.6.16.1	X	X			X		X	X	
External bending moment	3.5.16	4.6.17							X		
Salt spray (corrosion)	3.5.17	4.6.18	X							X	
Temperature life	3.5.18	4.6.19		X				X		X	
Temperature life with contact retention (class L only)	3.5.18.1	4.6.19.1								X	
Insulation resistance (elevated temp.)	3.5.15.2	4.6.16.2		X				X		X	
Ozone exposure (unmated)	3.5.19	4.6.20			X						
Magnetic permeability	3.5.20	4.6.21			X						
Insulation resistance (ambient temp.)	3.5.15.1	4.6.16.1			X				X		
Insert retention (class F and T)	3.5.21.1	4.6.22.1	X								
Insert retention (class G and L)	3.5.21.2	4.6.22.2								X	
Operating force	3.5.5	4.6.6	X	X	X	X	X	X	X	X	
Dielectric withstanding voltage (unmated at sea level)	3.5.8	4.6.9.1	X	X	X			X	X	X	
Air leakage	3.5.22.1	4.6.23.1		X	X	X					
Air leakage (class G and L)	3.5.22.2	4.6.23.2								X	
Water immersion	3.5.23	4.6.24		X	X	X			X	X	
Contact retention	3.5.6										
(non-removable contacts)		4.6.7.1	X	X	X			X			
(class L only)		4.6.7.3.2								X	
Insulation resistance (ambient temp.)	3.5.15.1	4.6.16.1		X		X	X	X		X	
Dielectric withstanding voltage (mated at altitude)	3.5.8	4.6.9.2		X	X	X		X		X	
Dielectric withstanding voltage (unmated at altitude)	3.5.8	4.6.9.2	X	X				X		X	
Contact resistance	3.5.25	4.6.26		X							
Shell to shell continuity	3.5.9.2	4.6.10.2	X	X				X			
EMI/RFI shielding effectiveness	3.5.9.1	4.6.10.1									X
Examination of product	3.1.3.3 thru 3.4.6, 3.6, 3.7	4.6.1	X	X	X	X	X	X	X	X	

1/ See 4.6.10.3.

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TABLE XXIIA. Qualification and group B inspection for dielectric retention systems (removable contact type connectors, series 1, 2, 3, and 4, class A, C, E, F, L, P, T and W).

Title	Requirement paragraph	Test paragraph	Test sample group								
			1	2	3	4	5	6	7	8	9
Examination of product	3.1, 3.3 thru 3.4.6, 3.6, 3.7	4.6.1	X	X	X	X	X	X	X	X	X
Gage location and retention	3.5.28	4.6.29	X	X	X	X	X	X	X	X	X
Pin stability (series 3 and 4 only)	3.5.29	4.6.30	X	X						X	X
Retention system fluid exposure	3.5.10.1	4.6.11.1	X	X	X	X	X	X	X		
Contact insertion and removal force	3.5.3										
Series 1 and 2 removable contacts		4.6.4.1	X	X	X	X	X	X	X		
Series 3 and 4 removable contacts		4.6.4.2	X	X	X	X	X	X	X		
Insertion-removal tool abuse (series 3 and 4)	3.5.27	4.6.28								X	
Contact retention	3.5.6										
Series 1 and 2 removable contacts		4.6.7.2	X	X	X	X	X	X	X	X	
Series 3 and 4 removable contacts		4.6.7.3	X	X	X	X	X	X	X	X	
Contact walk-out	3.5.30	4.6.31									X
Examination of product	3.1, 3.3 thru 3.4.6, 3.6, 3.7	4.6.1	X	X	X	X	X	X	X	X	X

TABLE XXIIB. Qualification and group B inspection, fluid immersion testing (series 1, 2, 3 and 4, class A, B, C, D, F, G, L, T and W).

Title	Requirement paragraph	Test paragraph	Test sample number (per para. 4.4.1.7) 1 thru 12
Examination of product	3.1, 3.3 thru 3.4.6, 3.6, 3.7	4.6.1	X
Fluid immersion	3.5.10	4.6.11	X
Operating force	3.5.5	4.6.6	X
Insulation resistance (ambient temp.)	3.5.15.1	4.6.16.1	X
Dielectric withstanding voltage (unmated at sea level)	3.5.8	4.6.9.1	X
Examination of product	3.1, 3.3 thru 3.4.6, 3.6, 3.7	4.6.1	X

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TABLE XXIII. Intermateability test, Series 1 with Series 3, and Series 2 with Series 4.

Title	Requirement Para.	Test Para.	Test Sample Number			
			1	2	3	4
Examination of Product	3.1, 3.3 thru 3.4.6, 3.6, 3.7	4.6.1	X	X	X	X
Shell to Shell Continuity	3.5.9.2	4.6.10.2	X		X	
Operating Force	3.5.5	4.6.6	X	X	X	X
Altitude Immersion	3.5.24	4.6.25	X	X	X	X
Vibration (conditioning)	3.5.12.1	4.6.13.1	X		X	
Shock (conditioning)	3.5.13.1	4.6.14.1	X		X	
Dielectric Withstanding Voltage (mated at altitude)	3.5.8	4.6.9.2	X	X	X	X
Durability	3.5.11	4.6.12	X		X	
Operating Force	3.5.5	4.6.6	X		X	
Examination of Product	3.1, 3.3 thru 3.4.6, 3.6, 3.7	4.6.1	X	X	X	X

TABLE XXIV. Qualification and group B inspection for connector accessories.

Examination or test	Requirement paragraph	Test paragraph	Test Sample No.	
			1	2
Protective covers, stowage receptacles and cable clamps				
Examination of product	3.1,3.3 thru 3.4.6,3.6, and 3.7	4.6.1	X	X
Magnetic permeability	3.5.20	4.6.21	X	X
Operating force	3.5.5	4.6.6	X	X
Vibration	3.5.12	4.6.13.1	X	X
Moisture resistance	3.5.14	4.6.15	X	X
Cover chains, tensile strength	3.5.26	4.6.27	X	X
Water immersion	3.5.23	4.6.24	X	X
Air leakage	3.5.22.1	4.6.23.1	X	X
Shell to shell continuity (protective covers & stowing receptacles only)	3.5.9.2	4.6.10.2	X	X
Examination of product	3.1,3.3 thru 3.4.6,3.6 and 3.7	4.6.1	X	X
Connector assemblies, J adapter				
Examination of product	3.1,3.3 thru 3.4.6,3.6 and 3.7	4.6.1	X	X
Magnetic permeability	3.5.20	4.6.21	X	X
External bending moment	3.5.16	4.6.17	X	X
Thermal shock	3.5.7	4.6.8	X	X
Vibration	3.5.12	4.6.13	X	X
Moisture resistance	3.5.14	4.6.15	X	X
Salt spray (corrosion)	3.5.17	4.6.18	X	X
Water immersion	3.5.23	4.6.24	X	X
Air leakage	3.5.22.1	4.6.23.1	X	X
Shielding effectiveness	3.5.9.1	4.6.10.1	X	X
Fluid immersion	3.5.10	4.6.11	X	X

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TABLE XXV. Qualification of additional connectors.

Title	Requirement paragraph	Test paragraph
Examination of product	3.1,3.3 thru 3.4.6,3.6 and 3.7	4.6.1
Contact protection, mating		
Series 1 and 3	3.5.1.1	4.6.2
Series 2 and 4	3.5.1.2	4.6.2
Operating force	3.5.5	4.6.6
Contact insertion and removal	3.5.3	
Series 1 and 2		4.6.4.1
Series 3 and 4		4.6.4.2
Contact retention	3.5.6	
Series 1,2,3 and 4, non-removable contacts		4.6.7.1
Series 1 and 2, removable contacts		4.6.7.2
Series 3 and 4, removable contacts		4.6.7.3
Dielectric withstanding voltage (unmated at sea level)	3.5.8	4.6.9.1
Durability	3.5.11	4.6.12
External bending moment	3.5.16	4.6.17
Examination of product	3.1,3.3 thru 3.4.6,3.6 and 3.7	4.6.1

4.5 Quality conformance inspection.

4.5.1 Inspection for delivery. Inspection of product for delivery shall consist of groups A and B inspection.

4.5.1.1 Inspection lot. An inspection lot, as far as practicable, shall consist of all connectors covered by one specification sheet, produced under essentially the same conditions, and offered for inspection at one time.

4.5.1.2 Disposition of sample units. Sample units which have been subjected to the group A inspection may be delivered on the contract or order. Sample units which have been subjected to group B inspection shall not be delivered on the contract or order.

4.5.1.3 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table XXVI , and shall be made on the same set of sample units in the order shown. In process control of component parts, unrelated to lot sizes of finished connectors, may be used in lieu of examination of these components in the finished connectors to assure performance of these component parts. Hermetically sealed receptacles (class B, G, H, and L) must undergo 100 percent inspection to assure compliance with the requirements of 3.5.22.2 (air leakage).

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TABLE XXVI. Group A inspection.

Examination or test	Connector type		Accessories	Requirement paragraph	Test paragraph
	Removable contacts	Nonremovable contacts			
Examination of product		X	X	3.1, 3.3 thru 3.4.6, and 3.7	4.6.1
Contact retention 1/				3.5.6	
Series 1 and 2	X				4.6.7.2
Series 3 and 4	X				4.6.7.3
Insulation resistance		X		3.5.15	4.6.16
Dielectric withstanding voltage (unmated)		X		3.5.8	4.6.9.3
Dielectric withstanding voltage (unmated)	X			3.5.8	4.6.9.4

1/ 100% of contact cavities shall be tested for retention. Displacement measurements not required.

4.5.1.3.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level shall be 1 percent.

4.5.1.3.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection, and shall not thereafter be tendered for acceptance unless the former rejection or requirement of correction is disclosed. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.5.1.4 Group B inspection. Group B inspection shall consist of the applicable examinations and tests specified in table XVIII, XIX, XX, XXI, or XXII. Shipment shall not be held up pending results of this inspection.

4.5.1.4.1 Sampling plan. Sample units of one size, insert configuration, and each class representative of production at the time of selection shall be selected at 6-month intervals. Upon passing this inspection, the supplier may select sample units every 12 months. If the second level of sampling is passed two successive times, the supplier may select sample units every 24 months. In the event of a failure, sampling shall revert to the 6-month interval.

4.6 Methods of examination and test.

4.6.1 Examination of product. The connectors, accessories and piece parts shall be examined to insure conformance with the specification and the applicable specification sheets. In-process control of component parts, unrelated to lot sizes of furnished connectors, may be utilized in lieu of examination of these components in the finished connectors to assure conformance of these component parts. 100% visual examination of each retention cavity for molding discrepancies shall be a required in-process control for all removable contact connectors.

4.6.2 Contact protection (see 3.5.1). The front edge of the shell of the plug connector shall be scooped within the front aperture made by the shell of the mating receptacle. The front edges of both shells shall be in contact during this test.

4.6.3 Probe damage (see 3.5.2). Socket contacts shall be tested in accordance with the probe damage test of MIL-C-39029 using a probe damage test fixture conforming to MS3321. Solder contacts may be tested installed in the connectors.

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4.6.4 Contact insertion and removal force (see 3.5.3).

4.6.4.1 Contact insertion and removal force (series 1 and 2, crimp removable contact connectors only). All contacts shall be inserted and removed once using the appropriate military tool (MS3323, MS3342, and MS3343). Insertion and removal forces shall be measured and recorded on a minimum of 20 percent but not less than three of the contacts. All contacts shall be reinserted. After the contacts are locked in place, counterpart connectors shall be mated and unmated once and the torque values recorded. The contact retention mechanism shall then be unlocked and the contacts for which measurements were recorded shall be removed and reinserted nine more times (with the other contacts in place) using the appropriate military tool. The contact insertion and removal forces shall again be measured during the fifth and tenth insertion cycles. After the contacts are again locked, the torque to couple and uncouple the counterpart connectors shall again be recorded. The forces and torques shall comply with the requirements of table IX (torque) and table VII-A (forces), as applicable.

4.6.4.2 Contact insertion and removal forces (series 3 and 4, crimp removable contacts only). Twenty percent but not less than three of the contacts in each connector shall be removed and inserted 10 times using the appropriate tool (table VI). The forces required to remove and insert the contacts shall be measured on the first and tenth removal and insertion.

4.6.5 Contact engaging and separating force (see 3.5.4). Socket contacts shall be tested in accordance with the contact engagement and separation test of MIL-C-39029.

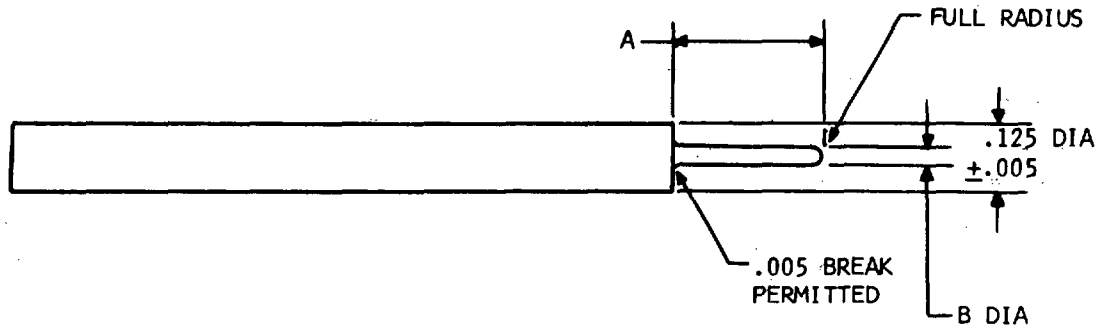
4.6.6 Operating force (see 3.5.5). Receptacles, plugs, dummy stowage receptacles, and protective covers shall be engaged with and disengaged from counterpart connectors; the forces or torques which must be applied to the coupling rings in order to facilitate full coupling and uncoupling shall be measured and recorded. The force or torque application shall be applied at a uniform rate of approximately one inch-pound per second.

4.6.7 Contact retention (see 3.5.6).

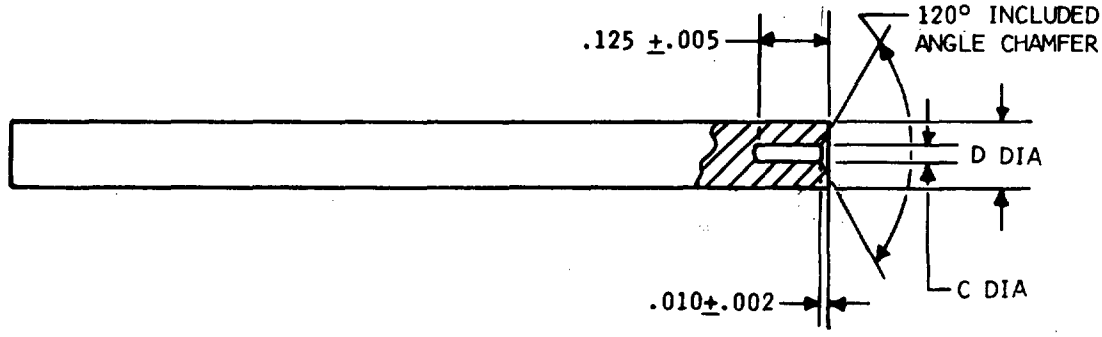
4.6.7.1 Contact retention (non-removable contacts). Axial loads in accordance with table X shall be applied to the mating end of individual contacts. The load shall be applied after the slack in contact retention has been taken up and shall be applied at a rate of approximately 1-pound per second until the specified load has been reached. The displacement of the contacts shall be measured under load after the load has been applied for a minimum period of 10 seconds. Test probes per figure 14 shall be used to apply the load.

4.6.7.2 Contact retention (series 1 and 2, removable contacts). Series 1 and 2 connectors with removable contacts shall be tested per 4.6.7.1 with the contact lock ring on the connector tightened to the lock position.

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SOCKET CONTACT RETENTION PROBE



PIN CONTACT RETENTION PROBE

CONTACT SIZE	A ±.010	B ±.001	C ±.001	D ±.005
23	.300	.026	.029	.125
20	.410	.039	.043	.125
16	.410	.061	.065	.125
12	.410	.093	.097	.156

FIGURE 14. CONTACT RETENTION PROBES.

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4.6.7.3 Contact retention (series 3 and 4, removable contacts).

4.6.7.3.1 Contact retention (series 3 and 4, class A and F). Axial loads in accordance with table X shall be applied to the mating end of each individual contact. Test probes per figure 14 shall be used to apply the load. The load shall be applied after the slack in the contact position has been taken up and shall be applied at a rate of approximately 1 pound per second until the specified load is reached. The displacement of the contacts shall be measured under load after the load has been applied for 10 seconds minimum.

4.6.7.3.2 Contact retention (series 3 and 4, class L). Twenty percent but not less than three of the contacts in each connector shall be tested. A contact with a solid rod soldered into the crimp barrel shall be substituted for the wired contact in each cavity to be tested. An axial load in accordance with table X shall be applied to the test contact rod in a manner tending to pull the contact from the cavity. The load shall be applied after the slack in the contact position has been taken up and shall be applied at a rate of approximately 1 pound per second until the specified load is reached. The displacement of the contacts shall be measured under load after the load has been applied for 10 seconds minimum. (Non-removable contacts of the class L connectors shall be tested per 4.6.7.1).

4.6.8 Thermal shock (see 3.5.7). Unmated connectors or unmated connectors with adapters shall be tested in accordance with method 1003 of MIL-STD-1344. Test condition F shall be used for class E, H, and P connectors. Test condition F shall be used for F, G, D, L, T and W connectors except that the upper temperature shall be 175°C and test condition C shall be used for class A, B, and C connectors. At the completion of the last cycle, the connectors shall be returned to room temperature for further inspection.

4.6.9 Dielectric withstanding voltage (see 3.5.8).

4.6.9.1 Dielectric withstanding voltage (sea level). Mated and unmated connectors shall be tested in accordance with method 3001 of MIL-STD-1344. The applicable test voltages (table XII) shall be applied between all adjacent contacts and between the shell and each contact closest to the shell. If an insert possesses more than one service rating, similar connections shall be made for the different test voltages as necessary.

4.6.9.2 Dielectric withstanding voltage (altitude). The connectors shall be tested in accordance with method 105 of MIL-STD-202 at the altitudes specified in table XII. After 30 minutes at the simulated altitude, the connectors shall be tested as specified in 4.6.9.1. Only the engaging faces of classes B, G, D, L, and H connectors shall be subjected to the high altitude, and the rear faces shall be suitably protected.

4.6.9.3 Dielectric withstanding voltage (non-removable contacts) - group A inspection. Unmated connectors shall be tested in accordance with 4.6.9.1, except that the period of application of voltage shall be 1 second minimum.

4.6.9.4 Dielectric withstanding voltage (removable contacts) - group A inspection. Unmated connectors shall be tested in accordance with 4.6.9.1 except test voltages shall be as specified for mated connectors per table XII, simulated contacts may be used, and the period of voltage application shall be 1 second minimum. (When connectors with pin contact inserts are being tested, the simulated contacts shall not protrude from the insert).

4.6.10 EMI/RFI shielding effectiveness (see 3.5.9).

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4.6.10.1 Shielding effectiveness (see 3.5.9.1). The electromagnetic susceptibility of mated pairs of shells shall be measured in a tri-axial radio frequency leakage tester. The RF leakage from the coaxial line through the shells of the connector pairs into the outer coaxial geometry shall be measured at the frequency shown in table XIII (accuracy of frequency ±5 percent). The level of detected signal power indicated by a tuned radio frequency field intensity meter is a result of RF leakage from the mated pair of shells. The test arrangement shall be as shown in figures 15 and 16.

4.6.10.2 Shell to shell continuity (see 3.5.9.2). The dc resistance of wired, mated, assembled connectors shall be measured between the points shown in figure 17. The applied potential shall be 1.5 volts d.c. maximum. The resistance shall be inserted in the circuit to limit the current to 0.100 +0.010 amperes. Probes with spherical ends of 0.05 inch minimum shall be used to make the voltage measurements and shall be placed on the simulated panel at the extreme edge of the plug as shown in figure 17. The probe shall not puncture or otherwise damage the connector.

4.6.10.3 Shell spring finger forces (see 3.5.9.3). An axial load shall be applied to engage the shell with spring fingers under test with a counterpart gage connector. The shells and gage shall not include coupling rings or inserts. The force necessary to engage and separate shall be within the values specified in 3.5.9.3.

4.6.10.3.1 Gage receptacle shell. The gage receptacle shell mating dimensions shall be in accordance with figures 9A and 9B except that the "A" diameter shall be at the maximum limit.

4.6.10.3.2 Gage plug shell. The gage plug shell mating dimensions shall be in accordance with figures 7A and 7B except that the "A" diameter shall be at the minimum limit.

4.6.11 Fluid immersion (see 3.5.10). Counterpart connectors shall be fully immersed as specified in table XXVII for the required periods. At least one pair of mating counterpart connectors shall be immersed in each fluid. After removal from the fluid, each pair of connectors shall remain for one hour in free air at room conditions in a position to allow the fluid to drain from the insert faces. Subsequent testing shall be performed on connectors mated with the same mating connectors used previously in testing.

4.6.11.1 Retention system fluid exposure. Connectors shall be unmated and contacts shall be removed. Connectors shall be immersed in the following fluids (one sample per fluid) for 20 hours at room temperature.

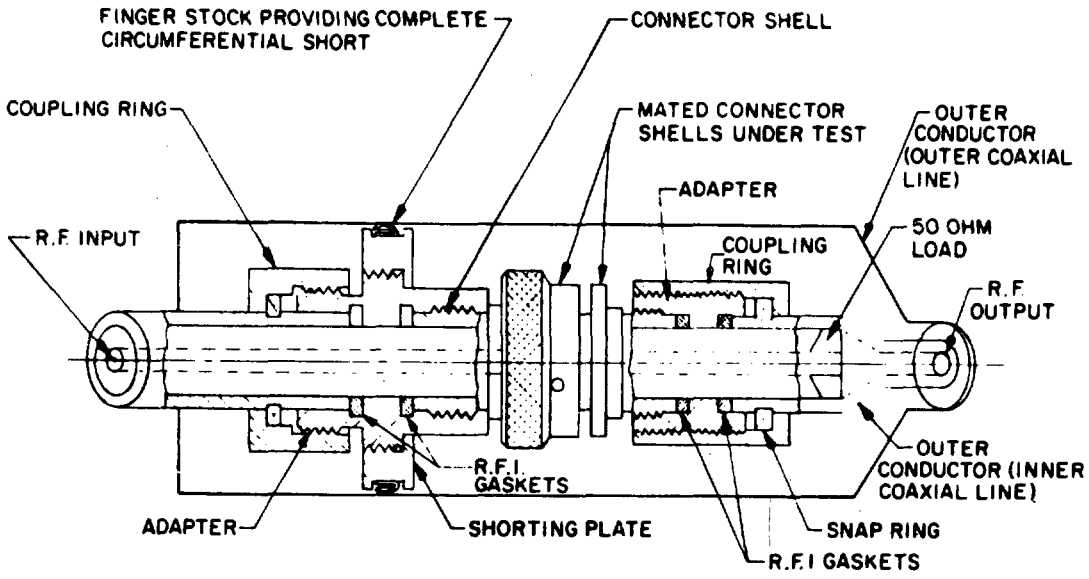
Sample no.	Test fluid
1	MIL-T-5624 (grade JP-5)
2	Coolanol 25
3	MIL-H-5606
4	MIL-L-7808
5	MIL-L-23699
6	Methyl alcohol
7	Isopropyl alcohol

After removal, excess fluid shall be allowed to drain from the connectors for 4 hours and the contacts shall be reinstalled. The connectors shall then be subjected to subsequent tests specified in table XXII.

4.6.12 Durability (see 3.5.11).

4.6.12.1 Durability (class A, B, C, D, E, F, G, H, L, P, and T). Wired, assembled counterpart connectors shall be mated and unmated 500 times at a maximum rate of 300 cycles per hour with the coupling rings operated in a manner to simulate actual service in that plugs and receptacles shall be completely separated during each cycle.

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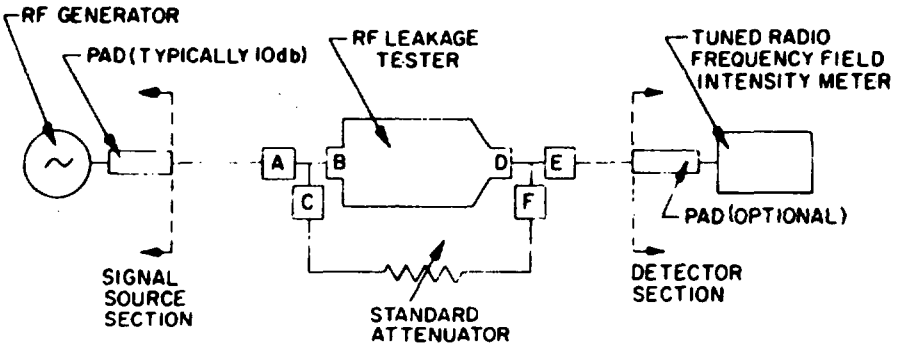
The mated pairs of connector shells shall be connected to the center coaxial line as shown in figure 14 and tested as shown in figure 15. Attachment of the connector shells to the outer conductor of the inner coaxial line shall be accomplished using R.F. gaskets to prevent leakage through the adapters.

Tester design requirements and terminations:

a. The characteristic impedances of the coaxial lines comprising the leakage tester shall be a nominal 50 ohms.
b. The VSWR (Zo = 50 ohms) looking into B shall be 1.5 or less when the mated pairs of connector shells are removed and replaced by a 50 ohm line section.
c. The outer coaxial line must not support higher order modes, especially the transverse electric 11 (TE_{11}) mode, at frequencies below 1,400 MHz.
d. The input and output VSWR of the standard attenuator (see figure 15) should be less than 1.5 in the 20 to 100 db attenuation range.
e. The output impedance of the signal source and the input impedance of the detector (see figure 15) shall be a nominal 50 ohms having a maximum allowable VSWR of 1.5.
f. Connectors A, B, C, D, E and F (see figure 15) should be of a low leakage type which exhibit R.F. leakage attenuation of >100 db.

FIGURE 15 Tri-axial RF leakage tester.

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A standard attenuator capable of providing at least 100 db (accuracy ±3 db) of attenuation in one db step shall be used to measure the RF leakage.

1. Tune the detector to the generator signal frequency by connecting C to A and F to E, using approximately 85 db attenuation in the standard attenuator.
2. Disconnect C from A and F from E and couple A to B and D to E. Adjust frequency slightly, if necessary, to maximize output indication on the tuned Radio Field Intensity Meter. Record output level.
3. Disconnect A from B and D from E. Connect C to A and F to E within one minute, so as to prevent excessive drift of voltage in the signal source and of calibration in the Tuned Radio Frequency Field Intensity Meter. Adjust the standard attenuator until the output level recorded in step 2 in reached. The value of the attenuation provided by the standard attenuator is a measure of the RF leakage attenuation of the mated pair of shells, measured in db of power.

FIGURE 16 Test procedure for tri-axial RF leakage tester.

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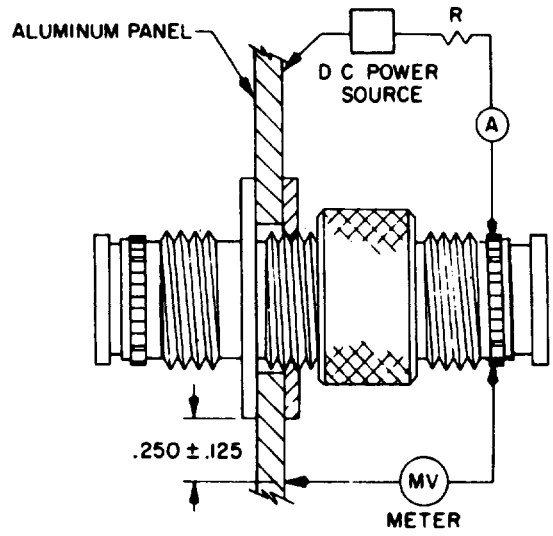


FIGURE 17 Shell to shell continuity test diagram.

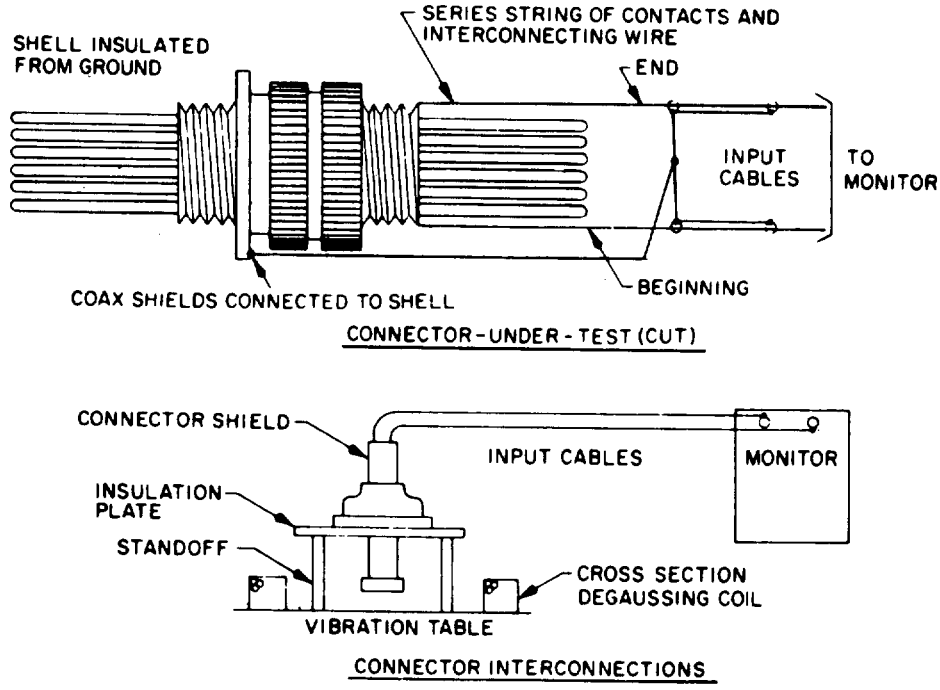


FIGURE 18 Application of resistance change monitor.

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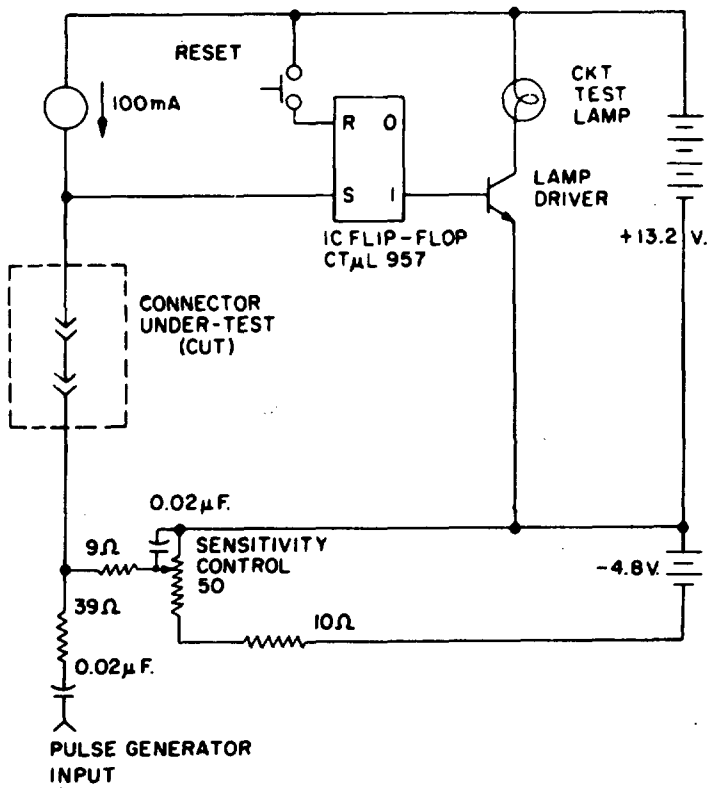


FIGURE 19 Contact resistance change monitor circuit.

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4.6.12.2 Durability (class W). Wired, assembled counterpart connectors shall be mated and unmated 500 times at a maximum rate of 300 cycles per hour with the coupling rings operated in a manner to simulate actual service in that plugs and receptacles shall be completely separated during each cycle. The assemblies shall be mated and unmated 50 cycles followed by the salt spray test. After the salt spray test, the remaining 450 mating and unmating cycles shall be performed (see table XX).

4.6.13 Vibration.

4.6.13.1 Vibration (conditioning) (see 3.5.12.1). The connector assembly shall be mounted, as specified herein and vibrated in accordance with method 2005, test condition IV, MIL-STD-1344. In addition, the vibration shall be conducted at the low temperature extreme and the high temperature extreme as shown in table XI.

4.6.13.2 Vibration (resistance change) (see 3.5.12.2). The connectors shall be subjected to the vibration specified in 4.6.13.1. In addition, all contacts shall be wired in series with at least 100 milliamperes of current allowed to flow, and there shall be a minimum of 10 percent but not less than 3 contact pairs in the circuit (see figure 19). A suitable instrument shall be employed to monitor the increase in contact resistance (see figure 18). The duration of vibration at extreme temperatures shall be 25 percent of the duration specified for the standard temperature condition.

4.6.13.3 Connector mounting. Each receptacle shall be mounted on a suitable fixture, which in turn shall be attached to a vibration table. A suitable sensor shall monitor the vibration of the receptacle at a point on or near the receptacle. A counterpart plug shall be engaged with the receptacle and held by normal locking means without the use of safety wire. The wire bundles or cables for those samples being monitored for resistance change shall be 6 inches maximum from the rear of the connectors, and shall be clamped to non-vibrating points at least 8 inches from the rear of the connector.

4.6.14 Shock (see 3.5.13).

4.6.14.1 Shock (conditioning) (see 3.5.13.1). Mated connectors shall be mounted as specified (see 4.6.13.1) and subjected to the shock test of method 2004, test condition A, of MIL-STD-1344 except that only one shock shall be applied in each direction of the three major axes of the connector.

4.6.12.2 Shock (resistance change) (see 3.5.13.2). Mated connector shall be mounted as specified in 4.6.13.1, and wired and monitored for resistance change as specified in 4.6.13.2. Shock shall be applied as specified in 4.6.14.1.

4.6.15 Moisture resistance (see 3.5.14). The mated and wired connectors shall be subjected to a moisture resistance test in accordance with method 106 of MIL-STD-202, with the following details and exceptions:

a. Step 7b, vibration not required.
b. There shall be no drip loops in the wires.
c. Wires shall be brought out of the chamber through vaportight seals.
d. There shall be no open wire splices in the chamber.
e. After a minimum of 3 hours at step 7 of the final cycle, while connectors are still subjected to high humidity, the insulation resistance shall be measured.
f. The class B, D, G, H, and L, and nonremovable contact receptacles shall have their rear portion suitably protected from the humidity exposure.

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TABLE XXVII. Fluids for fluid immersion test.

Sample no.	Test fluid specification	Test procedure
A. Class A, B, C, D, F, G, L, T and W connectors (see 4.4.1.7)		
1	MIL-L-7808	Immerse unmated connectors in fluid at 120° ±3°C for five minutes.
2	MIL-L-23699	Remove connectors and allow to drain for one hour at room temperature. Fluid shall be drained from all recesses. Mate connectors and expose to 125° ±3°C in an air circulating oven for 6 hours. Remove connectors and allow to stabilize at room temperature for one hour. Repeat procedure for a total of seven cycles.
3 4	MIL-H-5606 Hydraulic fluid 1/	Same as for MIL-L-7808 except expose mated connectors to 104° ±3°C in an air circulating oven for 6 hours.
5	MIL-T-5624 (grade JP-5)	Same as MIL-L-7808 except fluid shall be at room temperature.
6 7	MIL-A-8243 MIL-C-25769 (10% solution)	Same as MIL-L-7808 except fluid temperature shall be 65° ±3°C and connectors shall be mated during fluid immersion.
8	Coolant-dielectric fluid synthetic silicate ester base 2/	Unmated connectors shall be pre-conditioned at 175°C for thirty (30) minutes. Immerse connectors fully in room temperature fluid for 1 minute. Remove connectors and allow to stabilize at room temperature for one hour. Fluid shall be drained from all recesses.
9	MIL-G-3056 (type 1)	The wired assembled unmated connectors shall be immersed in the fluid for a minimum of 5 minutes, removed from the fluid and exposed to free air for 24 ±2 hours. This conditioning cycle shall be repeated until the connector has been subjected to 5 complete cycles; for a maximum of 2 cycles the exposure to free air may be extended to 50 hours.
10	Solvent (a) specified in method 215 of MIL-STD-202	Same as for MIL-G-3056.
11	Solvent (b) specified in method 215 of MIL-STD-202	Same as for MIL-G-3056.
12	Solvent (c) specified in method 215 of MIL-STD-202	Same as for MIL-G-3056.
B. Class E, H, and P connectors (see 4.4.1.1)		
4	MIL-H-5606	Immerse unmated connectors at room temperature for 20 hours; one connector in each fluid.
5	MIL-L-23699	Remove connectors and allow to drain for one hour at room temperature. Subsequent testing to be performed.

1/ M2-V Chevron oil, ST0145LB0001 or equivalent.
2/ Coolanol 25 or equivalent.

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4.6.16 Insulation resistance.

4.6.16.1 Insulation resistance at ambient temperature (see 3.5.15.1). Unmated connectors shall be tested in accordance with method 3003 of MIL-STD-1344. The contacts selected shall be those having the closest spacing between measurement points.

4.6.16.2 Insulation resistance at elevated temperature (see 3.5.15.2).

4.6.16.2.1 Insulation resistance at elevated temperature (class E, H, and P connectors). The insulation resistance shall be measured in accordance with 4.6.16.1, except that the connectors shall have been exposed to an ambient temperature of 150°C for 1,000 hours. The measurements shall be made at the end of the temperature life (see 4.6.19) 1,000 hours, while the connectors are at elevated temperatures (see 3.5.15.2).

4.6.16.2.2 Insulation resistance at elevated temperature (class D, G, L, and T connectors). The insulation resistance shall be measured in accordance with 4.6.16.1, except that the connectors shall have been exposed to an ambient temperature of 175°C for 1,000 hours. The measurements shall be made at the end of the temperature life (see 4.6.19) 1,000 hours, while the connectors are at elevated temperature (see 3.5.15.2).

4.6.16.2.3 Insulation resistance at elevated temperature (class A, B, and C connectors). The insulation resistance shall be measured in accordance with 4.6.16.1, except that the connectors shall have been exposed to an ambient temperature of 200°C for 1,000 hours. The measurements shall be made at the end of temperature life (see 4.6.19) 1,000 hours, while the connectors are at elevated temperature (see 3.5.15.2).

4.6.17 External bending moment (see 3.5.16). The receptacle connector shall be mounted as in normal service to a rigid panel. Before mating the plug connector to the receptacle, the "J" adapter or test torque shall be attached as shown in figure 20. After mating the plug and receptacle connectors, the distance "L" from the point of load application "P", to the mounting panel shall be determined. The load to be applied at point "P" shall then be determined as the bending moment listed in table XV divided by the lever arm "L". This load shall be applied at a rate of approximately 10 pounds per second until the required load is achieved. The load so applied shall be held for a period of 1 minute after which the load shall be released. Continuity of the contacts shall be monitored during the test. The connector shall be removed, unmated, and inspected to 3X power magnification for damage or breakage.

4.6.18 Salt spray (corrosion) (see 3.5.17). Unmated connectors, shells, and accessories shall be subjected to a salt spray test in accordance with method 1001, test condition B, of MIL-STD-1344. The following details and exceptions shall apply: Class W connectors shall be exposed to 452 hours mated, followed by 48 hours unmated. Wire ends must be protected to prevent salt migration. The salt concentration shall be 5 percent. The specimens shall then be washed off with tap water and then dried in a circulating air oven at a temperature of 38° ±3°C for a period of not more than 12 hours, after which they shall be removed and inspected. Prior to all subsequent tests, corrosion tested connectors shall be engaged and disengaged for one cycle to remove free salt deposits. The rear portions of class B, D, G, H, and L receptacles shall be suitably protected from the salt spray exposure.

4.6.19 Temperature life (see 3.5.18). The mated connectors shall be subjected to the applicable high temperature specified in table XI for a period of 1,000 continuous hours. The internal temperature shall be monitored at a contact closest to the center of the connector.

4.6.19.1 Temperature life with contact retention. Connectors with removable contacts shall have one mating pair of contacts removed and replaced with contacts crimped to steel core copper wire (copperweld or equivalent) of the appropriate size. The axial location of these contacts shall be measured for conformance with the applicable dimensions shown in figures 4 thru 11, with a load of 2 pounds applied to seat the contact back against the retention device. The connector shall then be mounted in a fixture as shown in figure 21. A weight equal to 50% of the axial load specified in

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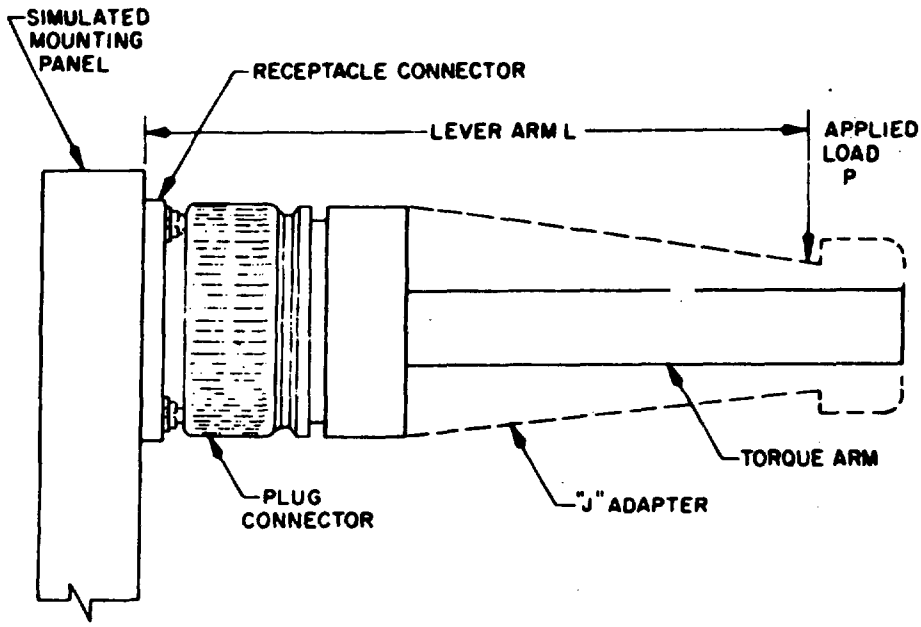
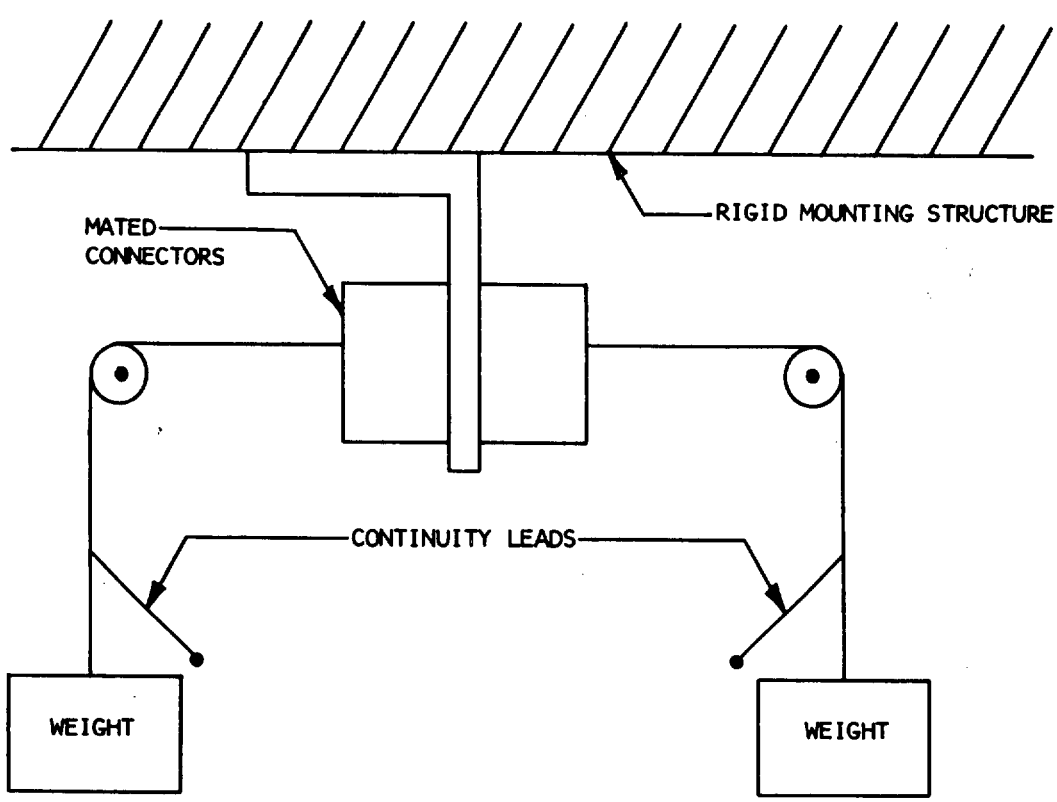


FIGURE 20 External bending moment test set-up.

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TEMPERATURE LIFE WITH CONTACT RETENTION

FIGURE 21

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table X for the applicable contact size shall be suspended freely from each steel core wire. A current of 100 ±10 milliamperes supplied from a 10.0 Vdc maximum power source shall be applied to the test contacts and a suitable instrument shall be used to monitor the circuit for discontinuity in excess of one microsecond. The connector, mounted as shown in figure 21, shall then be subjected to the temperature life test of 4.6.19. After the connectors return to ambient temperature they shall be unmated and the contact locations remeasured (figures 4 through 11 as applicable) with 2 pounds axial load applied to seat the contact back against the retention device. The tested contacts shall then be replaced with unwired contacts and sealing plugs, and those cavities shall be exempted from subsequent testing.

4.6.20 Ozone exposure (see 3.5.19). The unmated connectors shall be subjected to ozone having a concentration from 0.010 to 0.015 percent by volume for 2 hours at room temperature. At the end of the specified period, the samples shall be examined for signs of ozone deterioration as specified.

4.6.21 Magnetic permeability (class A, C, E, F, T and P) (see 3.5.20). Permeability shall be checked with an instrument conforming to MIL-I-17214. The connectors may be wired or unwired, as convenient, but shall not be carrying current.

4.6.22 Insert retention.

4.6.22.1 Insert retention (class A, C, F, T, E, P and W) (see 3.5.21.1). Inserts in connectors less removable grommets or any accessory shall be subjected to axial loads in each direction separately. Loading shall be accomplished by applying air pressure or equivalent load. The pressure shall be increased gradually at a rate of approximately 10 psi/second until the specified pressure is reached. The insert shall retain its normal position in the connector shall for at least 10 seconds at maximum specified pressure.

4.6.22.2 Insert retention (class B, D, G, H, and L) (see 3.5.21.2). Inserts in connectors less any accessories shall be subjected to axial in each direction separately. Loading shall be accomplished by applying hydraulic pressure or equivalent load. The pressure shall be increased gradually at a rate of 50 psi/second until the specified pressure is reached. The insert shall retain its normal position in the connector shell for at least 10 seconds at maximum specified pressure.

4.6.23 Air leakage.

4.6.23.1 Connectors with solder contacts and environmental sealing adapters (see 3.5.22.1). Connectors with solder contacts and sealing adapters shall be mounted in a manner suitable for application of a 30 psi pressure differential across the connectors. The leakage rates shall be measured at the low-temperature extreme in both directions after 30 minutes of exposure to the low-temperature extreme.

4.6.23.2 Class B, D, G, H, and L connectors (see 3.5.22.2). Class B, D, G, H, and L connectors shall be mounted in a manner suitable for application of 15 psi pressure across the connectors and tested in accordance with method 112, test condition C, of MIL-STD-202. The leakage rate shall be determined while pressurized air or gas, containing not less than 10 percent of helium by volume, is applied to the connector.

4.6.23.3 Stowage receptacle and protective covers (see 3.5.22.3). Stowage receptacles and protective covers shall be mated to connectors having either contacts or inserts removed so that a pressure of 30 psi can be applied against the insides of the protective covers or stowage receptacles.

4.6.24 Water immersion (non-removable contact connectors with J adapter) (see 3.5.23). The connectors shall be immersed in tap water to a depth of 6 feet for a period of 48 hours in accordance with the following:

a. Plugs with J adapters shall be assembled to test cyclinders (see table XXVI).

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b. Class A, F, and E receptacles shall be mounted by their normal mounting means with mounting flange gaskets to that water pressure is applied to only one side. One-half the sample shall be front-mounted and the remainder back-mounted (except the single-hole mounted receptacles).
c. Fifty percent of the connectors tested shall be mated and shall be such that mated receptacles will be subjected to water pressure from the mating side or mating plug and receptacle shall be completely immersed in water as applicable to meet conditions of this test. Insulation resistance of the mated connectors shall be measured at the end of the 48 hour period while still immersed.
d. Class A, F, and E receptacles, mated or unmated, shall be inspected for leakage through or around the insert and for leakage of the panel seals.
e. Fifty percent of the connector test samples shall be tested unmated and shall be such that receptacles are subjected to water pressure from both the unmated front and the rear of the connector. Upon completion of the test, the connectors shall be removed from the connectors by shaking them at room temperature. Insulation resistance shall be measured within one-half hour after the connector is removed from the water.
f. All mated connectors shall be inspected for internal leakage of water at the interface and cable housing.

TABLE XXVII. Test cylinder OD sizes.

Shell size	Diameter ±0.016 inch
8	0.214
10	0.250
14	0.462
16	0.596
18	0.646
20	0.812
22	0.947
24	1.062

4.6.25 Altitude immersion (see 3.5.24). Connectors shall be mated and unmated 50 times. The mated connectors shall then be immersed in a container of water at approximately 30°C and placed in a chamber. Wire ends, opposite the connector, shall be terminated within the altitude chamber, exposed to the chamber atmosphere but not submerged. The exposed wire ends shall not be sealed. A quantity of salt, 5 percent by weight, shall be added to make the water conductive. The chamber pressure shall be reduced to approximately 1.2 inch of mercury and maintained for 30 minutes. The chamber pressure shall then be returned to atmospheric pressure within one minute and while at the submerged condition, the insulation resistance requirements specified in 3.5.15.1 shall be met. This shall be considered to be one cycle. A period of recovery of fifteen minutes at atmospheric pressure is required following each altitude cycle. Two additional environmental cycles shall be run. At the end of the last cycle, the electrical test conducted at the end of the first cycle shall be repeated and a dielectric withstanding voltage as specified in 3.5.8 shall be applied between the same points as those used for insulation resistance measurements. The voltage shall be applied while the connectors are still immersed, but at sea level. Class B, D, G, H, and L connectors and nonremovable contact connectors shall have their rear portion excluded from the water immersion.

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4.6.26 Contact resistance (see 3.5.25). The contact resistance shall be measured in accordance with the contact resistance test of MIL-C-39029.

4.6.27 Cover chain tensile strength (see 3.5.26). The protective cover shall be securely held and a tensile static load of 50 pounds shall be applied to the end of the chain for 5 minutes in each direction as specified:

a. With the axis of the chain at right angles to the axis of the holding fastener.
b. With the axis of the chain in the same axis as that of the fastener.

4.6.28 Insertion/removal tool abuse (see 3.5.27). (Series 3 and 4 only) steps A, B, C, and D shall be performed. Separate groups of 5 contacts cavities each shall be used for each step.

a. Removal tool - The tool shall be inserted as if to remove a contact and a total of 3 pounds axial load shall be applied. With the force applied, the tool shall be rotated 180° and then removed also removing the contact. The contact shall be reinserted. These steps shall be repeated three times on each of the five contacts selected.
b. Insertion tool - The contact shall first be removed. With the insertion end of the tool, insert the contact and continue to press until an axial load of 3 pounds is applied. With the force applied, the tool shall be rotated 180° and then removed. These steps shall be repeated three times on each of the five contacts selected.
c. Insertion tool - The contact shall first be removed. With the insertion end of the tool, the contact shall be inserted and an axial load of 10 pounds applied to the tool. These steps shall be performed only once on each of the five contacts selected. A new tool shall be used for each contact.
d. Removal tool - The tool shall be inserted as if to remove contact and a total of 10 pounds axial load shall be applied. The tool shall then be removed also removing the contact. These steps shall be performed only once on each of the five contacts selected. A new tool shall be used for each contact.

NOTE: Should a tool become damaged during any of the testing it shall be replaced. Failure of a tool shall not constitute a test failure.

4.6.29 Gage location and retention (see 3.5.28). Applicable test gages in accordance with MS3462, MS3434, or MS3435 shall be installed in 3 randomly selected cavities of each connector. Remaining cavities shall have contacts in place. An axial loar of 2 pounds shall be applied to the engaging end of the gage and the location of the gage shall be measured in accordance with figures 4 through 11 as applicable. The axial load specified in table X shall then be applied to the test gage in the direction tending to displace it toward the rear of the connector. The load shall be applied at a rate of approximately 1 pound per second until the specified load has been reached. The displacement of the gage relative to the shell shall be measured under load after the load has been applied for 10 seconds minimum. Class L connectors shall be tested for gage retention only; location measurements do not apply.

4.6.30 Pin contact stability (see 3.5.29). (Series 3 and 4 only) three pin contact cavities in each connector shall be tested. The applicable test probe (MS3435) shall be inserted into the cavity until it is locked into place by the contact retaining mechanism. The load specified in table XXIX shall be applied perpendicular to the probe at the point of load application shown in figure 22. With the load applied, the position of the test probe tip shall be located. The applicable load shall then be applied in the opposite direction (180°) and the total displacement (splay) of the test probe tip shall be measured. See figure 22.

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TABLE XXIX. Pin contact stability.

Contact size	Total displacement (inch maximum)	Load (ounces minimum)
23	.025	4
20	.030	8
16	.048	16
12	.062	16

4.6.31 Contact walk-out (see 3.5.30). Two contacts in each plug and receptacle shall be tested. The contacts shall be crimped to stranded steel cable of an appropriate size and installed in the connector. The unmated connector shall be mounted in a test fixture as shown in figure 23. A 3 pound load shall be applied to the cable. One 360° rotation of the fixture with the connector mounted shall constitute one cycle. The connector shall be subjected to 100 cycles at a rate of 10 to 20 cycles per minute.

4.6.32 Interfacial seal compression (see 3.4.2.3). Each measurement of compression displacement of the interfacial seal shall be obtained in the following manner.

a. Pin contacts shall be installed and socket contacts shall not be installed in mating connectors under test.
b. Series 1 and 2 mating connectors shall be engaged without the aid of the coupling ring until the face of the pin insert touches the face of the socket insert (determined by feel). Series 3 and 4 mating connectors shall be engaged without the aid of the coupling ring until the raised seal barriers around the pin contacts touch the socket insert entry cones (determined by feel).
c. The overall length from the rear of the plug shell to the rear of the receptacle shell shall be measured at 2 or more locations equally spaced about the connector diameter.
d. The connectors shall be fully engaged by means of the coupling ring and the measurements repeated at the same locations as in step (c).
e. Measurements obtained in step (d) shall be subtracted from those obtained at corresponding locations in step (c). The resultant difference in measurements is the compression displacement of the interfacial seal.

5. PACKAGING

5.1 Preservation and packaging. Preservation shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

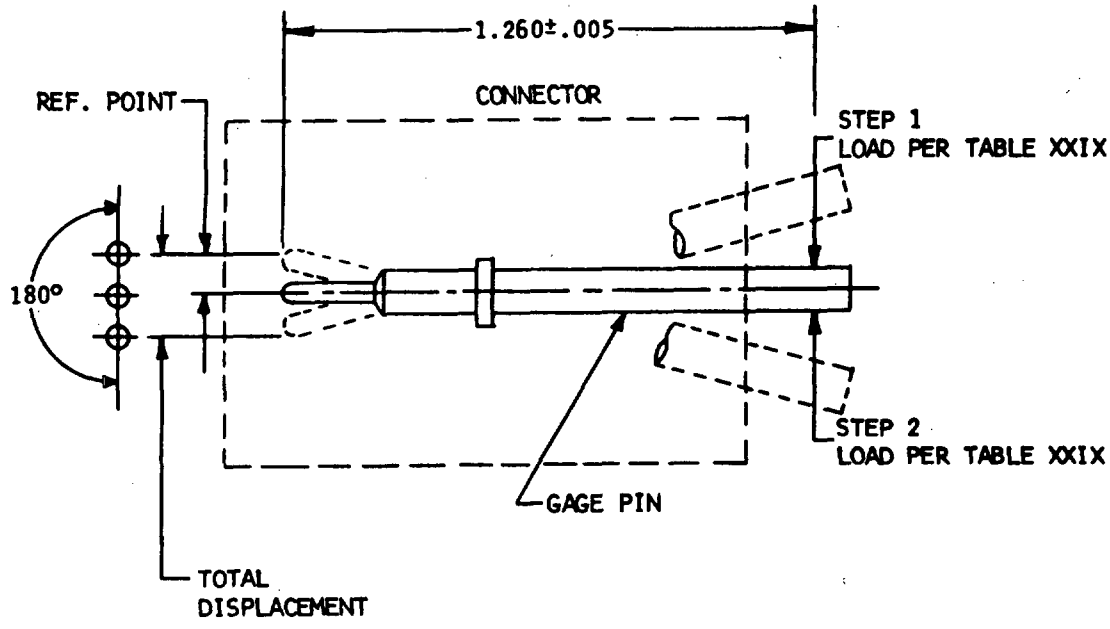
5.1.1.1 Cleaning. Connectors and accessories shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Connectors and accessories shall be dried in accordance with MIL-P-116. The procedures used shall not be injurious to the item.

5.1.1.3 Preservative application. None required.

5.1.1.4 Unit packaging. Unless otherwise specified (see 6.2), connectors, accessories, and connectors with accessory parts shall be individually packaged in accordance with MIL-P-116, submethod IA-8 insuring compliance with the general requirements paragraph under methods of preservation (unit protection) and the physical protection requirements paragraph therein.

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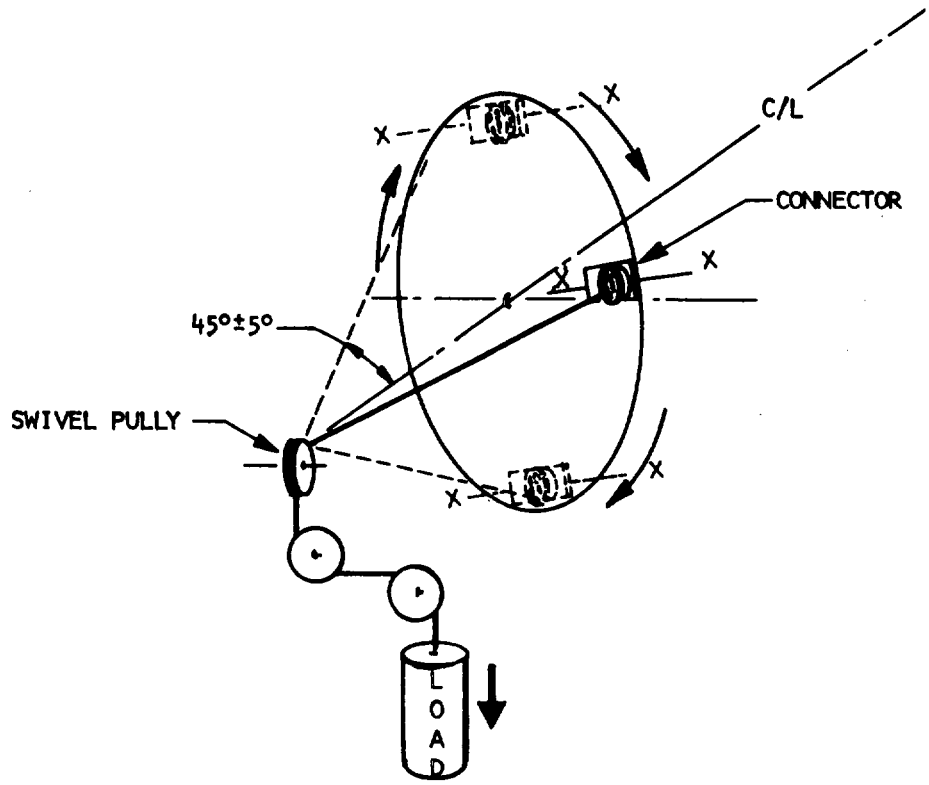


STEP 1 APPLY LOAD TO DETERMINE REFERENCE POINT.
STEP 2 APPLY LOAD IN OPPOSITE DIRECTION (180°) AND MEASURE TOTAL DEFLECTION.

PIN CONTACT STABILITY

FIGURE 22

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NOTE: CONNECTOR AXIS (REF X-X) TO REMAIN CONSTANT DURING ROTATION.

FIGURE 23. CONTACT WALK-OUT

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5.1.1.4.1 Removable contacts. Connectors with removable crimp type contacts shall not be packaged with the contacts installed in the connector inserts. The contacts shall be packaged in a close fitting rigid container formed of plastic, chipboard, or other suitable material. Neutral tissue paper, conforming to UU-P-553, shall be used to fill voids or cushion to prevent abrasion of the plating when the method itself does not accomplish fixed positioning. The unit package quantity shall be adjusted to the normal complement, plus one spare contact for connector arrangements having 26 contacts or less and two spares for arrangements over 26 contacts. Include in the connector unit package.

5.1.1.4.2 Grommet sealing plugs. A quantity of grommet sealing plugs in accordance with the applicable specification sheet (15 percent of the number of contacts, but not less than one) shall be included in the container with the contacts for both removable and nonremovable contact type connectors having grommets.

5.1.1.4.3 Accessory parts. When accessory parts such as grommet sealing plugs, adaptors, and hardware are included with each connector, these parts shall be separately cushioned and included within the unit package.

5.1.1.5 Intermediate packaging. Connectors and accessories, packaged as described in 5.1.1.4, shall be placed in intermediate containers conforming to PPP-B-566 or PPP-B-676. Intermediate containers shall be uniform in size and shape, shall be of minimum tare and cube, and shall contain multiples of five unit packages, not to exceed 50 unit packages. No intermediate packaging is required when the total quantity shipped to a single destination is less than 50 units.

5.1.2 Level C. Cleaned and dried connectors, accessories, and connectors with accessories shall be individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The packaged connectors and accessories shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirements in the appendix of PPP-B-636, closures and waterproofing shall be accomplished by sealing the center seams, corners and manufacturer's joint with tape, 2 inches minimum width, conforming to PPP-T-60, class 1, or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636, using non-metallic or tape banding only.

5.2.2 Level B. The packaged connectors and accessories shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix thereto. For Army procurement, fiberboard containers shall be class weather-resistant as specified in level A.

5.2.3 Level C. The packaged connectors and accessories shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

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5.2.4.1 Level A. Connectors and accessories, packed as specified in 5.2.1, shall be placed on pallets in conformance with MIL-STD-147, load type 1, with a fiberboard cap (storage aid 4) positioned over the load.

5.2.4.2 Level B. Connectors and accessories, packed as specified in 5.2.2, shall be palletized as specified in 5.2.4.1 except that fiberboard caps shall be class domestic. For Army procurement, caps shall be weather resistant as specified in level A.

5.2.4.3 Level C. Connectors and accessoreis, packed as specified in 5.2.3, shall be unitized with pallets and caps of the type, size and kind commonly used for the purpose and shall conform to the applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or order, each unit package, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2, and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Navy procurements. For Navy procurements, the use of polystyrene loose fill material (such as strips, strands and beads) is prohibited for packaging and packing applications.

6. NOTES

6.1 Intended use. The classes and types of connectors are intended for military application as follows (see 1.1.2.2):

a. Class E and P connectors are intended for use in environmental applications where the operating temperature range of -65 to 150°C is experienced. Class F and T are intended for use in environmental applications where the operating temperature range of -65 to 175°C is experienced and additional fluid resistance is needed. Class A and C are intended for use in environmental applications where the operating temperature range of -65 to 200°C is experienced and additional fluid resistance is needed.
b. All connectors are environment resisting with provisions to insure electrical continuity from plug to receptacle shell.
c. Class B, D, G, H, and L receptacles are intended for use in applications wherein fluid pressures must be contained by the connectors with a very low leakage rate across the walls or panels on which they are mounted.
d. Crimp contact connectors shall have contacts installed in all positions when the connector is wired. Sealing plugs shall be installed in the grommet holes when no wire is attached to the contact in grommet-sealed connectors.
e. If air leakage requirements are critical, a class A, F, and E solder-contact connector should be used.
f. To provide adequate moisture seal, wire diameters larger than minimum specified in table V for the applicable contact size must be used.
g. Where two or more wires are used in a wire barrel, moisture sealing is not obtainable.
h. Class W connectors provide a highly corrosion resistant shell.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Title, number, and date of the applicable specification sheet and the complete part number (see 1.2 and 3.1).
c. Levels of preservation and packaging and packing, and applicable marking (see section 5).
d. Flange gaskets as specified (see 3.4.5).
e. For indirect shipment, these connectors may be furnished without contacts.

6.2.1 Accessory hardware. Accessory hardware, such as covers, dummy stowage receptacles, or mounting hardware especially designed for these connectors, is defined in the applicable specification sheet. Flange gaskets will not be supplied with receptacles unless they are specified.

6.2.2 Crimp contacts. When specified in the contract or purchase order, connectors may be supplied less crimp contacts and/or grommet sealing plugs to permit procurement of these items in bulk. Crimp contacts may be ordered in bulk in accordance with MIL-C-39029.

6.2.3 Solder contacts. *Solder contacts are not to be ordered or qualified* separately from the connectors.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable qualified products list (QPL) whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification.

6.3.1 Application for qualification. The activity responsible for the QPL for MIL-C-81511 is the Naval Air Systems Command. The Naval Weapons Support Center, Crane, Indiana, has been designated by the Naval Air Systems Command as agent for establishment of the QPL. Requests for information pertaining to and applications for qualification should be addressed to: Commanding Officer, Naval Weapons Support Center, ATTN: Code 30633, Crane, IN 47522.

6.3.2 Upon application for qualification testing, the qualifying activity shall verify that the manufacturer has established written procedures to assure actual connector performance to the requirements of this specification. Emphasis shall be *on the molded plastic retention system piece parts prior to assembly into the con*nector. These procedures shall contain: (1) Analysis of minimum/maximum dimensions of contacts and molded plastic retention discs. (2) Acceptable levels for:

a. Plastic mixtures and chemical tests.
b. Pureness of materials.
c. Method for flash removal.

(3) (a) Contact retention tests on molded plastic retention disc based on analysis of number (1).
(b) Insertion and removal forces for pin and socket contacts in plastic disc.

When approved by the qualifying activity, these shall become the minimum requirements for molded plastic contact retention systems prior to assembly into the connectors for that manufacturer.

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6.4 Noncompliance. If a sample fails to pass group B inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group B inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the qualifying activity). Group A inspection may be reinstituted; however, final acceptance and shipment shall be withheld until the group B inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

6.5 Maximum recommended working voltage. The maximum recommended working voltage is shown in table XXX.

TABLE XXX. Maximum recommended working voltage, ac, rms.

Condition	Service rating I	Service rating II
Sea level	600	1,000
50,000 feet	300	450
70,000 feet	300	450
110,000 feet	300	450

6.6 Definitions.

6.6.1 Permanent. For purposes of this specification, when the word "permanent" or "permanently" is used in connection with the attachment or retention or material, it shall be construed to mean that descruction or rupture of material is required before removal can be accomplished.

6.6.2 Scoop-proof. Scoop proof refers to the impossibility of a mating plug connector inadvertently being cocked into a mating receptacle and damaging pins or electrically shorting the contacts.

6.7 Patent notice. The Government has a royalty-free license under the following listed patents and applications for patent, for the benefit of manufacturers of the patented item either for the Government or for use in equipment to be delivered to the Government:

U.S. Patent number	U.S. Patent application serial number
3,003,135	558,958
3,101,229	
3,727,172	
3,784,966	

6.8 Connector weights. Approximate weights for all classes of connectors are as shown in the applicable specification sheet.

6.9 Cognizance. This specification is under the cognizance of NAVAIR 533D1B3.

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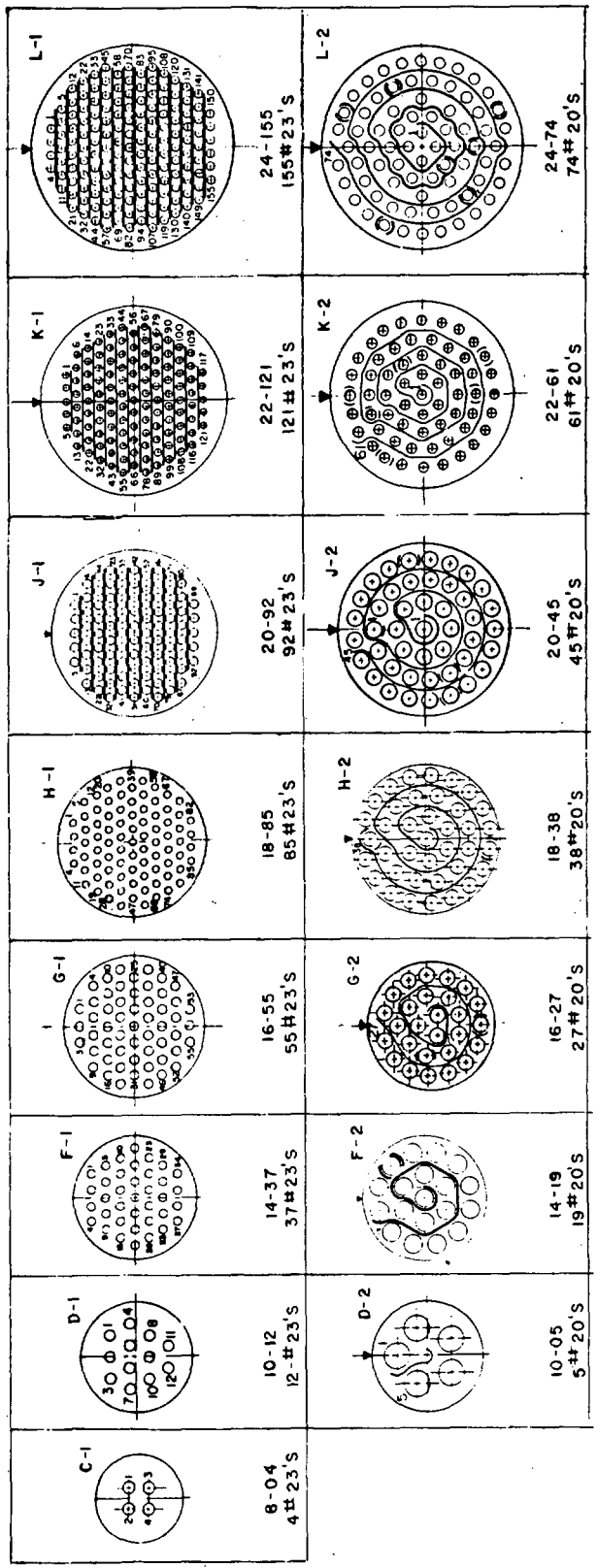


FIGURE 24. Insert arrangement summary.

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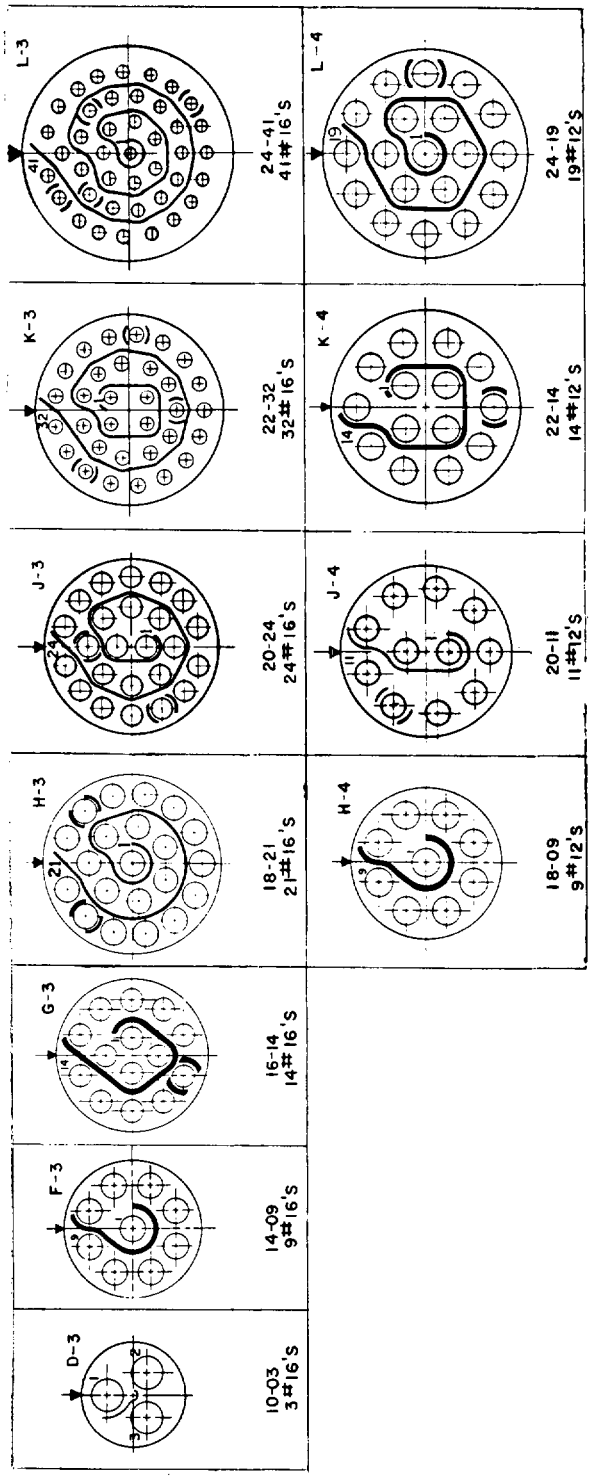


FIGURE 24. Insert arrangement summary - Continued.

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6.10 Insert arrangement summary. Figure 24 provides a summary of insert arrangements and shell sizes.

Custodians:
Navy - AS

Review activities:
Navy - SH, EC, OS
DLA - ES

User activities:
Navy - MC

Preparing activity:
Navy - AS

Agent:
DLA - ES

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